

**GREEN
CLIMATE
FUND**

Meeting of the Board
17– 20 October 2018
Manama, Bahrain
Provisional Agenda Item 17

GCF/B.21/10/Add.20

26 September 2018

Consideration of funding proposals – Addendum XX

Funding proposal package for FP098

Summary

This addendum contains the following three parts:

- a) A funding proposal summary titled “DBSA Climate Finance Facility” submitted by the Development Bank of Southern Africa;
- b) No-objection letters issued by the national designated authorities or focal points; and
- c) Environmental and social report(s) disclosure;

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled “DBSA Climate Finance Facility” submitted by the Development Bank of Southern Africa is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

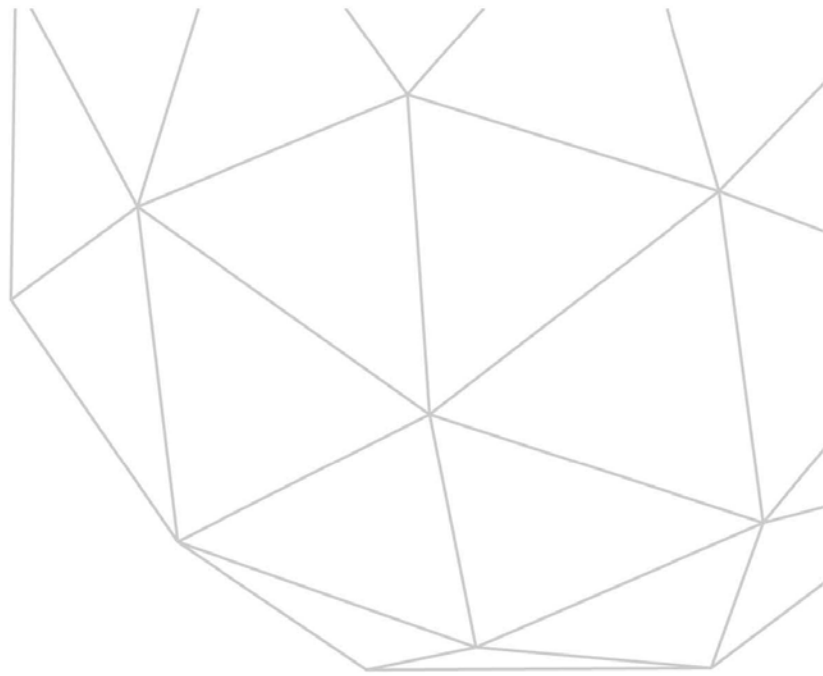
The funding proposal package for FP098 was formerly submitted for the Board's consideration at its twentieth meeting and it remains unchanged.

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Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Programme Title: DBSA Climate Finance Facility

Country/Region: South Africa & Southern Africa Development Community (SADC) Region

Accredited Entity: Development Bank of Southern Africa

Date of Submission: March 29, 2018

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Note to accredited entities on the use of the funding proposal template

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal, or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

Please submit the completed form to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“[FP]-[Agency Short Name]-[Date]-[Serial Number]”

LIST OF KEY ABBREVIATIONS

AE	ACCREDITED ENTITY
BCIC	BOARD CREDIT AND INVESTMENT COMMITTEE
CFF	CLIMATE FINANCE FACILITY
CFU	CLIMATE FINANCE UNIT
CEA	CONTROLLED ENVIRONMENT AGRICULTURE
CFL	COMPACT FLUORESCENT LAMPS
CGC	COALITION FOR GREEN CAPITAL
C&I	COMMERCIAL AND INDUSTRIAL
CPI	CLIMATE POLICY INITIATIVE
DBSA	DEVELOPMENT BANK OF SOUTHERN AFRICA
DEA	DEPARTMENT OF ENVIRONMENTAL AFFAIRS
DoE	DEPARTMENT OF ENERGY
DST	DEPARTMENT OF SCIENCE AND TECHNOLOGY
DWS	DEPARTMENT OF WATER AND SANITATION
EE	ENERGY EFFICIENCY
EV	ELECTRIC VEHICLE
EIA	ENVIRONMENTAL IMPACT ASSESSMENT
ERR	EARLY REVIEW REPORT
ESCO	ENERGY SERVICE PERFORMANCE CONTRACTORS (ESCO)
DFI	DEVELOPMENT FINANCE INSTITUTIONS
EIB	EUROPEAN INVESTMENT BANK
FOLU	FORESTRY AND OTHER LAND USE
GHG	GREENHOUSE GAS
IC	INVESTMENT COMMITTEE
INDC	INTENDED NATIONALLY DETERMINED CONTRIBUTIONS
IPP	INDEPENDENT POWER PRODUCER
FBL	FOOD POVERTY LINE
FDI	FOREIGN DIRECT INVESTMENT
GEF	GLOBAL ENVIRONMENT FACILITY
GOF	GLOBAL OUTCOMES FUND
GCF	GREEN CLIMATE FUND
IRENA	INTERNATIONAL RENEWABLE ENERGY AGENCY
JSE	JOHANNESBURG STOCK EXCHANGE
JIBAR	JOHANNESBURG INTERBANK AVERAGE RATE
LBPL	LOWER-BOUND POVERTY LINE
M&E	MONITORING AND EVALUATION
MSME	MICRO, SMALL TO MEDIUM ENTERPRISES
NAMA	NATIONALLY APPROPRIATE MITIGATION ACTIONS
NAP	NATIONAL ADAPTATION PLAN
NCD	NEGOTIABLE CERTIFICATES OF DEPOSITS
NDA	NATIONAL DESIGNATED AUTHORITY
NDB	NATIONAL DEVELOPMENT BANK
NDC	NATIONALLY DETERMINED CONTRIBUTIONS
NDP	NATIONAL DEVELOPMENT PLAN
NERSA	NATIONAL ENERGY REGULATOR OF SOUTH AFRICA
NOL	NO-OBJECTION LETTER
NPC	NATIONAL PLANNING COMMISSION
NSSD	NATIONAL STRATEGY FOR SUSTAINABLE DEVELOPMENT AND ACTION PLAN
OECD	ORGANIZATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT
OEU	OPERATIONS AND EVALUATION UNIT
PACE	PROPERTY ASSESSED CLEAN ENERGY
PAR	PROJECT APPRAISAL REPORT
PAYS	PAY AS YOU SAVE
PPU	PROJECT PREPARATION UNIT
PPA	POWER PURCHASE AGREEMENTS
PPP	PUBLIC-PRIVATE PARTNERSHIP

RE	RENEWABLE ENERGY
REIPPPP	RENEWABLE ENERGY INDEPENDENT POWER PRODUCER PROCUREMENT PROGRAMME
RFP	REQUEST FOR PROPOSALS
SA	SOUTH AFRICA
SADC	SOUTH AFRICA & SOUTHERN AFRICA DEVELOPMENT COMMUNITY
SAPP	SOUTHERN AFRICAN POWER POOL
SARB	SOUTH AFRICAN RESERVE BANK
SCF	SUPPLY CHAIN FINANCE
SME	SMALL AND MEDIUM-SIZED ENTERPRISES
SPIPPP	SMALL PROJECTS INDEPENDENT POWER PRODUCER PROCUREMENT PROGRAMME
SPU	STRUCTURED PRODUCTS UNIT
SSEG	SMALL SCALE EMBEDDED GENERATION
UPBL	UPPER-BOUND POVERTY LINE
USD	UNITED STATES DOLLAR
USTDA	UNITED STATES TRADE AND DEVELOPMENT AGENCY
WPA	WATER PURCHASE AGREEMENTS
ZAR	SOUTH AFRICAN RAND

A.1. Brief Project / Programme Information		
A.1.1. Project / programme title		DBSA Climate Finance Facility (CFF)
A.1.2. Project or programme		programme
A.1.3. Country (ies) / region		South Africa and Rand-based economies (Swaziland, Namibia, Lesotho)
A.1.4. National designated authority (ies)		South Africa Department of Environmental Affairs (DEA)
A.1.5. Accredited entity		Development Bank of Southern Africa (DBSA)
A.1.5.a. Access modality		☒ Direct ☐ International
A.1.6. Executing entity / beneficiary		Executing Entity: Development Bank of Southern Africa Beneficiary:
A.1.7. Project size category (Total investment, million USD)		☐ Micro (≤ 10) ☒ Medium ($50 < x \leq 250$) ☐ Small ($10 < x \leq 50$) ☐ Large (> 250)
A.1.8. Mitigation / adaptation focus		☐ Mitigation ☐ Adaptation ☒ Cross-cutting
A.1.9. Date of submission		March 20, 2018
A.1.10. Project contact details	Contact person, position	Olympus Manthata, Climate Finance Unit Head
	Organization	DBSA
	Email address	olympusm@dbsa.org
	Telephone number	27 11 313 5238
	Mailing address	1258 Lever Road Headway Hill Midrand 1685 , Gauteng South Africa

A.1.11. Results areas <i>(mark all that apply)</i>	
Reduced emissions from:	
☒	Energy access and power generation (E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.)
☒	Low emission transport (E.g. high-speed rail, rapid bus system, etc.)
☒	Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
☐	Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)
Increased resilience of:	
☐	Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)
☒	Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.)
☐	Infrastructure and built environment (E.g. sea walls, resilient road networks, etc.)
☒	Ecosystem and ecosystem services (E.g. ecosystem conservation and management, ecotourism, etc.)

A.2. Project / Programme Executive Summary (max 300 words)

Please provide a brief description of the proposed project/programme, including the objectives and primary measurable benefits (see [investment criteria in section E](#)). The detailed description can be elaborated in [section C](#).

The DBSA Climate Finance Facility Programme (“the Programme” or “CFF”) is a lending facility that aims to address market constraints, play a catalytic role with a blended finance approach, to increase climate related investments in the Southern African region. The lending facility will consist of credit enhancements focused on first loss or subordinated debt and tenor extensions to catalyze private sector climate investments. The CFF will be a first-of-its-kind application based on the Green Bank model, adapted for emerging market conditions. It offers globally significant proof-of-concept value to middle and lower income nations seeking to address market barriers and quickly scale up the high levels of private investment required by Paris climate commitments.

The CFF Programme will fill market gaps and crowd-in private investment, targeting commercially viable technologies that cannot currently attract market-rate capital at scale. It will focus on infrastructure projects that mitigate or adapt to climate change and utilize two main instruments: subordinated debt / first-loss and credit enhancements such as tenor extension to projects that are commercially viable but not currently being financed by the private sector banks. It should be noted that CFF will not fund projects that could be funded solely by commercial banks. Projects will be required to demonstrate that they are focused on technically and economically feasible transactions where there is market interest but limited capital availability due to specific financing gaps and barriers. Developers will also be required to show that project cannot be fully financed by the private sector. Projects will be required to show that they involve financial participation by one or more private sector financial parties and that it will have some transformative effect on markets in terms of scale, improved private sector participation, confidence in clean energy investments, or other aspects. The activities of the CFF are expected to result in reduction or avoidance of 29,727,942 tonnes of carbon dioxide equivalent (t CO₂ eq through support of climate mitigation measures alone during the lifetime of the programme. In terms of the potential adaptation impact through water efficiency, treatment and clean water outputs in the private sector, the CFF has the potential to create at least 132 jobs through installation and save 22 600 jobs through the installation of water systems (avoiding dismissal due to water shortages). The expected number of indirect beneficiaries is 410 963.

The DBSA's CFF will deliver outcomes related to each of the GCFs Investment Criteria including: 1) mitigation of climate change via expanding private investment in climate-friendly infrastructure; 2) providing a replicable model for building nation-specific capacity to scale up climate finance in support of Paris climate goals; 3) supporting at least eight of the UN Sustainable Development Goals supporting: access to clean water, renewable energy generation, economic growth, development of sustainable infrastructure, reduced inequality in terms of energy access and access to clean water, sustainable cities and communities, and taking climate action; 4) providing necessary financial capacity to meet socio-economic needs related to climate friendly and clean water infrastructure; 5) expanding national capacity to achieve established climate goals working through local stakeholders, and 6) increasing market efficiency through a leverage approach to scaling up private investment.

A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)	13/06/2018
Expected financial close (if applicable)	Within three months after relevant agreements and terms with GCF and other DFIs are finalized (Fall 2018)
Estimated implementation start and end date	Start: <u>01/11/2018</u> End: <u>01/11/2023</u> (5 years)
Project/programme lifespan	20 years (Construction and implementation).

B.1. Description of Financial Elements of the Project / Programme							
B.2. Project Financing Information							
	Financial Instrument	Amount	Currency	Tenor	Pricing		
(a) Total project financing	(a) = (b) + (c)	170.55...	million USD (\$)				
(b) GCF financing to recipient	(i) Senior Loans						
	(ii) Subordinated Loans	55.....	million USD (\$)				
	(iii) Equity		Options				
	(iv) Guarantees		Options				
	(v) Reimbursable grants *		Options				
	(vi) Grants *	0.61.....	million USD (\$)				
* Please provide economic and financial justification in section F.1 for the concessionality that GCF is expected to provide, particularly in the case of grants. Please specify difference in tenor and price between GCF financing and that of accredited entities. Please note that the level of concessionality should correspond to the level of the project/programme's expected performance against the investment criteria indicated in section E.							
	Total requested (i+ii+iii+iv+v+vi)	55.61.....	million USD (\$)				
(c) Co-financing to recipient	Financial Instrument	Amount	Currency	Name of Institution	Tenor	Pricing	Seniority
	Subordinated Loans	55	million USD (\$)	DBSA	(15) years max (15) years max		
	Subordinated Loans	59	million USD (\$)	Other*			
	Grant	0.61	million USD (\$)	DBSA			
	Grant	0.33	million USD (\$)	Convergence			
* Please provide a confirmation letter or a letter of commitment in section I issued by the co-financing institution.							
(d) Financial terms between GCF and AE (if applicable)	In cases where the accredited entity (AE) deploys the GCF financing directly to the recipient, (i.e. the GCF financing passes directly from the GCF to the recipient through the AE) r if the AE is the recipient itself, in the proposed financial instrument and terms as described in part (b), this subsection can be skipped. If there is a financial arrangement between the GCF and the AE, which entails a financial instrument and/or financial terms separate from the ones described in part (b), please fill out the table below to specify the proposed instrument and terms between the GCF and the AE.						
	Financial instrument	Amount	Currency	Tenor	Pricing		

			<u>Options</u>	() years	() %
Please provide a justification for the difference in the financial instrument and/or terms between what is provided by the AE to the recipient and what is requested from the GCF to the AE.					

B.3. Financial Markets Overview (if applicable)

Below is a summary of comparable data on the financial market overview for the CFF target countries (ZAR zone countries) with a specific focus on the commercial banking system, banking assets and interest rates in those countries. The financial market in South Africa is the largest market but with similar characteristics to the other ZAR zone countries as the major commercial banks that we have consulted in the design of the CFF operate throughout the 4 countries as shown below. In addition to the financial market overview, a macro-economic outlook is also important which substantiates the above view. The economies of the 4 countries are in many ways integrated and dependent on each other. Therefore even the market dynamic exhibit similarities even though there are different levels of development and the currencies are pegged to the South African “rand”. The value chains for the various sectors that the CFF intends to support are also significantly linked. In this regard, we have also attached as annexure, a summary of macro-economic overview of each of the CFF targeted countries which we encourage should be looked at in conjunction with the financial market overview below.

SOUTH AFRICA - FINANCIAL MARKET OVERVIEW

Banking assets

As the below table indicates, total banking assets total R5.16 trillion¹ as of December 2017.

Table 1: Bank and Mutual Bank overview from September 2017 – December 2017²

Line item	Sep, 2017	Oct, 2017	Nov, 2017	Dec, 2017
Bank and Mutual Banks (R millions)				
Total equity	442,066	439,946	443,962	454,744
Total assets	5,116,490	5,122,536	5,105,244	5,155,621
Banks (R millions)				
Required liquid assets	218,334	220,000	221,496	
Actual liquid asset holdings	481,188	482,664	494,045	

Debt Capital Market

Bond market³

“Over a period of 10 years, net debt issuance in the domestic primary bond market translated into an increase of 246% in the total outstanding nominal value of listed debt securities in issue on the JSE to R2.7 trillion as at the end of November 2017. The general government⁴ sector increased its dominance due to national government’s need to finance sizeable budget deficits. Banks, non-bank private corporations and public corporations maintained their contribution, while that of

¹ On 22 February 2018 \$1 = R11,68 (over the course of Feb the rand range was R12,12 to R11,57 to the dollar)

² South African Reserve Bank Quarterly Bulletin, December 2017

³ South African Reserve Bank Quarterly Bulletin, December 2017

⁴ Only national and local government issue bonds

private sector securitised bonds declined significantly following the global financial crisis. Over R2.6tr worth of bonds were issued (new) from 2011 – 2017 on the JSE.

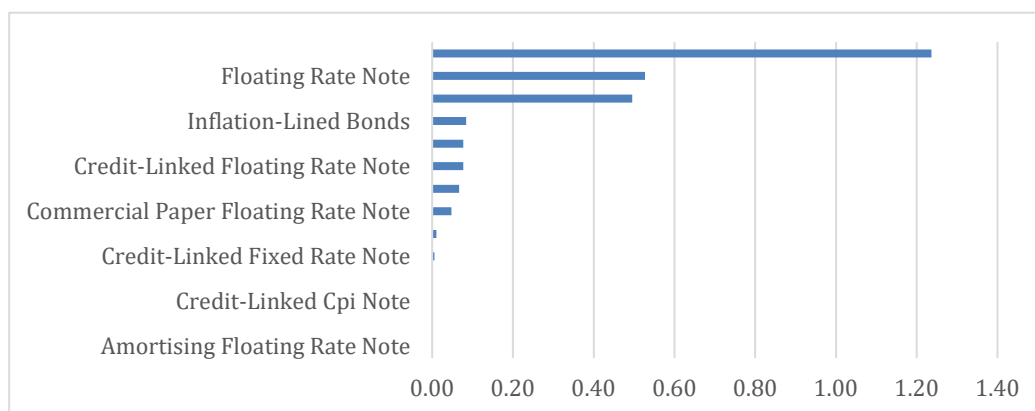


Figure 1: Bonds issued on the JSE between 2011 - 2017 (ZAR, trillion)⁵

Green Bonds

“South Africa’s first green bond (the first on the continent) was launched by the Industrial Development in 2012 – a R5bn bond for investment in clean energy infrastructure. The City of Johannesburg followed as the first local municipality to list a green bond on the JSE in 2014 – a R1,46 bn issuance to finance initiatives such as biogas to energy and the Solar Geyser Initiative (City of Johannesburg, 2014). More recently, the City of Cape Town’s R1bn Green Bond – which was launched in July 2017 – will fund projects aligned with the City’s Climate Change Strategy, including measures to secure long term water availability.”⁶ “The City of Cape Town green bond, which is already listed on the JSE’s bond market, was certified by the Climate Bonds Initiative, while international ratings agency Moody’s also awarded it an excellent GB 1 rating. The projects to be funded by the green bond are a mix of adaptation and mitigation initiatives and include the procurement of electric buses, energy efficiency in buildings and water management initiatives like water meter installations and replacements, water pressure management, and the upgrade of reservoirs. The City raised R1 billion for these projects when it issued the 10- year bond in July 2017.”⁷

Equity Market Capital

Share market overview⁸

Table 2: Market Capitalization of the Johannesburg Stock Exchange⁹

	Week Ended 26.01.2018	Week Ended 27.01.2017	% Change Year-on- Year
JSE Market Capitalisation (R bn)	15 833.11	14 085.71	12.41

⁵ Johannesburg Stock Exchange

⁶ GCX, <http://www.gcx.co.za/green-bonds-a-growing-green-market/>

⁷ Johannesburg Stock Exchange, <https://www.jse.co.za/articles/Pages/JSE-launches-Green-Bond-segment-to-fund-low-carbon-projects.aspx>

⁸ South African Reserve Bank Quarterly Bulletin, December 2017

⁹ JSE Weekly Statistics, Week Ended 26 January 2018

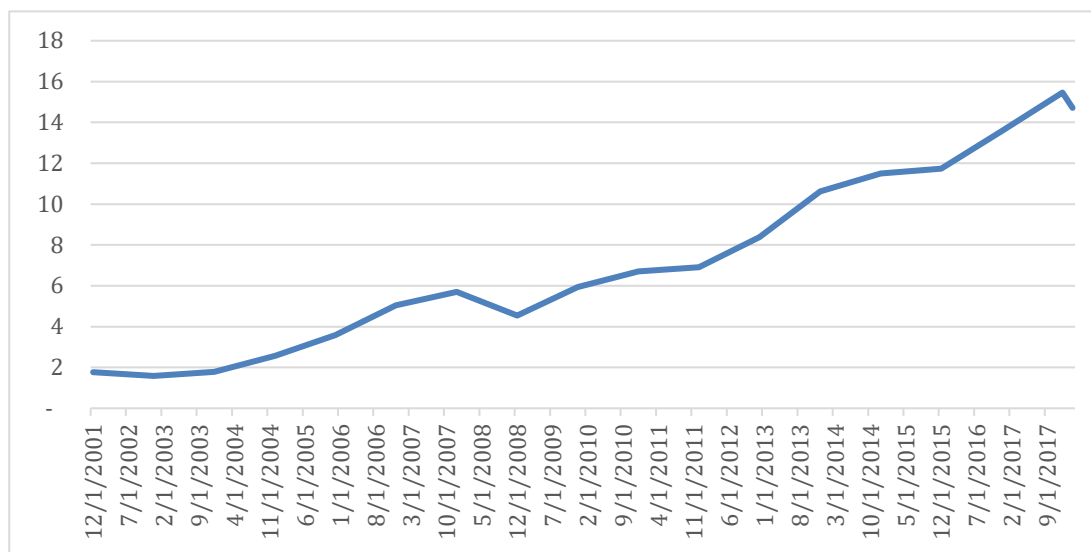


Figure 2: Evolution of JSE Market Cap (ZAR, trillion)¹⁰

Private Equity Survey Results (2016)¹¹

Southern Africa's private equity industry, including both government and private funds, had R171.8 billion in funds under management (FUM) at 31 December 2016, an increase from R158.5 billion at 31 December 2015. This represents a compound annual growth rate of 11.4% since 1999, when the survey first began. Funds returned to investors in 2016 totalled R18.3 billion, a 123.2% increase from the R8.2 billion returned to investors during 2015.

Venture Capital Survey Results (2016)

The reported value of venture capital (VC) investments made during 2016 was R872 million (2015: R372 million), an increase of 134%. The total number of investments increased from 93 in 2015 to 114 in 2016, an increase of 23%.

Current Rates

Table 3: Current rates overview (South African Reserve Bank)

Indicator	Key	Value	Last Period
Money Market Rates	(6)		
Repo rate		6.75	2018/02/02
Sabor	(5)	6.743	2018/02/01
Overnight FX rate	(5)	6.95	2018/02/01
Treasury bills - 91 day (tender rates)		7.21	2018/02/01

¹⁰ Johannesburg Stock Exchange

¹¹ SAVCA Private Equity Survey (2016), <http://savca.co.za/wp-content/uploads/2017/06/SAVCA-2017-Private-Equity-Industry-Survey-electronic.pdf>

Treasury bills - 182 day (tender rates)		7.28	2018/02/01
Treasury bills - 273 day (tender rates)		7.19	2018/02/01
Treasury bills - 364 day (tender rates)		7.09	2018/02/01
NCD's - 3 months (closing rates)	(4)	7.11	2018/02/01
NCD's - 6 months (closing rates)	(4)	7.58	2018/02/01
NCD's - 12 months (closing rates)	(4)	7.92	2018/02/01
Jibar - 3 months		7.13	2018/02/01
Prime lending rate (predominant rate)		10.25	2018/02/02
Capital Market Rates	(7)		
6.75% 2021 (R208) (closing yields)		7.32	2018/02/01
10,5% 2026 (R186) (closing yields)		8.44	2018/02/01
3-5 years (daily average bond yields)		7.37	2018/01/31
5-10 years (daily average bond yields)		7.78	2018/01/31
10 years and longer (daily average bond yields)		9	2018/01/31

Key

1. Source: JSE Limited.
2. The weighted average exchange rate of the rand is based on trade between South Africa and its twenty most important trading partners. Index: 2010=100.(See Article in June 2014 Quarterly Bulletin for various weights).
3. Weighted average of the banks' daily rates at approximately 10:30am.
4. As from 1 April 2004, the rate reflected related to negotiable certificates of deposit (NCD's) (instead of promissory notes).
5. Replacing the Saonia+, Saonia, Forex Forwards and carry rate from 27 March 2007.
6. Bankers Acceptances are no longer issued or traded. The last BA matured on 13 September 2013 and as a consequence publication of the BA rate has been discontinued.
7. The R157 government bond matured on 15 September 2014 and is therefore no longer published. The rate on the R208 bond is published in its place.

Corporate bonds¹²

For corporate bonds with a 5 year maturity (issued between 2011 – 2017), the average rate was 8.94% (approximately 200 issuances)

NAMIBIA – FINANCIAL MARKET OVERVIEW

Namibia has one of the most sophisticated, diverse, and developed financial systems in Africa. Most of the country's financial institutions are privately owned and maintain strong links with South African institutions as noted below. The

¹² Johannesburg Stock Exchange

Banking Sector in Namibia comprises of eleven banking institutions, which can be categorised as follows: 5 commercial banks (including the newly authorized Bank BIC); a small medium enterprise bank; an E-bank, two micro-finance banking institutions, a branch of foreign banking institution and a Representative Office. These banking institutions are the primary mobilisers of funds from the public and the main sources of financing, which support business operations and economic activities in Namibia.

Amongst the authorised banking Institutions, the top four (excluding Bank of Namibia) are South African banks as shown below. Another South African bank (ABSA) also has a representative office in the country.

- [First National Bank Namibia Limited](#)
- [Bank Windhoek Namibia Limited](#)
- [Standard Bank Namibia Limited](#)
- [Nedbank Namibia Limited](#)
- Bank BIC Namibia Limited
- Trustco Bank Namibia Limited
- Letshego Bank Namibia Limited (micro-finance banking institution)
- Banco Atlántico (branch of foreign banking institution)
- ABSA Representative Office

Banking Assets

Namibia's banking sector's liquid assets increased by N\$5 billion, standing at roughly N\$15 billion during the last quarter of 2017.

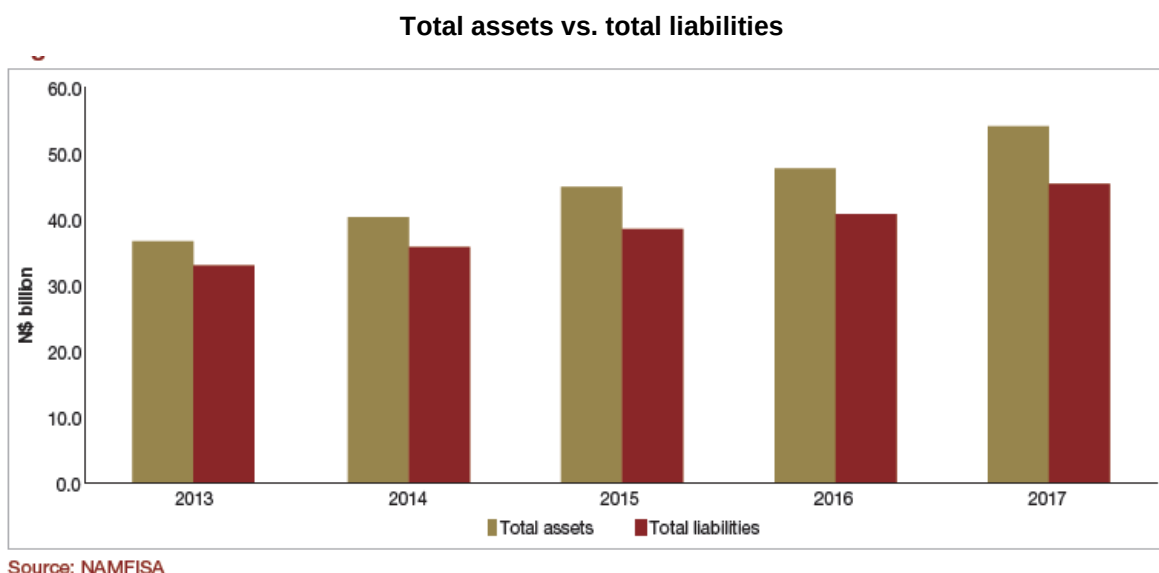
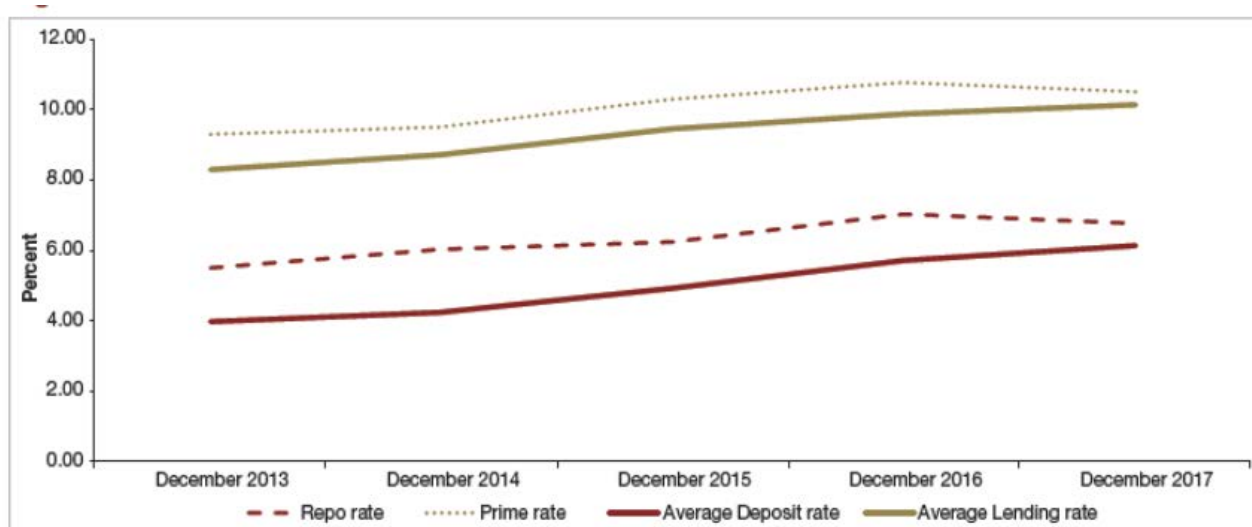


Figure 3: Total Assets versus Liabilities in Namibia

As illustrated in Figure 3, the industry remained solvent as its total assets consistently exceeded total liabilities. During the period under review, the sector remained solvent as it held excess assets to the tune of N\$8.7 billion. Further, the industry solvency position improved due to an increase in excess assets by N\$2.1 billion translating to a 27.4 percent increase during 2017.

Interest Rates



Source: Bank of Namibia

Figure 4: Historic Interest Trends in Namibia

Historically, interest rates in Namibia are relatively still low and continue to facilitate the servicing of loans. In addition, local banks remain in a favourable position as their assets could reprice faster than their liabilities should interest rates start to increase, which may aid financial stability according to the Bank of Namibia.

LESOTHO – FINANCIAL MARKET OVERVIEW

Similar to Namibia, Lesotho's banking industry is comprised of four banks, of which the big three are subsidiaries of South African banks. Lesotho's banking system remained sound throughout 2016. The banks remained adequately capitalised and continued to maintain a good quality of assets. The level of liquidity within the banking system remained fairly adequate to withstand shocks and the industry remained satisfactorily profitable. Banks' exposure to foreign exchange risk has decreased during the review period manifesting decreased sensitivity to market risk. Credit extension to the private sector declined significantly during 2016 in an environment of low economic growth.

Banking Assets

The industry's total assets amounted to M13.2 billion in December 2016, largely dominated by foreign subsidiaries which accounted for 97 per cent. The banking sector's total assets continued to be largely concentrated in the top-two banks, with a significant proportion of 80.5 per cent in December 2016. The first two quarters of 2016 saw the industry's total assets increasing due to growth in financial intermediation, while contraction in total assets was observed in the second half of the year, due to a decline in balances with South African banks and investment securities.

Source: Central Bank of Lesotho

Interest Rates

Lesotho's Central Bank does not define a benchmark interest rate. The main interest rate is the 91-day Treasury bill. The benchmark interest rate in Lesotho was last recorded at 6.27 percent. Interest Rate in Lesotho averaged 12.07 percent from 1980 until 2017, reaching an all time high of 20.42 percent in December of 1990 and a record low of 5.18 percent in December of 2013. The MPC decided to increase the CBL rate from 6.75 per cent to 7.00 per cent per annum in January 2018.

Source: Central Bank of Lesotho

SWAZILAND – FINANCIAL MARKET OVERVIEW

The banking sector, composed by 4 commercial banks remains well capitalized and profitable and dominates the Swazi financial system, with South African banks accounting for the lion's share of the market.

Banking assets

The banking sector assets grew by 5.0 per cent to E17.9 billion during the year ended March 2017 from E17.0 billion at the end of March 2016. Net loans and advances consisted of 62.0 per cent of the banking sector assets, rising marginally from 61.0 per cent reported at the end of March 2016. In line with the Bank's mandate of promoting a sound and safe banking system and ensuring in the domestic banking sector, the Central Bank increased the minimum liquidity requirement for commercial banks with effect from 1st July 2016 while the reserve requirement remained unaltered.

Source: Central Bank of Swaziland Annual Report 2016/2017

Interest rates

During the 2016/17 financial year, the structure of Interest rates was revised upwards on two occasions; following the tight monetary policy stance pursued by the Bank of Swaziland. Therefore, in May 2017, the policy (discount) rate increased by 50 basis points from 6.5 per cent in April 2016 to 7.0 per cent and in January 2017 by an additional 25 basis points to 7.25 per cent at the end of March 2017. Similarly, the prime lending rate increased from 10.0 per cent in March 2016 to 10.5 per cent and further to 10.75 per cent in March 2017.

Average treasury-bill rates increased by a higher margin of 154 basis points from 6.65 per cent in March 2016 to 8.19 per cent in March 2017 and turned around to be more than the Republic of South Africa (RSA) treasury-bill rates since September 2016. The spread between the local and RSA treasury bill rates widened considerably from 49 basis points to 102 basis points. This was attributed to the government's efforts to attract investors amidst the cash shortfall challenges during the year under review.

Source: Central Bank of Swaziland Annual Report 2016/2017

Concluding Remarks

As can be seen from the data above for all the ZAR zone countries, the financial market context is similar in respect of interest rate trends. Furthermore, the banking system is similar which is corroborated by the significant presence of South African banks in these markets. Section B3 albeit for South Africa aims to show some of the reasons why the barriers for climate investments in particular exists based on the financial market context albeit for South Africa. In light of the similarities between the financial markets noted above for CFF targeted countries and the supplementary data provided for the other ZAR zone countries, the financial market in the CFF target countries has significant capital and assets (ie banking sector total assets USD> 400bln) which can be channeled to climate finance. However climate finance is limited due to barriers and lack of appropriate financial mechanisms.

In our extensive consultation with the major commercial banks with operations in all the 4 target countries, the following were identified as key barriers for investment in climate related projects:

- **Tenor:** Tenor extension was seen by the commercial banks as the primary intervention that will unlock investment in climate mitigation and adaptation projects. The driver for this barrier is mainly regulatory constraints i.e. Basel III which impacts all the 4 countries. Commercial banks across all the CFF target countries cannot provide tenors > 7-8yrs as confirmed in our stakeholder engagements. **The proposed CFF tenor extension of up to 15 years is therefore seen as an essential intervention by the CFF in unlocking this barrier for the CFF targeted sectors.**
- **High interest rates:** High interest rates is also a barrier to investment in climate mitigation and adaptation. With this intervention in place, projects will be co-funded between the CFF funders and the commercial banks. The GCF concessionality would make it possible for the projects to benefit from a favorable blended rate which would be significantly lower than what is currently offered in the market (as per the data provided). Based on the envisaged concessionality from the GCF, the CFF could offer rates as low as 8.65% to projects (current rates

could range from JIBAR plus 5% to 7%). This rate blended with the current commercial bank rates would be quite competitive in the market.

- **Perceived high investment risk of climate mitigation and adaptation projects:** The ability of the CFF to offer subordinated debt/first loss will help unlock the market.

In conclusion, the current observation is that investment in climate mitigation and adaptation is a small fraction of the total investment portfolio of the various target countries. Although actual data with regards to the exact figures is not readily available, it was confirmed through our consultation with the various stakeholders including commercial banks that the above statement applies to the 4 target countries. In this regard, the CFF is seen as a key intervention to address this investment gap.

Please fill out applicable sub-sections and provide additional information if necessary, as these requirements may vary depending on the nature of the project / programme.

C.1. Strategic Context

Please describe relevant national, sub-national, regional, global, political, and/or economic factors that help to contextualize the proposal, including existing national and sector policies and strategies.

The strategic context has a particular focus on South Africa since the pipeline is expected to be dominated by projects in South Africa during the initial phase of the implementation of CFF. The latter part of this section provides a context to the other ZAR based countries in Southern Africa which includes Namibia, Swaziland and Lesotho. In recent years South Africa has faced significant mega-challenges in the water and energy sectors. Security of supply and affordability of electricity and water both present challenges for the South African Economy. The 'water crises' challenge was ranked by the World Economic Forum (WEF) as the third highest risk for doing business in South Africa in 2017 'energy price shock' is not far behind (WEF 2017). In order to overcome the infrastructure backlog and meet the sustainable development goals (SDGs) South Africa needs to invest some 2,4% of its GDP ¹³ in infrastructure. Innovative finance, public private partnerships, enabling legislation and the mobilisation of private capital are all needed to address climate related challenges. In the South African context specifically, the need for patient, long tenor capital is required to crowd in significant private sector investment.

South Africa's National Strategic Planning:

South Africa's primary government planning document is the National Development Plan (NDP) which was launched in 2012. The transition to a lower carbon and climate resilient society is articulated in the National Development Plan (NDP). The NDP advocates a transition to an environmentally sustainable, climate change resilient, low-carbon economy and just society, which will be enabled by:

- Coordinated planning and investment in infrastructure and services that take account of climate change and other environmental pressures;
- Implemented adaptation and national development strategies;
- Focus on becoming a zero-waste society;
- Growth in the renewable energy sector;
- Domestic manufacturing of renewable energy technologies coupled with job creation;
- Reducing the country's carbon emissions;
- Conservation and restoration of protected areas through policy and regulatory frameworks for land use; and
- Public investment in the development of resilient and environmentally sustainable strategies.

The National Development Plan recognises that climate change will have a significant impact on the water sector. The National Climate Change Response White Paper states that the water sector is a critical component to climate change mitigation and adaptation. The National Climate Change Response Strategy for the Water Sector (DWS, 2014) emphasises good water management which is a critical foundation for adaptation to water-related climate change impacts. South Africa published its Intended Nationally Determined Contribution in September 2015 (DEA, 2015) in preparation for climate change talks at the Conference of the Parties (COP) 21 in Paris. Due to South Africa's unique developmental challenges, the document focused on eliminating poverty while also working towards reducing greenhouse gas emissions.

The adaptation goals were developed taking cognisance of the NDP sectoral goals and timelines. Carbon taxes 2017: In the first draft Carbon Tax Bill, it was envisaged that a carbon tax proposed by the National Treasury will be implemented, commencing in 2017 at a rate of R120 per ton of carbon dioxide equivalent (CO₂e) on direct emissions, increasing by 10% per annum until 2020. Tax-free allowances of between 60% and 95% would be provided, based on trade exposure, fugitive emissions, carbon budgets compliance and other factors (National Treasury 2016). The National Treasury published the Second Draft Carbon Tax Bill in December 2017. It will be used for introduction in Parliament, as well as for public comment and convening of public hearings by Parliament early in 2018. Following this process, a revised Bill is expected to be formally tabled in Parliament by mid-2018.

¹³ Global Infrastructure Outlook, G20, Oxford Economics

Table 4: Summary of the markets of interest for CFF and the primary drivers in South Africa

Market	Description	Primary driver	Policy	Incentives
Energy: solar PV and commercial and industrial energy efficiency (C&I EE)	Coal dominates the South African energy mix. Opportunity in particular for transition into renewable energy (RE) for electricity. Need for innovative finance to drive investment in private infrastructure.	Cost competitiveness of RE technology. Current economic viability.	Strong policy support	Niche incentives, primarily tax rebates.
Waste to Energy	Replacing coal with waste-to-energy.	Niche opportunities for cost and carbon advantages.	Some policy support	
Water	SA is drought prone. Publicly provided infrastructure at the 1:50 year level of assurance is failing businesses in the private sector. Innovative finance can increase investment in private infrastructure.	Assurance of supply driver. Increased resilience to climate related droughts and price shocks.	Strong policy support	No direct incentives for private investment in water.
Rural and peri-urban alternative service delivery	Solar Energy, Water and Waste to Energy in the rural or peri-urban environment is a large underserved market. Cost competitive solutions will require innovative finance.	Social welfare driver. Increasing access to services drives wealth creation and an ability to pay for services.	Strong policy support.	Typically seen as a Government responsibility; but incentives and grants also available.

Table 5: Size of CFF target market sectors in South Africa^{14,15}

Market	Current Market Size	Potential Market Size (CFF Pipeline)	Technology Costs	Price Trajectory	CO2/Resiliency impact
Energy: Solar PV	R280 mil ¹⁶	R25 bn (2030)	R10 k-R15 k per kW	Rapid Decrease	1.03 kg reduced CO ₂ emissions per kWh
Energy: C&I EE	R750 mil	R41 bn (2030)	R1 per kWh/annum	Flat	1.03 kg reduced CO ₂ emissions per kWh

¹⁴ Green Cape Market Study, 2018

¹⁵ See Illustrative CFF pipeline as attached to application and section on CFF origination channels

¹⁶ Biogas cost per MW is 3-4 times that of PV, which results in a bigger market size, given the smaller installed MW cost of PV. Given the base date of 2015, there was a relatively small uptake of PV in the market. Only as of 2016-18 is 165MW being installed per annum in the C&I PV space. Also, REIPPP, initiated in 2010, provided a major boost for biogas project development, although many of these projects are now stranded and looking for private off takers (Green Cape).

Water Efficiency and Treatment	N/A	R16 bn (2030)	R30 k per kL/day	30% decrease over 15 years.	Increased resilience to water shortages and security of water supply
PV Without Grid Connection	Private Market R100 – 500 million	R20bn (2030) (if grant funded)	Paraffin: ~R10-15/litre Candles: ~R2/candle Solar PV: ~R15/W	variable	Variable, depending on fuel displaced
Rural Biomass to Energy	R500 mil	R2 bn (2030)	R42 k per kW	Marginal Decrease	1.03 kg reduced CO ₂ emissions per kWh
Rural Water (CEA)	R128M	R1 bn (2030)	Price decreases expected for new tech. Cost of water low; driver of uptake not based on water costs alone		

Table 6: Job creation potential of CFF target markets in South Africa
(A)

Market	Potential Job creation			Number of Businesses Impacted
	Total	Male	Female	
Solar PV	28,525	15,689	12,836	15,740
C&I EE	24,958	13,727	11,231	137,942
Waste to Energy	1,065	586	479	86
Water efficiency and treatment	1,103	607	496	1,398

A number of current projects were investigated through stakeholder discussions and desktop research to determine a Job/unit (MW, ML/Day, million invested) value. The Job/unit has been multiplied with the 2030 forecasted capacity (installed capacity, ZAR invested) to determine the 2030 potential jobs. A average size per project was also determined to provide an indication of the number of businesses that would be impacted by 2030.

(B)

Market	2030			
	Capacity	Unit	Jobs	/Unit
Solar PV	2,500	MW	11.41	/MW
C&I EE	R 41,596	million s	0.6	/million invested
Waste to Energy	55.10	MW	19.4	/MW
Water efficiency and treatment	838.85	ML/Day	1.3	/ML/Day

Table 7: Current incentives applicable to CFF market focus areas

Section 12 tax incentive	Name of incentive	Description
Section 12B	Deduction in respect of certain machinery, plant, implements, utensils and articles used in farming or production of renewable energy	Allows taxpayers to claim a capital allowance for qualifying movable assets, owned by the taxpayer and brought into use for the first time for the production of electricity from renewable energy for activities performed in the carrying on of any trade and in the production of income. The rate of the allowance was amended on 1 January 2016 from 50/30/20 basis over three years to one year (100%).
Section 12I	Additional investment and training allowances in respect of industrial policy projects	As of 1 January 2015, greenfield "industrial policy projects" above R50 million may deduct an additional investment allowance equal to 55% of the cost of a manufacturing asset or up to R900 million.
Section 12J	Deductions in respect of expenditure incurred in exchange for issue of venture capital company shares	Allows investors to get a tax rebate, to the value of their investment in the year that it is made, if they invest in a qualifying venture capital company (VCC) that in turn invests in qualifying SMEs. This investment will then remain tax free as long as it is held for five years.
Section 12L	Deduction in respect of energy efficiency savings	As of 1 November 2013, taxpayers are allowed to claim a deduction for most forms of energy-efficiency savings that result from activities performed in the carrying on of any trade and in the production of income. For assessments between 1 November 2013 and 28 February 2015, the deduction is calculated at 45 cents per kWh; for assessments commencing on or after 1 March 2015, the deduction is calculated at 95 cents per kWh.

Source: <http://www.investmentincentives.co.za/>

Energy context: Renewable Energy and Energy Efficiency

In South Africa, the energy situation includes a recent history of power shortages, rising electricity prices, high carbon emissions and energy security concerns.¹⁷ Currently 94% of South Africa's electricity is generated from coal¹⁸ and over 30% of South African gasoline and diesel needs are supplied by liquefied coal¹⁹ and this dependence on coal presents a fundamental challenge to reducing GHG emissions and achieving climate targets and NDCs. South Africa's 2010 total GHG emissions including forestry and other land use (FOLU) were 518.2 million tonnes carbon dioxide equivalent. In the same period, the country's total GHG emissions excluding FOLU were 544.3 million tonnes tCO₂e. Energy related emissions (fossil fuel combustion, transport and fugitive emissions), dominated South Africa's emissions profile, and contributed 75.1% of the total 2000-2010 emissions. The energy intensity of the South African economy has resulted in an emissions profile that differs substantially from that of other developing countries at a similar stage of development. Energy demand is forecast to double from roughly 44 GW to 80 GW by 2025 and best-case scenarios based on current policy anticipates that only about 42% (18 GW) of the required new build will be provided by renewables.²⁰ Estimated

¹⁷ https://thegiin.org/assets/documents/pub/Southern%20Africa/GIIN_SouthernAfrica.pdf

¹⁸ IEA, 2013

¹⁹ World Coal Association, 2015

²⁰ South Africa Department of Energy

costs for realizing South Africa's NDCs include \$349 billion over 2010 to 2050 for decarbonized electricity, over \$8 billion per year for renewable electricity, \$40 billion per year for next generation vehicles, \$.5 billion per year for public transportation infrastructure, and an additional \$7 billion in short term investment over the next five years for adaptation.²¹

²² Consequently, it is clear that significant additional low-carbon investment is needed to meet South Africa's climate goals and support international agreements intended to limit global warming to below 2°C.

In 2010, South Africa established the Integrated Resource Plan ("IRP") which is focused on achieving long term security of supply and sustainability through an energy mix including coal base-load energy, nuclear base-load energy, renewable energy and gas base-load energy. The primary objective of the IRP 2010 is to determine long term electricity demand and detail how this demand should be met in terms of generating capacity, mix of technologies, and the timing and envisaged investment costs from 2010 up to 2030. The plan prioritised deployment and installation of renewables to accelerate the establishment of a local industry. In addition to all existing and committed power plants, the plan included the deployment of 17.8 GW of RE by 2030. It also served as input and guide to other national development planning functions such as economic development, funding, environmental and social policy formulation.

South Africa has abundant renewable energy sources and renewable energy resource maps have shown that the country has some of the most viable wind and solar resources in the world. The country has an estimated technical capacity of 70 000 MW which could potentially contribute to 20 per cent of the country's electricity needs by 2025 (SAWEA, 2010). The IRP, which is seen as South Africa's blueprint for the energy mix in the period up to 2030, indicated the government's clear intention to not only diversify its energy mix away from the conventional fossil fuel-fired power generation, but also to take advantage of the possibilities relating to the green economy to mitigate the challenges of climate change, create new industries and job creation. Central to the South African energy policy paradigm shift has been the need to diversify energy sources, addressing the challenge of increasing demand for electricity, reducing carbon emissions mostly from coal generation as well as the need to ensure security of supply and access to energy.

National Energy Efficiency Strategy (NEES) 2005, 2008, post-2015: The aim of the original NEES (2005) was 'to explore the potential for improved energy utilisation through reducing the nation's energy intensity (thus reducing greenhouse gas emissions), and decoupling economic growth from energy demand. A recent amendment to the Electricity Regulation Act has removed the requirement for licencing of generation facilities smaller than 1MW. Municipalities around the country are putting in place feed-in tariffs for embedded generation, allowing excess energy to be sold back to the grid. These movements, together with indications that distributed generation will be included in South Africa's updated Integrated Resource Plan, the primary document that determine South Africa's electricity mix, highlights a strong push towards a cleaner distributed energy system.

Water context: Treatment and Management

South Africa is ranked as the 30th driest country in the world. It is a highly water-stressed country, with extreme climate and rainfall fluctuations (WRI 2015). Despite being a water-scarce country, consumption is around 233 litres/capita/day (l/c/d), compared to the international benchmark of around 180 l/c/d (DWS 2017a).²³ South Africa has a reliable yield (i.e. the supply from current infrastructure) of around 15 billion m³/year (at 98% assurance of supply), of which the majority is from surface water (68%) and return flows that support surface water (13%).

The Western Cape, Eastern Cape, Northern Cape, Free State and parts of KwaZulu-Natal are currently drought-stricken and water scarcity is a significant challenge. On 9 February 2018 the drought in the Western Cape was declared a national disaster. The Western Cape has been the worst affected and has been declared a disaster area since May 2017. Most municipalities in the province have implemented water restrictions, which have become increasingly stringent over the course of the drought.

Current usage is estimated to be 15-16 billion m³/year, and in many water supply systems the water usage exceeds the reliable yield (DWS 2017b). This essentially means that whilst the water supply sources can meet this increased usage,

²¹ Energy Services: Market Intelligence Report 2016

²² IEA

²³ These figures are based on the system input volume divided by the population served. The system input volume includes commercial and industrial demand and water losses through infrastructure leaks etc.

the assurance of supply is less than 98%, meaning water shortages are likely to be experienced more often than two of every 100 years.

Reconciliation strategy studies forecast future demand and then provide recommendations on how to reconcile the gap between demand and supply by developing new water resources or reducing demand. These recommendations form the long-term reconciliation strategies of municipalities, and have been developed for all the major water use areas in SA (including the Western Cape Water Supply System). Typically, the strategies include plans for water conservation and water demand management programmes, groundwater projects and other augmentations schemes that serve to diversify supply and increase resilience.

Waste context: Waste to energy and recycling

South Africa's current waste economy is estimated to be worth ZAR15 billion and provides 29,833 people with employment. However, a GreenCape report (2017) suggests that R17 billion worth of resources could be unlocked if 100% of the identified 13 waste streams could be recycled. The report also suggests that municipalities should look into innovative and alternative ways to fund support functions in the waste sector in order to explore its true revenue potential. "This includes partnerships with the private sector, where provincial and national government play a role in the implementation of extended producer responsibility (EPR) and allow for the levies charged to assist with some of the infrastructural and operational demands. According to the report, legislation has been passed to help unlock the potential R17 billion of material currently being landfilled that could be recycled. The Western Cape provincial government in South Africa has plans to implement public-private partnerships that will potentially attract a further R1.3 billion and create approximately 1,600 jobs in the next five years. The report highlights current opportunities for businesses and investors in the Western Cape's waste sector which are primarily focused on recyclables (plastics), organics, e-waste, and construction and demolition waste. Opportunities were identified across the value chain in the collection, sorting, processing and treatment of waste. These opportunities are possible provided there is access to waste, at least one market for recovered materials, and a viable business case for the recovery of materials.

South Africa is faced with crises in the provision of Energy and Water services. The National Development Plan is the country's primary planning document. The NDP frames the required response to these crises to be delivered through a climate sensitive lens. Infrastructure build requirements for South African are some 2,4% of GDP - this cannot be met by the public purse alone. Innovative financing programmes such as the CFF, and private to private infrastructure provision, will all play a key role in the future of South Africa's infrastructure development.

SOUTHERN AFRICA CONTEXT – NAMIBIA, LESOTHO AND SWAZILAND

Energy sector in Southern Africa

Countries in the Southern Africa region face large climate mitigation and adaptation challenges. Electricity is mostly fossil fuel-based, millions of people lack any access to electricity, many lack access to clean water supplies, and high economic reliance on agriculture leaves many people exposed to climate risks.²⁴ Based on significant increases in electricity costs over the past ten years and anticipation that they will again double in the next 5 years,²⁵ the private sector is focused on expanding self-generation as a critical element of producing adequate and affordable sources of power. Further, best-case scenarios based on current policy anticipates that only about 42% (18 GW) of the required new generation in South Africa - which is the major supplier of power via Eskom to the SAPP serving Southern Africa - will be provided by renewables.²⁶ The growing green agenda within the private sector and southern Africa, provides additional impetus for the private sector to expand self-generation with a focus on renewables. The CFF will be well-timed and highly relevant to supporting this emerging drive for private sector and low-carbon power in South and Southern Africa.

²⁴ <http://www.sadc.int/themes/infrastructure/en/>

²⁵ South Africa Department of Energy

²⁶ South Africa Department of Energy

The Southern African Development Community (SADC) region, access to electricity is minimal with only 5% of rural areas in the region having access to power. Overall, the SADC falls behind other regions in Africa regarding access to electricity. While 24 % of the SADC region's residents have access, 36% of the Eastern Africa Power Pool area's residents are connected, as are 44 % of the Western African Power Pool's residents. Recent power shortages in South Africa also impacted the Southern Africa Power Pool ("SAPP"), as the majority of the region is reliant on South Africa for power supply which accounts for approximately 75% of the existing generation capacity across the SAPP.²⁷ In general low tariffs, poor project preparation, issues with Power Purchase Agreements, and absent regulatory frameworks stunt investment and financing in the energy sector. Further, weak infrastructure and foreign commitments restrict use of the region's abundant petroleum and natural gas resources and pricing and infrastructure hurdles undermine development of the region's renewable energy potential.²⁸

Lesotho

Electricity in Lesotho is supplied by Muela's hydroelectric power station (72 MW), with some small hydro projects also providing generation capacity. However, peak demand has reached up to 140 MW in the past, forcing Lesotho to meet the deficit with more expensive imports from Mozambique and South Africa. Based on 2013 data, Lesotho's national electrification rate reached 17%, (8% in rural areas, 43% in urban areas).

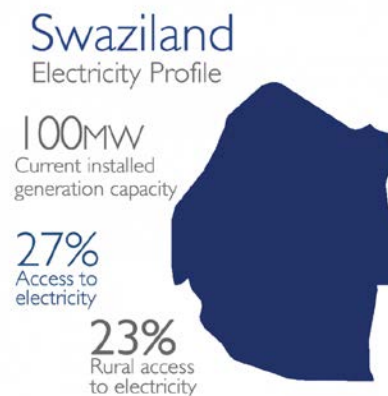
Figure 5: Power sector in Lesotho
Source: USAID Power Africa Fact Sheet



Swaziland

The Swaziland Electricity Company, the state-owned utility, operates four hydropower stations with a combined capacity of 60 MW. These stations represent 15-17% of the total electricity consumed in the country. The rest is imported, mostly from South Africa and Mozambique, as a part of the Southern African Power Pool (SAPP). The country's available renewable energy resources, like biomass and solar, are currently being assessed, with the goal of complementing existing hydropower generation and reducing reliance on electricity imports. Based on 2013 data, Swaziland's national electrification rate reached 27%, (23% in rural areas, 40% in urban areas).

Figure 6: Power sector in Swaziland
Source: USAID Power Africa Fact Sheet



Namibia

The power sector in Namibia has undertaken a number of reforms aimed at attracting IPPs by providing a stable investment environment. Such reforms include the horizontal unbundling of regional distribution companies and the establishment of transparent tariff setting procedures, all overseen by the sector regulator, the Electricity Control Board (ECB). While the country's generation mix is comprised primarily of hydropower, the majority of electricity is imported from Southern Africa Power Pool (SAPP). Recently, the Government of Namibia has taken steps to increase generation from other generation sources through a renewable energy feed-in-tariff (REFIT) for a total of 70 MWs and an upcoming solar procurement for 37 MWs. With the support of USAID, the REFIT was adopted by ECB to grant 14 licenses to IPP developers for 5 MW projects by establishing a predictable enabling environment for renewable energy integration.

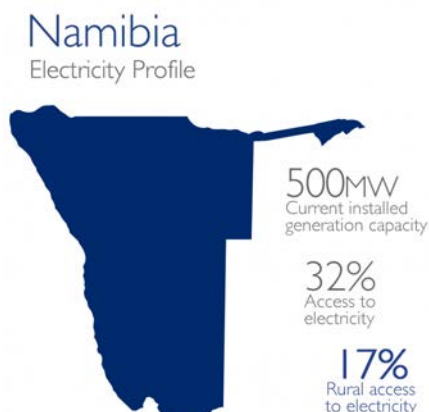


Figure 7: Power sector in Namibia
Source: USAID Power Africa Fact Sheet

Water sector in Southern Africa

Southern African countries also lack access to clean drinking water. Due to a lack of infrastructure, only 61% of the region's population have access to safe drinking water and 39% have access to adequate sanitation facilities. Overall, as a changing climate alters global water distribution, water usage is becoming an increasing concern across the region. With a reliance on agriculture, Southern Africa needs efficient water infrastructure that supports the basic needs of its population and overall economic development.²⁹ The water context for countries being targeted by the CFF is discussed below.

Swaziland

Surrounded on three sides by South Africa and bordering Mozambique to the east, Swaziland is a small country with a population of around 1.3 million people. In its mountains, valleys, savannah and jungle, four distinct climates shape people's lives. Almost one in three people do not have clean water and two in five have nowhere to go to the toilet. This situation has a big impact on health and contributes to an average national life expectancy of just 49 years. Swaziland also has the world's highest proportion of people living with HIV/AIDS – almost a third of the population. They are extremely vulnerable to diarrhoeal diseases caused by a lack of toilets and hygiene, and need clean water for medication to be effective.

In recent years, a serious water shortage gripped the city of Mbabane, Swaziland's administrative capital due to the El Nino phenomenon. The rainfall season from January-March 2017 and November-December 2017 has alleviated some of the impacts of the drought, reaching normal to above normal levels across most of the country, however prolonged drought conditions continue to be felt in Shiselweni and Lubombo regions affecting food and water security for a significant part of the population. At the beginning of the emergency response in February 2016, 350,000 people were affected by the prolonged drought, including 189,000 children. The 2017 Vulnerability Assessment indicated that 137,380 people (16 per cent of the population), including 74,185 children, remain food insecure. Source: UNICEF Report (2017)

Lesotho

The Lesotho Vulnerability Assessment Committee (LVAC) results in June 2016 showed that 679,437 people in the rural areas of Lesotho are in need of humanitarian assistance between June 2016 and May 2017. The most immediate identified humanitarian needs caused by the El Nino-induced drought are food and water. The LVAC found that about 17 per cent of households were using water from unprotected sources. In Maseru, Mokhotlong and Thaba-Tseka districts, 22-32 per cent of people were reported to be using water from unprotected sources. The vulnerability caused by El Nino compounded existing high rates of poverty, and HIV infection rates which are among the highest in the world.

Namibia

The Permanent Secretary in the Ministry of Agriculture, Water and Forestry recently noted recently in early 2018 that "if no sufficient rain is received, water bodies such as rivers, dams and groundwater aquifers could be very low or even dry up". The water situation in Namibia can be described as manageable but demanding in terms of ensuring availability of

²⁹ SADC, <http://www.sadc.int/themes/infrastructure/water-sanitation/>

the required quantity and quality. Plans are being implemented ranging from upgrading water transfer schemes such as Karst to Windhoek, and development of new schemes such as the expansion of the Windhoek aquifer (drilling of numerous boreholes in Windhoek). Other plans he mentioned include feasibility studies for major projects such as Kavango link and coastal water supply and the implementation of trans-boundary water projects such as the Kunene trans-boundary water supply project and the Stampriet groundwater project.

The context discussed above for CFF targeted countries in the Southern African regions presents several opportunities for potential projects which could be supported through the CFF in the targeted sectors especially renewable energy and water.

C.2. Project / Programme Objective against Baseline

Describe the baseline scenario (i.e. emissions baseline, climate vulnerability baseline, key barriers, challenges and/or policies) and the outcomes and the impact that the project/programme will aim to achieve in improving the baseline scenario.

Background

The CFF will focus on specific climate investment opportunities based on the country needs and sectorial priorities identified in the Intended Nationally Determined Contributions (INDCs) for each CFF targeted country (see annexures 1A-D).. The attached INDCs highlights emission baselines in all the key sectors including energy (RE and energy efficiency), waste, transport and water. The CFF will align and adapt its investment focus in accordance with the primary mitigation and adaption interventions outlined in the respective INDCs for each target country. In Lesotho for example, the country's GHG emissions are minimal due to its predominant dependence on hydropower. The proportional contribution of three key sectors is agriculture (63%), energy (31%) and waste management (6%). In the case of Lesotho, the CFF investment focus will take into account Lesotho's socio-economic circumstances i.e an economy dependent on natural resources, a low but growing energy sector and industrial sector that is still in its infancy. It is noteworthy that the Lesotho Energy Policy 2015 envisions that energy shall be universally accessible and affordable in a sustainable manner, with minimal negative impact on the environment and sets goals to reduce in particular fuelwood usage in the national energy consumption including other fossil fuels. Emissions from energy sector make a total of 1,079.43 Gg CO₂eq mostly from residential fuel combustion (51%) followed by combustion of liquid fossil fuels in the transport subsector (29%). Residential emissions emanate from the use of biomass, coal, Liquid Petroleum Gas (LPG) and paraffin. CO₂ is the major contributor, making about 75% of total sectoral emissions. Energy sector emissions showed a consistently increasing trend reaching 30% between 1994 and 2000.

Furthermore energy efficiency has large mitigation potential in the residential sector in Lesotho. Households commonly use incandescent electric bulbs for lighting. Replacement of these bulbs with Compact Fluorescent Lamps (CFLs) can save as much as 80% of electricity used for lighting. If 40,000 households are using electricity for lighting, and each installs two (2) CFLs replacing incandescent bulbs, the reduction potential in a CFL programme would be about 3,700 tCO₂e per year based on an average saving of 50 kWh per bulb per year (using the SAPP grid emission factor of 0.92). However, all these efforts are conditional to financial support.

The waste sector in Lesotho is divided into two distinct sectors; the solid waste management (CH₄ emissions) and waste water handling (CH₄ and N₂O emissions). The sector was assessed for emissions from domestic, commercial and industrial waste. The total emissions from this sector are 199.63 Gg CO₂eq over 80% of the CO₂eq emissions is methane. The bulk of these emissions are from industrial waste water handling. Overall, GHG emissions in this sector have doubled since 1994. The sector emissions are driven by the increasing per capita solid waste generation among population especially in the urban areas. Deposal of solid waste to land with relatively deeper and sanitary landfill sites is becoming common practice in urban waste management."

As can be seen from the above example for Lesotho, the INDC presents significant opportunities for CFF to support various private sector climate mitigation inventions across several areas in all the ZAR based countries being earmarked by CFF. The following sections discusses climate vulnerability baselines for South Africa which is deemed to have the largest climate impact potential as noted earlier.

Climate vulnerability baseline³⁰

South Africa's population was estimated to be 55 million in 2014 (StatsSA, 2015a). South Africa has a positive population growth rate (1.65% in 2014/15; StatsSA, 2015a) suggesting that total GHG emissions will increase unless mitigation measures are pursued. Migration is expected to increase the populations of urban areas in Gauteng and Western Cape which could lead to increased pressure on service delivery and competition for resources, exacerbated by climate change (DEA, 2014a). The majority of South Africa's population is vulnerable to the impacts of climate change, with a large segment of the population dependent on free basic services, either as a result of age, disease and/or poverty.³¹ Given that the current vulnerability of human settlements is determined by access to basic services, the type of dwelling, health and age, economic factors, land tenure status and social grants, informal settlements and rural communities are typically the most vulnerable to climate change. Political and economic ideologies are considered to be root causes of vulnerability as they influence the distribution and allocation of resources (van Huyssteen, et al., 2013). Women³² in rural areas are particularly vulnerable to the effects of climate change than men in the same areas, as their roles typically include food production, household water supply and the procurement of energy for heating and cooking (Babugura, 2010).

Addressing the current risks and vulnerabilities of human settlements could potentially result in some of the best returns in terms of reducing climate vulnerabilities, as they are directly linked to human wellbeing and are thus critical for social and economic development. This means that efforts to address climate change vulnerability with regards to human settlements will, by implication, require implementation of adaptation options in other sectors. In terms of the economy, a review of the potential impact of climate change on the economy of South Africa noted that the greatest risk was from impacts to dryland (i.e. rain-fed) agriculture and transport infrastructure (DEA, 2015b; Cullis et al, 2015). Consideration of the economic impacts in different parts of the country noted that it was in particular the rural areas that were most at risk from the potential economic impacts of climate change.

Emission / consumption baselines

Energy

South Africa's total GHG emissions³³ (excluding forestry and other land uses) are 539 112 Gg CO₂eq in 2012. The energy sector in South Africa continued to be the main contributor of GHG emissions (>75%), providing 77.3% in 2000 and increasing to 79.4% in 2012. The agriculture sector was also highlighted as significant, contributing 9.6% to the total GHG emissions in 2012. Within the energy sector, the total GHG emissions increased by 25.0% between 2000 and 2012, and produced a total accumulated GHG estimate of 5 121 939 Gg CO₂eq over the 12-year period. The main source of emissions in this sector is CO₂ from fossil fuel combustion and majority of emissions are from energy industries (63.2%), followed by 10.7% from transport and 9.1% from manufacturing industries and construction. The residential and commercial sectors are both heavily reliant on electricity for meeting energy needs, contributing a total of 211 817 Gg CO₂eq and 189 211 Gg CO₂eq of emissions, respectively. South Africa's primary energy mix is dominated by fossil fuels at over 85% of the energy mix, with the bulk of the remaining energy coming from burning wood. Electricity generation follows a similar trend with around 92% of electricity generation coming from fossil fuels.

³⁰ The statistics comes directly from the South Africa's Draft Third National Communication (TNC) under the United National Framework Convention for Climate Change (DEA, 2017). Chapter 3 provides detail on climate vulnerability in South Africa, including baselines. For the sake of this report, poverty was used as a proxy for vulnerability,

³¹ South Africa has a large portion of the population classified as young (30.2% aged younger than 15) or old (8% aged 60 years or older), and has a relatively high HIV prevalence rate (11.2% of the total population). About 30% of the population is aged younger than 15 years (StatsSA, 2014a). Children are one of the social groups most vulnerable to climate change due to their physical and emotional immaturity. Some of the ailments which are the leading causes of death in children (e.g. malaria) are particularly sensitive to climate change (UNICEF, 2011).

³² Approximately 51% of the population is female (StatsSA, 2015a).

³³ The National Greenhouse Gas Inventory for South Africa provides baselines for the year 2012 and GHG emission trends since the year 2000 by gas and by sector, including energy and agriculture. The inventory was compiled in accordance with the IPCC 2006 Guidelines for National GHG Inventories.

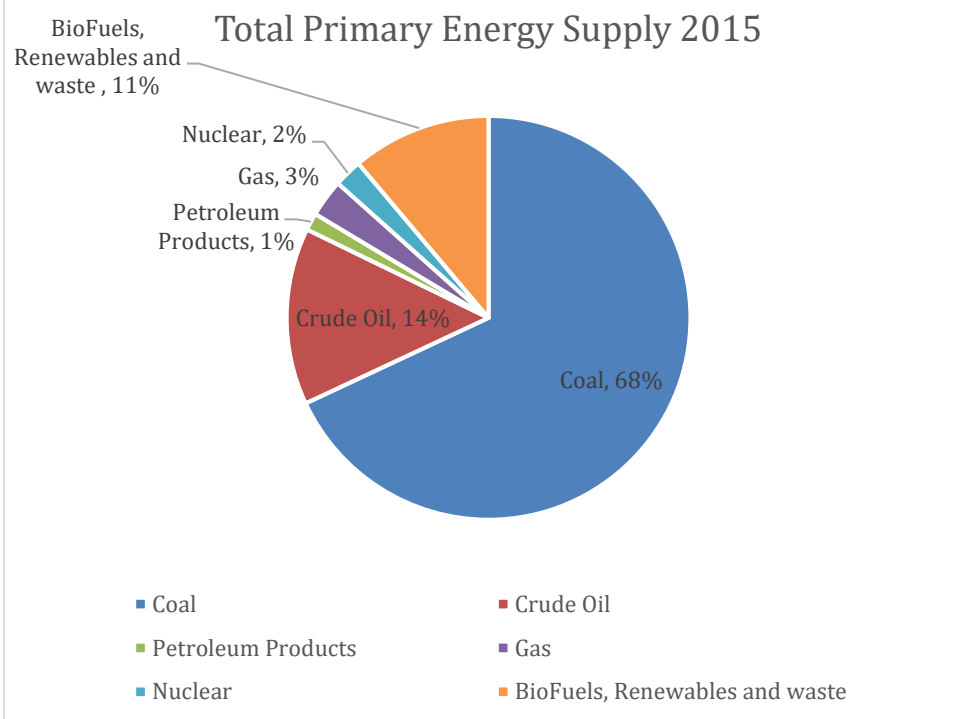


Figure 8: Primary Energy use in South Africa

Energy use in South Africa is primarily from industrial, residential and transport sectors (as seen in 7). Limited requirement for heating in South Africa means that there is very little district heating or cooling, and almost all building heating and cooling comes from electricity.

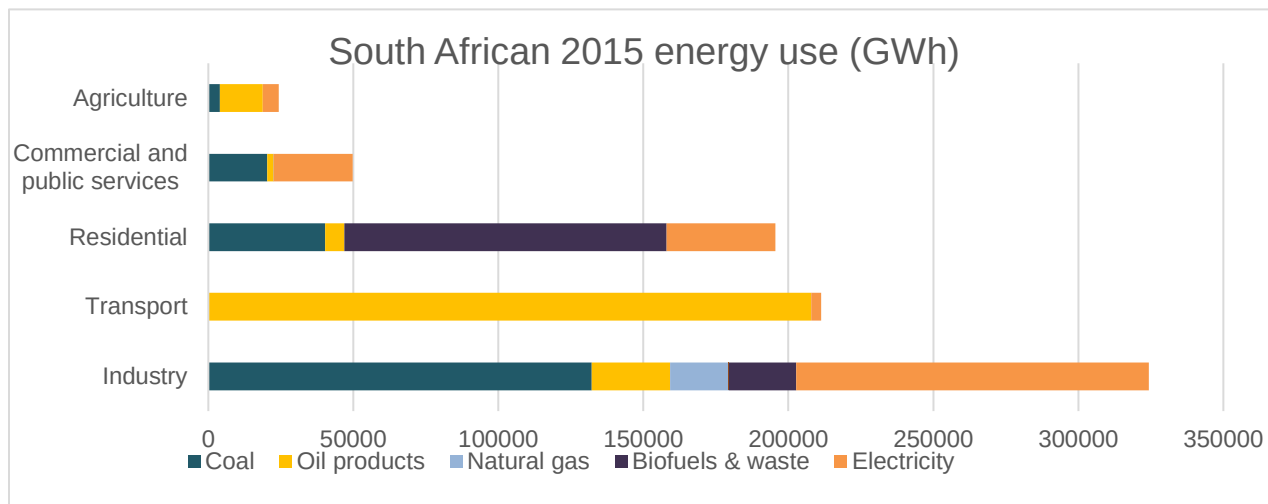


Figure 9: Energy use by sector

The figures clearly illustrate how dependent South Africa is on coal to meet its energy and particularly its needs, respectively.

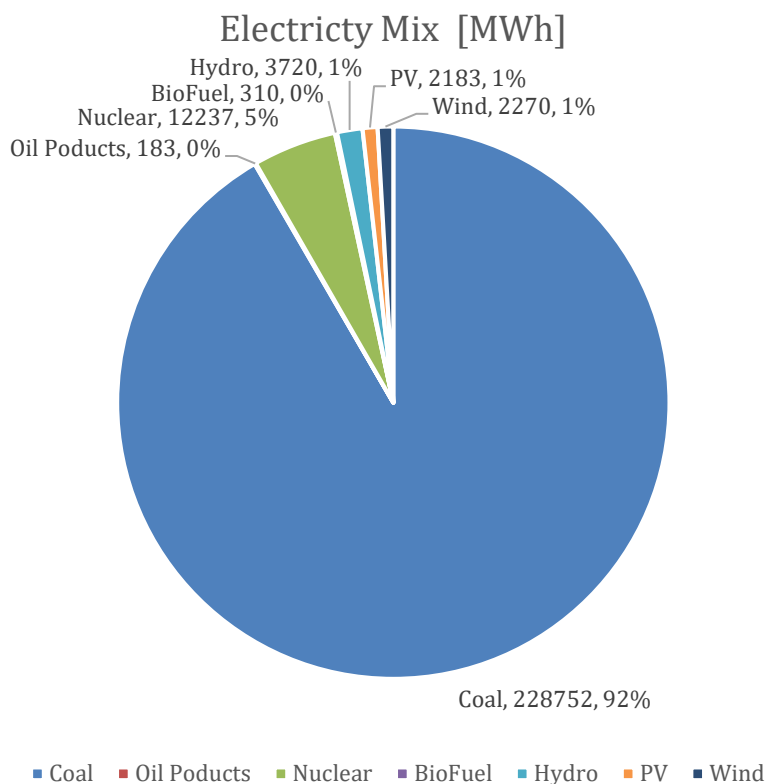


Figure 10: Electricity Generation Mix in South Africa

Electricity prices have been steadily rising in recent years and have approximately doubled since 2010. The rapid price increase for electricity has led to a decrease in electricity consumption per capita (10). South Africa has had a target inflation range of 3-6%. For the years in the chart below the average inflation is about 4,8%.

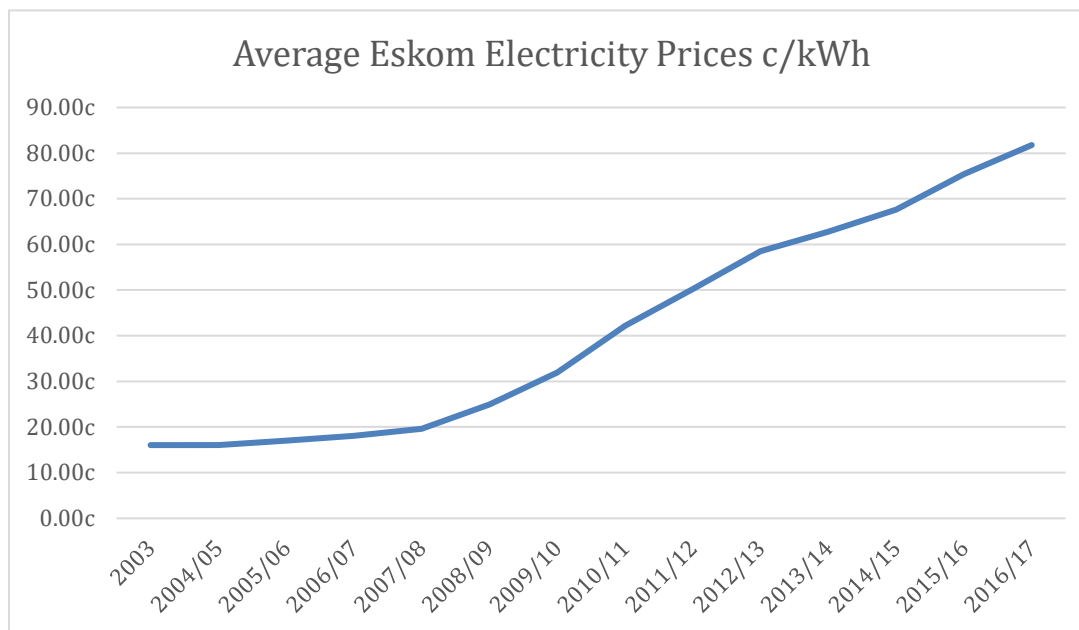


Figure 11: Energy Price Trends in South Africa

Figure 12 below shows the power consumption per capita. There is a clear trend of decreasing usage since 2007. This could be explained by the price increases and a drive towards efficiency; however, this could also be explained through suppressed demand as a result of curtailments in supply. These 'load-shedding' incidents or rolling blackouts were driven by the system operator to prevent the electricity system from collapsing.

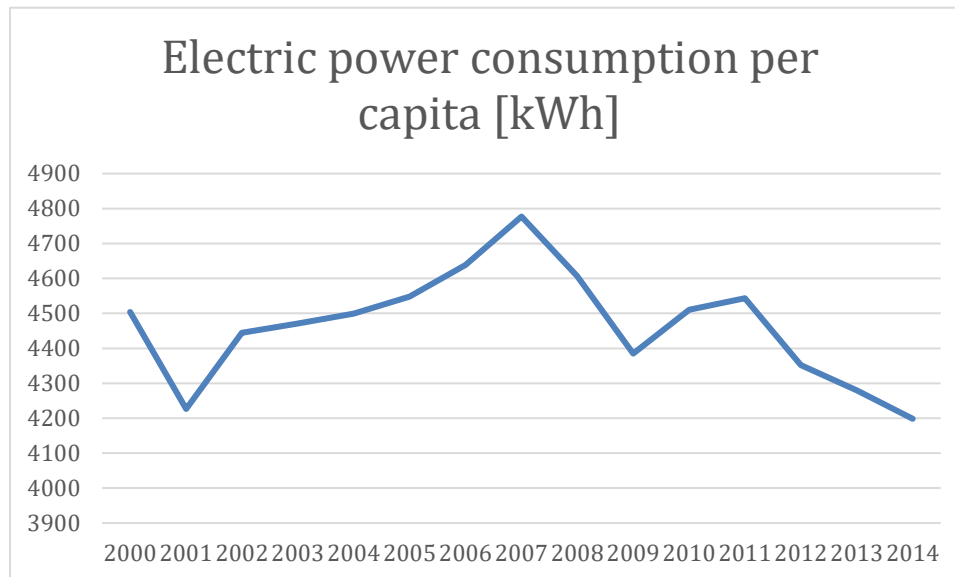


Figure 12 Electricity power consumption per capita in South Africa (<https://data.worldbank.org>)

Finally, electricity spending per capita per annum has increased well above inflation. The consequence of these drastic price increases is that many businesses and domestic consumers have become conscious of electricity prices, and are exploring the opportunities to reduce this cost. Historically, with extremely cheap electricity, services and light manufacturing had little incentive to explore energy efficiency or alternative electricity generation. However, a combination of supply curtailments and increasing cost are driving this market.

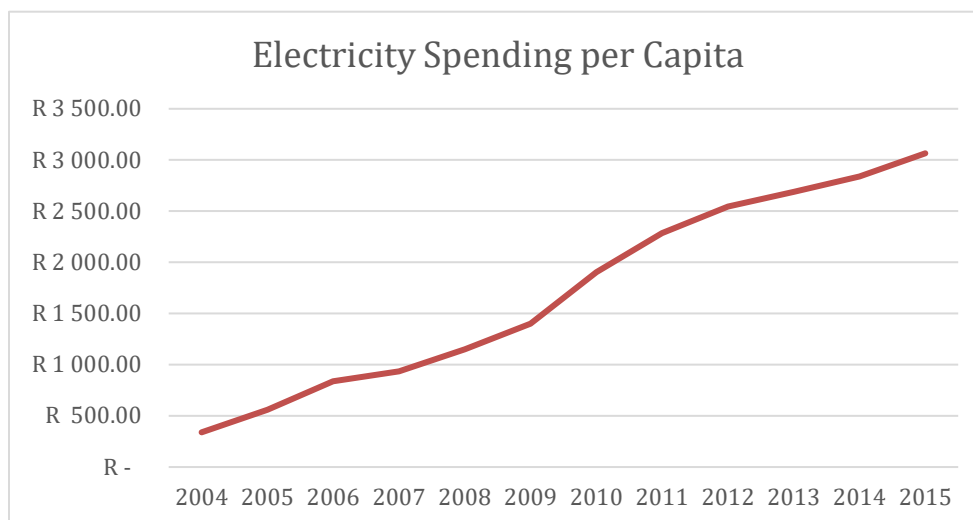


Figure 13: Electricity Spending per Capita (<https://data.worldbank.org>)

Water

South Africa has high levels of variability in rainfall from year to year, resulting in frequent floods and droughts. As a water scarce country, the river systems and aquifers are highly used and developed, and many are already highly degraded. Current water usage is estimated to be between 15 and 16 billion m³ per annum, split between agriculture (62%), municipal (27%), mining (3%), industry (3%), energy (2%) and afforestation (3%). Recent assessments suggest additional water resources need to be developed, with 28% of towns having inadequate water resources, and 50% of towns not

currently implementing water conservation and demand management (WC/WDM).³⁴ Water is unevenly distributed across the country and not always available where needed, resulting in restrictions. There is also extreme inequality in access to water for productive purposes, arising out of the apartheid legacy. Many rural settlements still have insufficient water resources to meet their basic water demands and further groundwater and surface water resource developments are necessary. These challenges are enhanced by evidence of deteriorating water quality of South Africa's major river systems, water storage reservoirs and ground water resources – the core water supply systems that underpin social and economic development in South Africa.

Market Barriers, Challenges and Opportunities by Sector

Solar PV

There is a large market opportunity in the rooftop PV space. While the latest IRP is still under finalization due to political shift moving away from nuclear, based on the IRP update developed in 2013 South Africa's generation mix should support at least 1GW/year of new build solar. It is expected that 500MW of this will come from the rooftop PV space. A database of recorded installations shows that currently only 112MW of rooftop solar was installed in 2017 (this number could be higher as not all installations are captured).³⁵

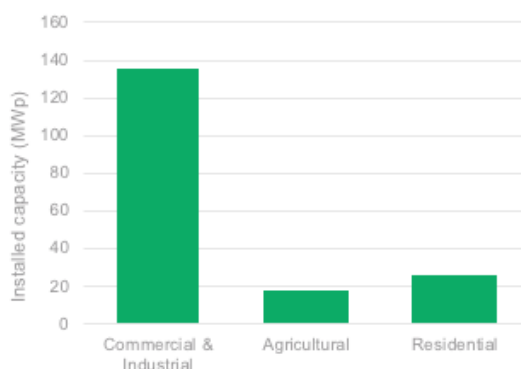
Corporate & Industrial solar

The C&I solar currently represents 70% of the rooftop solar installations. Due to larger installation sizes the cost for solar in the C&I and agricultural space is significantly lower than in the residential space.

Table 8: Solar PV price benchmark

System size	Capital cost of system (cost per kW _p)	ESCo Power Purchase Agreement tariff (price per kWh)
< 100 kW _p	R 13 500 – R 16 000	R 1.20 – R 1.45
> 100kW _p and < 500 kW _p	R 11 500 – R 14 000	R 1.05 – R 1.25
> 500 kW _p	R 10 500 – R 13 000	R 0.90 – R 1.15

Source: GreenCape [GreenCape ES MIR 2018]



Source: [GreenCape ES MIR 2018]

Figure14: Distribution of solar PV installations across end-user segments in South Africa

³⁴ In future some Water Management Areas will need to develop additional local water resource bulk infrastructure. Comprehensive water resource assessments, (or reconciliation studies), in 13 key demand/economic areas, have been completed along with other water resource assessments, (the so-called All Towns Studies), in 905 towns.

³⁵ GreenCape ES Market Intelligence Report 2018

Residential Solar

The urban residential solar market, often referred to as Small Scale Embedded Generation (SSEG), has the potential, with increased regulatory certainty, to be very important in South Africa in coming years. However, in its early stage, it is not envisioned that the CFF will focus on residential solar for two reasons:

- 1) The small ticket size for SSEG projects is likely to be below the CFF's minimum size threshold. This may change in the future as more projects become viable and project pipelines/portfolios grow, and CFF could explore an aggregation, or portfolio financing option to reach appropriate ticket size.
- 2) There are currently uncertainties in market rules and regulations that limit investment in the sector, making financing (CFF or otherwise) challenging until more issues are resolved.

Currently, in South Africa, there is no guaranteed period for continued payment to residential solar SSEG projects. The payment for electricity being fed back onto the grid could end or change in the future, resulting in high levels of uncertainty. That price uncertainty is a risk faced by SSEG customers, as many still rely on the feed-in tariff to create a financially feasible business case. The uncertainties around SSEG projects to ability to sell back to the grid include local municipality regulations, as well as the National Energy Regulator of South Africa (NERSA) rules for SSEG which are meant to be finalized soon in 2018. Further, the fact that only a small percentage of energy is self-consumed in the residential market due to the nature of the load makes these settling rules particularly important.

Available Finance

For directly financed projects, the risk profile and credit worthiness of the offtaker (also the owner of the system) is currently considered, rather than the value of the asset. Banks are using vehicle finance models to finance PV and rates are often significantly above market lending rates (14-18%). Finance is generally available for tenors of 5-7 years, significantly less than asset life. It is also worth noting that the expected norm for payback on an intervention is 3 years. Companies typically do not like investing over a longer period than this or signing contracts for longer than 3 years. This is partly driven by the rate at which legislation changes in South Africa.

Tariffs

There is a large range of different tariffs across this space, and while the average tariff is high enough that it looks like there should be an easy business case, this is not a true reflection of the market. The feed-in tariff for PV is significantly lower than consumption tariffs, and as such the size of systems is limited to the consumption of the off-taker. This means that higher consumption customers with the right load profile can build larger systems and enjoy lower system costs. However larger consumption users are generally on tariffs with lower energy charges and higher demand charges.

Eskom Time of Use Structure - Eskom's time of use tariffs are structured with a high and low season, and peak, offpeak and standard rates for both high and low season. Peak times are morning (7-10am) and evening (6-8pm). Offpeak is before those times and on weekends, with standard time between the peaks. High season is in the 3 winter months from June –August and tariffs are higher for these months. This structure does not benefit solar PV as the bulk of PV production falls outside of peak times and high season months.

Wheeling - While several of the large municipalities in South Africa offer wheeling, most municipalities do not, and while this is in the process of development, wheeling fees will mean that companies will be competing with the Eskom tariff and not their sale tariff.

Tariff certainty - There is no guarantee on the structure of consumption tariffs from year to year. It is not uncommon for municipalities to force customers to change tariff types or discontinue a tariff. For example, large power users are slowly being forced to move to time of use tariffs. This means that while there is currently a business case for PV at a PPA of R1.10/kwh for a customer on a tariff with a charge of R1.20/kwh, switching to a time of use tariff means that while the customers average tariff will not change the value of energy in daylight hours is significantly reduced. Municipalities are moving tariffs to be more cost reflective with the bulk of their costs coming in the form of energy purchases from Eskom. For this reason, long term large scale uptake of PV needs to compete with Eskom's wholesale tariffs. The CFF will focus on longer tenors and driving down debt costs which will allow for PPA rates that are lower (with lower escalators) than Eskom prices. Targeted credit enhancing investments from CFF can help improve project economics to increase project uptake.

C&I Energy Efficiency

The energy efficiency market is still largely untapped, despite the existence of multiple subsidies and tax incentives and no major policy barriers. One of the reasons for the limited interest in this space is the relative savings of interventions relative to the revenue of companies for many EE technologies. There is a cost to investigating projects and for most companies in South Africa energy is not sufficient enough of a cost centre to investigate energy efficiency. In response to this challenge, pilot initiatives undertaken in South Africa such as the Private Sector Energy Efficiency (PSEE) Programme provided energy efficiency services to 1148 companies, and further builds on the NAMA facility bid for funding of energy efficiency services for municipal buildings. The CFF could complement such initiatives introducing a financing assistance component to increase the implementation of capital projects.

Key risks and barriers in the sector include:

- Awareness - This market is limited primarily by awareness, and ability to understand and access incentives. There is more available finance and incentives than are being accessed. "There is an extremely low appointment rate of Energy Managers (7%), a poor adoption rate of Energy Targets amongst firms (31%), a generally low awareness of EE tax incentives offered by government, and amongst the high energy users, there is a low adoption rate of Energy Management systems."(Singh 2017).
- M&V costs to access subsidies - High costs of energy audits to determine feasibility of projects and high M&V costs for accessing incentives also limit the market growth. Another reason for the limited uptake in this space is the relative savings of interventions relative to the revenue of companies for many EE technologies. For example, lighting retrofits have paybacks under a year, while deeper energy retrofit have longer payback and are more poorly understood by the market.
- Opportunity Cost - There is a cost to investigating projects and for most companies in South Africa energy is not sufficient enough of a cost centre to investigate energy efficiency. Energy Savings Companies (ESCOs) can help lower this barrier to uptake by easy and "turnkey" free audits to undercover projects that can be financed via an Energy Savings Performance Contract. However, with energy costs low for many C&I customers, there can be limited impetus to take this first step without a proactive energy manager searching for project opportunities.
- Debt tenor - For deeper energy retrofits with longer paybacks, it can be incredibly difficult for ESCOs to convince companies to sign contract for periods exceeding ~5years. This has limited the ability for long-term energy projects to be implemented. There is a view that the renewal of 5 year contracts is highly likely, but as ESCOs are relatively new in South Africa and as such there is no precedent, it can be hard to secure commercial financing for ESCOs to do these projects. There exists a large opportunity for finance that is able to take the risk of funding a project that is contracted for a shorter duration than the tenor of debt, and useful life of the asset.

Two key areas where the useful life is long enough that are starting to see traction are efficient steam generation and refrigeration. These projects usually have a useful life of around 15 years, and while they can be financed over 5 years effectively there is scope to improve their attractiveness through longer term financing that is able to take on the risk of non-renewal of contracts.

Urban Water

Industrial companies are increasingly investigating the reuse of their effluent (wastewater) streams. Internationally, there is a general move towards zero liquid discharge, and several industries in South Africa already reclaim and reuse significant amounts of wastewater, such as the mining and sugar sectors. In most cases, the effluent requires treatment before it can be reused. There are opportunities for wastewater treatment companies to design, supply, install and maintain these systems.

Given the current drought in the Western and Eastern Capes, there is a drive to install water supply options that mitigate drought risk. In many instances, the decision making for water infrastructure is driven by the fear as well as scarcity's potential impacts on production, rather than a simplistic business case based on water savings alone. The water tariff in South Africa has increased at well above inflation, more than doubling over the last 8 years. While this is extremely significant in percentage terms, the overall cost of water in South African remains very low. The primary driver in the South African context is the need to increase the assurance of supply. South Africa is one of the 30 driest country in the world. Additionally, in 2018, three of the nine regions of South Africa were declared disaster areas for drought. The drought as a driver has caused many private sector companies to recognise that the risk associated with water shortages or big price shocks is not commensurate with the level of focus and investment. Public infrastructure is planned at a 1:50

year assurance, meaning that restrictions are anticipated every 50 years. However, with regions of the country facing a 1 in 400 year drought – the sentiment from the private sector is to supplement the level of supply assurance with privately funded infrastructure. Increasing assurance of supply will require investment in infrastructure that may not have a sufficient business case in the initial years. Exploring a finance product that can sculpt repayments to unlock these opportunities is a large opportunity for the CFF.

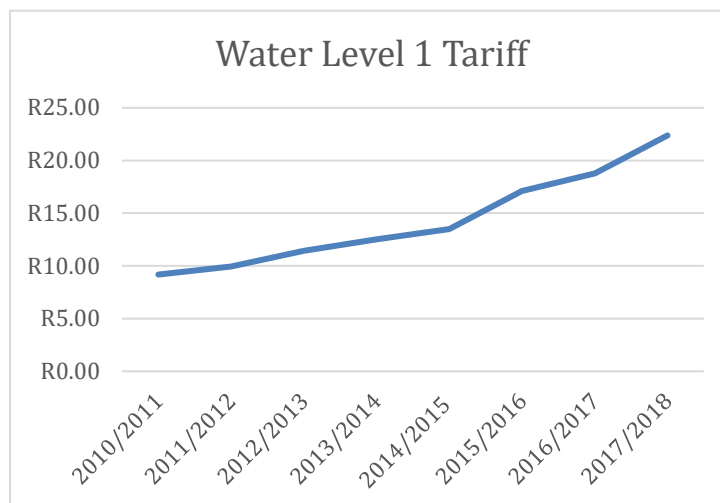


Figure 15:

Water costs are generally low, so water upgrades often have paybacks between 5-9 years. This requires either longer term financing, or a willingness of C&I customers to pay higher annual debt service (i.e. potentially above what they were paying previously for water alone) in exchange for increased resiliency in the face of future scarcity. Risk mitigation and financing from the CFF, in the form of first loss and extended tenors, can help unlock more commercial financing for projects, and at terms that lead to easier customer uptake.

Some key risks in the embedded water sector include:

- **Tariff Risk** - Water tariffs are raised significantly in many area's currently due to drought, but there is no certainty on the future trajectory of these tariffs, or what municipal tariff response will be to decreased demand post drought. As more expensive technologies are brought onto the municipal grids to protect against future drought (e.g. salinization) there is a potential for water tariffs to rise significantly.
- **High capital costs** - High capital costs are one of the main barriers to the adoption of wastewater treatment technologies in the industrial sector, and financial incentives are fairly limited across the province. The capital costs of reuse projects vary significantly, depending on a number of factors, including the quality of the wastewater and the required level of treatment. Treating organic wastewater to potable standards requires significant capital investments and can cost in the order of R20-R120 million per project, with typical project sizes ranging from 0.2 – 1 million litres per day (MLD) produced. A system treating inorganic effluent to potable standards can cost in the order of R10-15 million per MLD produced.
- **Brine disposal** - Depending on the level and type of treatment, a saline brine waste stream may remain after the recovery of water. This presents a challenge for many companies, as most municipalities do not permit the direct discharge of brine into sewerage lines. Companies may then need to incur additional costs to further concentrate the waste stream (e.g. using eutectic freeze technology or evaporative processes).
- **Treatment standards and testing** - There are some legislative barriers for water quality. These are reasonably strict in the South African context and are designed to prevent the spread of water borne disease. These requirements can also add to the ongoing projects costs and performance uncertainties that can limit uptake and drive up risk-adjusted financing costs.
- **Land availability** - Some industrial companies do not have the space to accommodate water reuse infrastructure. Particularly in existing industrial parks or estates.

- Public perception - While reuse is becoming increasingly accepted, some industrial companies are wary of the risk of negative public perception relating to the reuse of effluent.
- Awareness - A lack of knowledge or awareness of water reuse technologies is common among smaller industrial companies, and acts as a barrier to adopting these technology.
- Credit of companies - The credit worthiness of companies pursuing water projects also can be an issue, especially when the business case is largely driven by the risk of loss of supply.

Due to the drought conditions, the private sector in the Western Cape is beginning to investigate and invest in water efficiency interventions and alternative water solutions, including treatment. However, despite the risks of supply shortages, there has been a relatively low uptake of industrial effluent reuse, particularly for treatment to potable standards.

For projects that have high levels of complexity, as well as risks related to environmental performance, brine disposal, or are otherwise technologies or business models seen as risky or with limited track record in South Africa, the CFF can seek to lead the market by providing debt into projects that require additional leverage, as well as offering credit enhancements such as first loss and extended tenor to help facilitate commercial investment at appropriate risk-adjusted pricing. In this way, the CFF can support increased market confidence by creating focus and offering first-mover capital into these new markets.

Urban/Peri-urban and rural solar PV without current grid connection

Electrification in areas lacking a grid connection will typically replace illegal grid connections, candles, paraffin and wood. According to Impact Amplifiers report Energy Provision at the Base of the Pyramid It is estimated that 3.5 million (15%) South Africans are lacking access to a grid connection. About 31% of the South African population lives in rural areas of the country. In these areas, more than 60% of households have no access to electricity. The backlog of connections is projected to grow by 100,000 households per year for the near future.

The pressure to address energy poverty is particularly acute at local government level, where local authorities are responsible for the bulk of electricity distribution to residential households. In the midst of sprawling, and at times unplanned, human settlement growth, extending electrical infrastructure to all its constituents has become a growing and extremely costly challenge for local authorities.

Assuming this total market can be addressed by solar PV services on an individual household basis (i.e. sales of kWh or PV services and not hardware sales) this represents a current annual available market of R2.1 billion. It is expected that this market could grow by as much as 100000 households and R100 million per year. This is a low margin high volume business that is heavily dependent on accessing large numbers of customers.

South Africa represents a difficult yet addressable market under certain conditions. Although not the prime market in Africa there are some key opportunities/drivers that are unlocking this market. These include governmental policies promoting renewable energy, use of subsidies or pro-poor policy, increasing municipal backlogs, local entrepreneurs taking initiative and feed-in tariff for alternative generation.

Governmental policies promoting RE

The electricity supply crisis in South Africa has been stabilised over the past few months and the risk of load-shedding has been largely reduced. Municipalities now have space to look at sustainable energy solutions that will provide an affordable, reliable and secure supply of electricity into the future. Rapidly increasing electricity prices, significant decreases in the price of renewable energy and the economic impact of load shedding have created a large demand for viable alternatives. Based on current spatial growth patterns, the total capital costs of extending infrastructure for the provision of mandated municipal services over the next 10 years is projected at more than R24 billion. It is currently costs a municipality +/- R30 000 per household to extend the municipal grid to the growing urban population and this cost is only projected to increase over the next few years.

The clean energy revolution presents South Africa with an exciting opportunity to tackle numerous issues linked to the universal provision of safe and affordable energy to all citizens. Traditional energy development can be leapfrogged by transitioning directly to a more sustainable technology, while avoiding the dirty energy legacy of most developed countries. The use of subsidies or pro-poor policy allows every indigent customer in South Africa to access +-R100 worth of free electricity (free basic electricity) per month. Free Basic Electricity was launched by the Department in 2003, with the aim

to support indigent households in meeting their basic energy needs. However the programme suffers in most areas because of limited grid availability. After seeing this, the Department realised the need to support indigent households that reside in un-electrified areas with free basic alternative energy. In this regard, the policy on Free Basic Alternative Energy is intended to provide indigent households with alternative energy where electricity is not available.

Financially sustainable socio-technical interventions

Although the current inability of local governments to provide basic services to all urban dwellers represents a significant failure, it also offers a major urban opportunity for the private sector to become part of the solution (The backlog of connections is projected to grow by 100,000 households per year for the near future). Despite the fact that a growing number of entrepreneurs are demonstrating that profitable or technically sound or socially minded ventures (e.g. solar microgrids) can indeed be built in these growing low-income markets very few, if any, have successfully shown a venture combining all three elements.

Barriers

Promising alternatives have emerged with the promotion of renewable energy products throughout South Africa. Micro-energy systems have the benefit of generating energy locally and avoiding transmission costs and technical issues associated with larger grid systems. However, significant barriers to low income electrification and the adoption of renewable technologies exist. These barriers, defined as 'conditions that prevent investment from occurring' are summarized below. Many of the different barriers overlap and some are hotly contested. However, the widely accepted barriers are:

- **Technology risk**
The technology solutions implemented in SA to-date were not mature enough, leading to high maintenance and operating costs. Being technology agnostic is critical to ensure the best solution for diverse markets i.e. technology needs to be context specific.
- **High upfront capital costs and lack of co-investments (rural_development)**
High upfront costs and low fees make it difficult to see a return on investment. However, operational and administrative costs can be outweighed by the revenues generated from low-income households with low energy consumption.
Development-funded projects frequently run as "pilot sites" that are focused on bringing basic services such as lighting and are thus not scalable or sizeable enough to attract commercial investment and stimulate the rural economy.
- **Administrative costs in small off-grid systems and lack of standardization.** Development and admin costs are high in comparison to the size of the projects by due to a lack of consistency between projects, as different models are tested, making project aggregation difficult.
- **Users' low awareness of technology potential.** Key barriers in this space are primarily based on the aspirational standards of consumers, which remain Eskom's grid-connected electricity. The risk potential that embracing a medium independent PV system now would disincentives Eskom from laying a larger transmission to their village in the near future is high in the minds of the lower LSM consumers.
- **Other barriers include problems in local participation and theft, political interference in national utilities, lack of social networks on which to base intervention, lack of complementary investment and cross-sector coordination and the lack of a national legal framework.**

As the offgrid market matures, legal frameworks evolve and private-sector entrepreneurs develop new business models at sufficient scale, the CFF will be able to evaluate these projects and assess if catalytic or first loss investment can help spur additional commercial investment. The CFF would seek to focus on projects with private sector offtakers, and where standardization and aggregation can get projects to a scale where commercial investment is viable.

Waste Management and Recycling

Based on the waste industry's landscape outlined in a recent GreenCape Market Intelligence report (2017), there is much potential in the following sectors:

- **Paper, glass and plastic:** In terms of paper, glass and plastics, the report indicates that opportunities in the recycling market will be largely driven by mechanical recycling. This recycling is only economically viable in large volumes, requiring large capital investment. Growth in this sector would largely be driven by the PROs, material

converters such as Mpact, Amcor, Nampack, Consol, and Golden Era, and producers such as Consol, ArcelorMittal, and Sasol etc. with support from national and local government.

- e-waste: According to the e-Waste Association of South Africa, the country produces approximately 322,000 tonnes of electronic waste (e-waste) per annum. Since 2009, e-waste recyclers have increased from two large companies to over 20 companies which are located across the country and provide a variety of services from refurbishment to metal extraction. Collectively, these companies process 45,000 tonnes of e-waste which accounts for only 12% of national waste.
- Organic waste: The Western Cape generates 2.9 million tonnes of organic waste per year. The provincial government is planning to ban all organics from landfill within ten years, with a reduction of 50% over the next five years. If the organic fraction (the leachable / methane producing fraction) can be diverted, there would be no need for strict landfill construction requirements according to the reports.

The above status quo presents several opportunities:

- save municipalities money
- make organic waste available for beneficiation
- target the separation of organics from the waste stream. This will result in the remaining waste largely being dry recyclable waste with less risk of contamination.
- Construction and demolition waste: Builders' rubble is largely landfilled in South Africa, in spite of its potential for re-use and the high financial and societal costs of landfilling. One of the biggest opportunities in the recovery, processing and application of builders' rubble lies in the uptake of material into new applications, e.g. roads. This holds prospects on the supply side for the crushing industry and on the demand side in road material for both the public and private sectors.

Business opportunities for smaller entrepreneurs in the waste management and recycling sector include:

- Textile processing
- Cardboard core processing
- Cardboard core containers
- On- and off- site paper pulp / effluent technology
- Contaminated bentonite processing
- Foundry sand value add
- Paper recycler/dehydration of waste streams

Rural Energy

Businesses that run a heavy day time consumption pattern or require significant cooling represent the largest near-term potential. These are agri-packhouses, large farms, mining, water and liquid waste facilities. Technology intervention potential for these sectors include – among others – on-site or adjacently located biogas and solar PV, outlined below

Biogas

The South African biogas industry is considered to be in a nascent or infant state, as there is a low rate of uptake and general inexperience in designing, constructing and operating of biogas facilities. However, more uptake of biogas technology and development of local expertise would assist the industry in maturing to a level where biogas technology and industry is considered established, making it easier to access commercial finance at appropriate rates. The drivers that support and assist the South African biogas industry in maturing include economic, environmental, social and legislative factors. The need for the management of organic waste, along with increasing costs of disposal and the recognition of the potential for on-site energy use, are some of the key drivers for the uptake of biogas technology. Common challenges include lignocellulosic contaminants, odour, digestate management, grid feed-in prices and waste collection and separation. Furthermore, the current lack of operational skills and capacity within South Africa has presented itself as a key challenge that highlights the need for training and local capacity, building on all aspects of biogas project development (including design, construction, operation and maintenance). As a result emerging business models are seen as venture in nature, or lack the track record to make commercial financing viable. Biogas represents a very

small portion of the overall SA energy mix. In total³⁶, the market size is equivalent to 40MW with commercial and industrial representing less than 15MW. There are currently three key models that divide the biogas industry.

Table 9: Biogas models

System size	Project sizes (cost per kW _p) – R40 000	Revenue model	Financing	Project sites
< 500 kW _p	R 2 – R20 million	On site heat recovery, electricity generation and or gas supply	Site/developer collateral	Farms, abattoirs, malls, food processing
> 500kW _p	R 20 – 400 million	Combination of on-site use and off-site off take guarantees. Local wheeling agreements	Off-take guarantees with private client	Mega farms, centralised waste management facilities, feedstock alternatives
Utility	> R 100 million	PPA model with utility. Single off taker.	Off-take guarantee with utility	Mega farms, centralised waste management facilities, feedstock alternatives

Source: GreenCape, Market Intelligence Report 2017

Available finance

For directly financed projects, the risk profile and credit worthiness of the site owner/developer (also the owner of the system) is currently considered, rather than the value of the asset, since there is no second hand market or resale value for biogas plants. Banks require a minimum 3 year tail and do not consider much more than 60% debt for private funding. Funding is matched to turbine lifespan, which is generally 6000 hours before a major overhaul is required. This limits project lifespans to 7 and 14 years before the first overhaul and full engine replacement is required. Finance is generally available for tenors of 7-10 years, with rates varying on creditors risk profile (10 -15%+). For larger projects debt-to-equity ratio can extend to 70:30 with loan terms extending to 12 years.

Project Development Cycle Risk

Due to the nascent market for waste to energy, intricate business models (numerous risks) and long lead times for development (3-4 years), a number of development banks (IDC, DBSA) have supported the sector through the provision of grant funding for feasibility studies and concessionary finance to decrease the cost of debt. Project lifespans are heavily impacted by Environmental Impact Assessment (EIA) studies (6-24 months) and commissioning/testing (6-12 months) of plant.

Licensing

³⁶ Contracted, under construction and operational

DEVELOPMENTAL / IMPLEMENTATION		ENVIRONMENTAL	
Subdivision of Agricultural Land Approval	Department of Agriculture, Forestry and Fisheries	Water Use Licence	Department of Water Affairs and Sanitation
6 to 12 months		1 to 4 years	
Building Plan Approval	Municipality	Atmospheric Emissions Licence	District Municipality (or Province where this function has been delegated to the Province)
30- to 90 days			
Construction Work Permit	Provincial Director of the Department of Labour	1 to 2 years	Department of Environmental Affairs
		Environmental Authorisation	
30 days		1 to 3 years	South African Heritage Resources Association
Land Use Approval	Municipality	Heritage Approval	
1 year			
Notification of Major Hazard Installation	Chief Inspector & Provincial Director of the Department of Labour & local authority	3 to 6 months	Department of Environmental Affairs for hazardous waste and
		Waste Management Licence	
120 days	n/a		
Appointment of Health and Safety Representative		1 to 3 years	Provincial Waste Management Authorities in the case of general waste
		Biodiversity Permit	
Establishment of Health and Safety Committee	n/a		
Notification of Construction work to Department of Labour	Provincial Director of the Department of Labour	4 to 6 months (valid for 12 months)	
7 days			

Figure 16

Added to an EIA or basic assessment, there is also a need to complete a waste management license, air quality license and a potential water use license. These licenses cross numerous departments and can take extensive amounts of time and costs to complete.

Eskom Time of Use Structure

Eskom's time of use tariffs are structured with a high and low season, and peak, off peak and standard rates for both high and low season. Peak times are morning (7-10am) and evening (6-8pm). Offpeak is before those times and on weekends, with standard time between the peaks. High season is in the 3 winter months from June –August and tariffs are higher for these months. This structure improves the business model for biogas, but many of the plants are located in rural locations with competitive tariffs and have to incur the cost of wheeling on both the Eskom and Municipal (no standard wheeling framework) network to provide to an off-site customer.

Multiple business cases

Given the current high IRR requirements, debt tenors, development lead times, commissioning periods and technology variations, a viable project tariff is currently between the R1.4-R1.5/kWh mark, making biogas a less competitive business model in the electricity space. Currently, to make the business case work, the project either needs to sell electricity at a premium or guarantee multiple revenue streams (heat recovery, gas, landfill avoidance/savings). The reliance on multiple revenue streams and feedstock quality and quantity pushes the IRR requirement to a range of 20-25% for equity holders. Due to lack of familiarity in the commercial financing market, debt and equity providers can be hesitant to underwrite based on relatively small revenue streams with lack of track record (e.g. heat recovery, landfill avoidance).

Due to the uncertainties, risks, lack of track record, and difficulty underwriting multiple revenue streams, uptake and investment in biogas projects has been limited. This low market penetration in South Africa persists despite high uptake in other mature markets in other countries. The CFF can act as a first mover to help absorb risks in the biogas sector as

the market matures, and provide more certainty to private investors by offering first loss capital and demonstrating alternative underwriting approaches that factor in multiple revenue streams.

PV for diesel replacement for rural C&I customers

There is a very strong business case for the replacement of diesel generation with PV / PV with energy storage. Key examples of these opportunities are solar power irrigation, game farms, mines farms and pack-houses. The customers for these markets are well outside of urban areas where project developers are located, making it harder to share knowledge and develop projects. Little information is available on the number of rural diesel generator sets in use. Generally, these projects can be fairly small (e.g. game reserves) to medium sized (e.g. mining operation).

Barriers to uptakes in these projects vary, from small ticket size, to EPC contract issues, underlying user credit risk and need for longer tenors. The CFF, as a flexible program to lower risk profile of investment for commercial funders, has the potential to help develop this sector by providing credit enhancements and debt investments to projects in the sector.

Rural Water supply

There are opportunities for water efficiency in agriculture through precision agriculture and controlled environment agriculture (shade netting, plastic tunnel systems, hydroponic farms), but these are generally not driven by a pure "water saving" business case as the pricing for water is typically minimal. Currently, the primary driver is a lack of available water, preventing crop loss, and increasing crop size with constrained water allocation. There is growing potential market for conservation agriculture, which will save water, but is not driven by water savings, but rather crop productivity and increasing efficiency of limited water allocations.

Low water costs

Water costs in the agricultural sector are so low that they are hardly considered.

Technology Risk

These technologies are still relatively new in South Africa, and farmers want proof that the tech works (generally from the success of other farmers) before investing.

Knowledge

Poor feasibility studies, and components and systems that are not suitable for local conditions, are the main causes of failed projects in SA.

Capital Cost

The cost of some higher tech CEA technology can be prohibitively high, making farmers unwilling to take the technology risk.

For projects of a sufficient size, or where aggregation and standardization of offerings can drive scale, the CFF can potentially act as a catalytic investor in this sector, which currently suffers from lack of track record and perceived performance/technology risk.

It is clear from the current barriers highlighted above, the need for a facility such as CFF to address the lack of access to finance and lack of innovative funding models in the identified sectors is very apparent especially in Southern Africa where most countries in the region share similar challenges.

C.3. Project / Programme Description

Describe the main activities and the planned measures of the project/programme according to Components 1 and 2 as described in section B1. .

Component 1 - Operationalization and project management of CFF

The Development Bank of Southern Africa (DBSA) will implement the Climate Finance Facility Programme (CFF) with a dedicated operating unit within DBSA. The facility is designed based on the model of a “Green Bank” to rapidly scale up private investment in climate friendly projects in South Africa and Rand-based Southern Africa countries, with the possibility to extend to the broader Southern African Development Community (SADC) region. The CFF will address market constraints, playing a catalytic role with a blended finance approach, to increase climate related investment in the Southern African region. It will be a first-of-its-kind application of the Green Bank model, adapted for emerging market conditions. The Green Bank model is highly relevant to building nation-specific capacity to scale up climate finance in support of Paris climate goals. Formed as dedicated, locally based, public-purpose funded clean energy finance institutions, they provide a full range of credit enhancements to support clean energy market development. Green Banks use their capital to fill market gaps and crowd-in private capital, targeting commercially viable technologies that cannot currently attract market-rate capital at scale.

The detailed scope of activities to be undertaken by the DBSA Executive Management team, the CFF Portfolio Management team and various existing and relevant units within the DBSA eg. finance, legal, HR and IT during the set-up and implementation phase of CFF in accordance with the broad activity outline provided in section B.1 and summarized above would include the following:

Sub-Component 1.1 Unit establishment within DBSA: • Lead successful operationalization and launch of the CFF unit within the DBSA; • Design and put in place CFF portfolio management unit team; • Develop strategic plan, • Form CFF Steering Committee; • Oversee legal and accounting arrangements; • Create budget template, and bank accounts: and other tasks as required

Sub-Component 1.2 Development of CFF Operational Manual: • Manual will include detailed investment criteria, eligibility criteria for each sector targeted by CFF, monitoring and impact measurement guidelines; • The operational manual will stipulate how CFF investment and eligibility criteria will be fully integrated into the investment appraisal process of CFF; • Eligibility criteria will include a focus on the water sector.

Sub-Component 1.3 Annual Work Plan and Budget: • Develop budget plan that considers the financial sustainability of the facility and a balanced portfolio of programs;

Sub-Component 1.4 Project Pipeline & Financial Product Development: • Portfolio Management team to develop first round of CFF financial products; • Design and launch an open-ended Request for Proposal (RFP) process to support development of a robust project pipeline for the CFF; • Interface with DBSA origination channels and the new CPI “Climate Lab” to be hosted at DBSA; • Oversee ongoing market research to continually develop new opportunities, align programs to emerging markets, and maintain competitiveness with other financing options.

Sub-Component 1.5 Communications and Marketing: • Develop, manage, and maintain organizational brand and external presence of the CFF and execute a communications and outreach effort that promotes the objectives of the CFF working with the DBSA marketing team; • Lead direct outreach to project developers, commercial banks, DFI's and other relevant stakeholders; • Build the reputation and brand of the CFF with donors, foundations, corporations, the media and public officials and agencies; • Establish strategic partnerships with lenders, contractors, developers, government agencies, utilities, business and industry associations, and community groups.

Component 2 – Project Financing: providing credit enhancements and debt financing to climate change mitigation and adaptation projects

The CFF will focus on infrastructure projects that mitigate or adapt to climate change and will focus on two main instruments: subordinated debt / first-loss and credit enhancements such as tenor extension and other kinds of credit support to projects that are commercially viable but not bankable in the private sector. These instruments will de-risk and increase the bankability of climate investment, and crowd in significant private sector capital. The breakdown of

instruments deployed will be finalized as the operational plan is developed, and as the first projects enter the pipeline. The CFF will focus on relevant sectors including: off-grid projects, micro-grid/urban solar farm projects, industrial and commercial solar for self-generation and wheeling on the grid to MSMEs, industrial and commercial water projects, local Community water projects, and residential and commercial energy efficiency investments. Rather than establish a new Green Bank institution, as has been done in developed markets, the CFF will be formed as a unit within DBSA with its own dedicated operations. This project has globally significant proof-of-concept value to a range of middle and lower income nations seeking to address market barriers and quickly scale up blended finance approaches to support the high levels of private investment required by Paris commitments.

The activities to be undertaken by DBSA as Executing Entity of CFF include the following:

2.1 Mitigation financing: • DBSA will enter into loan agreements for mitigation projects (Renewable Energy, Waste to Energy, Energy Efficiency, Low Emission Transport)

2.2 Adaptation financing • DBSA will enter into loan agreements for adaptation projects (Water Efficiency, Water Treatment, New Clean Water)

By design, the CFF is a programme intended to scale up private investment in low carbon energy and clean water infrastructure. The CFF will operate based on Green Bank principles, as a public-purpose finance institution dedicated to addressing critical market gaps and driving public and private climate investment. This increased investment will help fill the current investment shortfall to achieve Southern Africa's NDCs and provide adequate supplies of sustainable clean drinking water. The CFF will be designed to maximize total investment, using limited public funds to leverage far greater private investment. By developing innovative finance and market development solutions, Green Banks address barriers that currently restrict clean energy market growth and effectively scale up private investment at ratios ranging from 2:1 to 10:1 depending on market sector and project specifics.

The CFF will provide a wide range of social, economic and environmental benefits for South and Southern Africa including the following:

- **Social Impacts:** Increased access to affordable power and clean drinking water to people across South and Southern Africa ; reduction in poverty and inequality through job creation and increasing access to energy and clean water for businesses and communities, education and community use; and support for sustainable cities and communities through innovation and development of sustainable infrastructure.
- **Economic Impacts:** Economic benefits stemming from increased access to power for MSMEs; economic benefits from job creation for construction and operation of distributed generation facilities in poor and underdeveloped areas of South Africa/SADC; support for economic growth through diversification of energy generation.
- **Environmental Impacts:** The CFF will have positive effects on climate change through reduction of GHGs and decrease of air emissions and other environmental pollutants. In addition, in a water-scarce country like South Africa, the projects will result in the reduction in water abstraction and consumption for electricity generation purposes.

For detailed discussion of the CFF mandate, structure, investment criteria, governance and origination channels, see section C.7 below. For illustrative project-level investments, see the Financial Model in Section I, Annex.

Provide information on how the activities are linked to objectives, outputs and outcomes that the project/programme intends to achieve. The objectives, outputs and outcomes should be consistent with the information reported in the logic framework in section H.

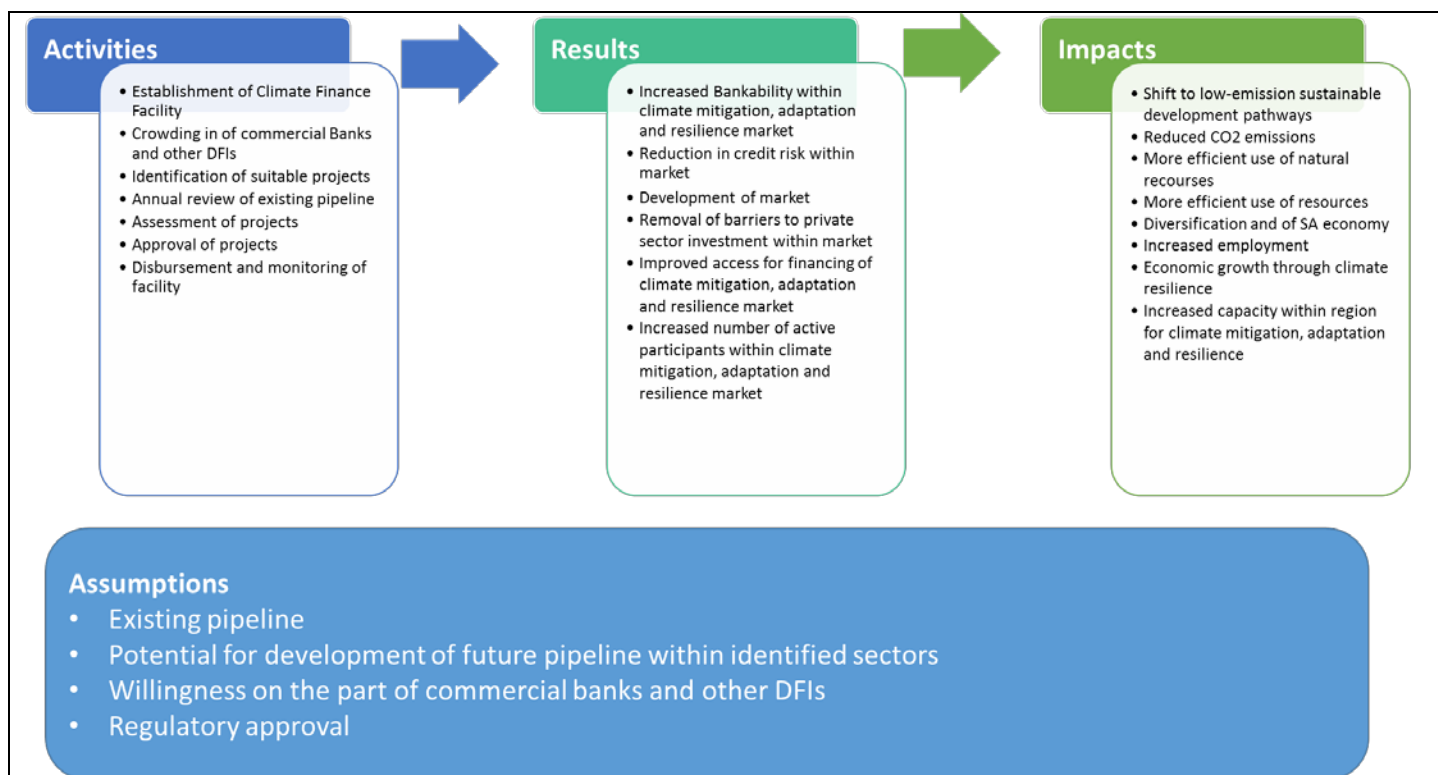


Figure 17: Theory of change

C.4. Background Information on Project / Programme Sponsor (Executing Entity)

Describe the quality of the management team, overall strategy and financial profile of the Sponsor (Executing Entity) and how it will support the project/programme in terms of equity investment, management, operations, production and marketing.

Project Sponsor: DBSA

The Development Bank of Southern Africa (DBSA) based in South Africa, is a national entity, specifically a development finance institution, with a mandate to finance both private and public sector activities at national and regional levels in Africa. DBSA provides sustainable infrastructure project preparation, finance and implementation support in order to improve the population's quality of life, accelerating the sustainable reduction of poverty and inequity, and promoting broad-based economic growth and regional economic integration. Climate and environmental finance is embedded in the DBSA strategy and the entity has played a key role in implementation of projects on transitioning South Africa to a green economy, including acting as the implementing agency for the Green Fund of South Africa and financing of the commendable independent power producer programme projects. Since its accreditation to the Global Environment Facility, the DBSA has developed and will soon be implementing several projects that fall within different initiatives such as the Sustainable Cities Integrated Approach (SC-IAP), U4E, and the small scale national REIPP programme. The current GEF-DBSA portfolio is further proof that the Bank has a good track record for implementing projects such as the CFF. In terms the DBSA's recent track record with regards to sustainability measures, the DBSA continues to work towards embedding sustainability in its core business.

As noted earlier, all CFF projects will first be reviewed and assessed by DBSA's Structured Project Unit and the CFF Operations Management Team within DBSA. DBSA shall establish a Project Steering/Advisory Committee or "Steercom" with representatives from DBSA and other investors in the CFF. The Steercom will provide oversight to the CFFs operations, ensure that proper due diligence and risk assessment has been performed, and that only projects that meet CFF eligibility in accordance with the operational manual are recommended to the DBSA Investment Committee for final approval. The DBSA Investment Committee will make the final decision on the approval of projects to be financed.

The DBSA's sustainability highlights for 2016 include:

- R17.1 billion (2015: R13.0 billion) infrastructure investment into core sectors of energy, water, transportation and telecommunication. Of this, R8.1 billion (2015: R5.4 billion) was invested with municipalities, a key provider of social infrastructure in South Africa
- 638 000 households are estimated to benefit from new or upgraded infrastructure through funding committed to municipalities
- Accelerated support for the implementation of social infrastructure
- Completion of 35 schools as part of the Accelerated School Infrastructure Delivery Initiative Programme, benefiting 17 900 learners in 2016. More than 36 000 learners have benefited since the inception of the programme
- 1,382 affordable houses completed
- 2,250 water and 203 electricity meters installed
- 500 municipal officials trained on water and equipment repairs and 269 traditional leaders and councilors trained in visionary leadership and integrated development planning
- During 2016, the DBSA committed no funding (2015: R4.2 billion) but disbursed R1.5 billion (2015: R2.4 billion) to the REIPPP and Peakers programmes. R209 million was disbursed in favour of B-BBEE enterprises.

The DBSA (As implementer of the National Green Fund)

DBSA also serves as the implementing agency of the Green Fund, a government national fund that aims to transition South Africa to a green economy. The fund provides financial resources to small scale initiatives, for example, renewable energy projects of a throughput of 1 – 5 MW would be eligible for funding. Projects funded by the fund are categorized under 3 funding windows; (i) Green Cities and Towns, (ii) Low Carbon Economy and (iii) Environment and Natural Resources. Capacity building and Research and Development initiatives are also eligible for financial assistance through the fund. Financial instruments offered include grants (payable and non-payable), debt and equity. Projects that are eligible for Green Fund financing under the 3 windows belong to a range of sectors including transport, water, renewable energy, ICT, energy efficiency, natural resources, biodiversity and sustainable land management. By virtue of implementing the Green Fund, DBSA's track record on implementing a climate finance facility is clearly proven.

CFF alignment with DBSA Climate Finance Strategy:

The DBSA sees climate change as central to addressing SDGs and as crucial to meeting its mission, vision and mandate. DBSA as an organ of state requires a cohesive, measurable and accountable climate change response grounded on and based on the organisations mandate, structures, policies, and practices. The DBSA climate financing strategy outlines the key opportunities, baselines indicators, targets and reporting frameworks to track DBSA progress in responding to climate change. It further provides a basis for developing the tools for the DBSA to report on achievements in respect of climate change associated development results.

DBSA has identified 3 key levers to drive interventions to proactively address climate change. Each lever has performance indicators and targets with progress monitored through a climate change dashboard. Annual green climate mapping enables DBSA to assess progress and report on achievements in meeting its commitments to climate change. The levers of DBSA Climate Change Policy Framework include:

- Financial Commitment: Promote leadership and increase volumes of climate finance
- Continuous improvement: Enhance due diligence to embed low carbon and resilient trajectories of countries in potential investments
- Selectivity of operations: contribute to upscaling financial and investment flows in support of climate change interventions.

DBSA aims to benchmark its progress against the indicators outlined in the chart below and to achieve a level 5 by 2022. The rating of organisations has been based on the DBSA team exposure to various institutions whilst compiling the strategy.

The DBSA climate financing strategy sets specific targets for the bank with regard **annual lending** for climate projects and programmes, the targets are further broken down into percentage lending into mitigation and adaptation. **The DBSA sees the CFF as a crucial intervention that will contribute significantly in meeting the set targets with regard to the Climate Financing Strategy.**

The DBSA is further working on integrating climate change reporting as an element of its Development Results Reporting Framework to provide an accurate portfolio breakdown of the DBSA commitment to climate change thereby enabling credible and accurate reporting to stakeholders including the GEF, GCF and the International Development Finance Club. The strategy is currently being finalised and has received very positive support from the DBSA board.

CFF Team The DBSA will implement the CFF Programme based on their track record as one of the strongest institutions in South Africa and the foremost African NDB in the Southern African region, with established financial expertise, sound organizational best practice, credibility with the international donor community, trust and credibility with all major banks and project investors. To support early stage development and launch of the CFF within DBSA, the DBSA has formed a partnership with the [Coalition for Green Capital](#) (CGC) based on their unique and proven capacity to support the formation of Green Banks with governmental and civil society partners, and provide on-going consulting and guidance to operating Green Banks. CGC is a Washington, DC-based 501(c)(3) non-profit organization that has been the leading creator, advocate, and expert on Green Banks since 2009. CGC directly supports the formation of Green Banks with government partners, and provides on-going consulting and guidance to operating Green Banks. Over years of experience, CGC has refined an effective Green Bank creation process to launch dedicated clean energy finance institutions, tailored to local market conditions. CGC is a recognized thought leader on Green Banks and innovative clean energy finance policy, speaking regularly around the world. Working across the U.S. and internationally, CGC has developed a global network of partners in policy, finance, development and economics, allowing the organization to tap a deep pool of expertise to support clients and governments.

CFF Operations Management Team	Project Role & Responsibilities
Jonathan First DBSA Lead Specialist/Product Incubator	Lead for creation of CFF. Structure, business plan, capitalization, interface with DBSA management and Board of Directors; interface with the new Climate Lab; Lead contact with commercial banks and Development Finance Institutions
Paul Currie DBSA CIO	Senior point of contact with DBSA Board of Directors for project oversight, management, capitalization, operation of CFF Steering/Advisory Committee.
Olympus Manthata DBSA Head of Climate Finance	Senior point of contact within DBSA for interface with the GCF
CFF Portfolio Management team	A team within the DBSA established to evaluate, process and administer projects and businesses that qualify for CFF funding. This team will prepare and submit the applications to be submitted to the Steercom and to the DBSA IC for approval. This team will have the necessary skills to evaluate all aspects of the proposed projects and businesses including technical, environmental, socio-economic and financial.
CFF Steering and Advisory Committee	A new high-level committee will be created within the DBSA to support operationalization of the CFF. Members of this team will support the following functions: CFF planning and operations, financial product development, pipeline development including RFP process, review project due diligence including impact assessments. Steercom will also serve as an advisory Committee of the CFF and Ensure that proper due diligence and risk assessment has been performed, and that projects meet CFF eligibility. Projects assessed and recommended by the Project Steering Committee will be sent to the DBSA IC for final approval.
DBSA Investment Committee	The DBSA Investment Committee will make the final decision on the approval of projects to be financed.

DBSA Project Team

Jonathan First, DBSA - Jonathan First is a lead specialist within the Structured Products Group and a member of the DBSA's Innovation Hub. He works with the DBSA's origination teams to conceptualize, develop, structure and distribute funding for large infrastructure projects in South Africa and sub-Saharan Africa in the sectors of water, energy, ICT and transport/logistics. He is also leading the DBSA's initiative that is working towards creating a Climate Finance Facility within DBSA to offer credit enhanced debt funding to catalyze blended finance in "green related" infrastructure projects and transactions. Mr. First qualified as solicitor in South Africa and then spent 22 years in investment banking in Johannesburg, Toronto and the City of London. He represented the Canadian Banks on the Brazil, Argentina and Mexico debt restructuring committees during the 1980s South American debt crises. Mr. First originated several debt syndications for South African private and SOE enterprises post the 1994 elections when South Africa could once again access the Debt Capital Markets.

Paul Currie, DBSA - Paul Currie is the Chief Investment Officer of the Development Bank of Southern Africa. Before joining the DBSA, he held several senior management positions at the Nedbank Group. These include Head of Risk: Investment Banking, and General Manager: Corporate Banking Credit. During his tenure as the Chief Risk Officer at the DBSA, Paul Currie has been responsible for the Bank's Credit Risk Monitoring and Governance function. Under his leadership, the Group Risk Assurance Division has re-engineered and rolled-out credit risk rating and project pricing methodologies, implemented policies and processes to ensure alignment with the Bank's requirements and strategic business initiatives, instituted the Enterprise-Wide Risk Management Framework, ensured that the DBSA is in full compliance with applicable regulations and supported the development of new products. He has enhanced the monitoring and reporting of Group Risk to Exco, Board and various rating agencies and stakeholders.

Olympus Manthata, DBSA - Olympus Manthata is currently of the Climate Finance Unit based at the Development Bank of Southern Africa. His role also includes the management of Climate Financing mechanisms that the DBSA is accredited to, such as the Global Environmental Facility (GEF) and the Green Climate Fund (GCF). Olympus has been instrumental in assisting the DBSA to obtain the GEF accreditation, as a first national agency to do so, and further contributed to the successful accreditation of the bank by the (GCF). He also contributed to the establishment and implementation of South Africa's National Green Fund. As the Fund Manager for the National Green Fund he oversaw the commitment and disbursement of over ZAR1 billion worth of debt, grant and mezzanine financing into high impact investment projects which are realizing significant social, economic and environmental returns. His 26 years' experience includes working as an Investment Professional within development finance arena, the technology innovation/commercialization space and various facets of the Engineering Environment as Business Development Manager, Engineering Manager and Commercial Manager. He is registered with the Engineering Council of South Africa as Professional Technologist: Pr. Tech Eng.

CFF Operations Management Team and CFF Project Steering/Advisory Committee – A new, high-level project team will be created by the DBSA to manage and implement the work of the CFF. This team will support the following range of functions.

- CFF Operationalization – Support successful operationalization and launch of the CFF Programme –, development of strategic plan, crafting bylaws and operational policies, legal arrangements, accounting arrangements, financial statement creation, budget template, bank accounts, business model, fiscal sponsorship, and other tasks as required.
- Product Development – Support development of first round of CFF financial products. Responsibilities will include stakeholder interviews, site visits, program briefings, lender and contractor engagement, development of program guides, program modeling, lender package, contractor package, trainings, and interface with DBSA origination channels and the new CPI "Climate Lab" to be hosted at DBSA.
- Communications & Marketing – Work with the DBSA marketing team to develop the CFF communications plan, logo, branding package, slides, mission and vision statement, website, social media, and direct outreach to project developers, commercial banks, DFI's and other relevant stakeholders.
- Project Pipeline Development – Support development of the CFF's project pipeline, working with the DBSA's Project Preparation Unit, Origination Team and Innovation Hub and with project developers, commercial banks, DFIs, and the new CPI "Climate Lab" to be hosted at DBSA.
- Market-Responsive Investment Window – Design and launch an open-ended Request for Proposal (RFP) process to support development of a robust project pipeline for the CFF – responsibilities will include developing, vetting, launching and supporting successful implementation of the RFP process.

CGC Project Team

Andrea L. Colnes, CGC - Andrea Colnes serves as International Director for CGC where she leads CGC's efforts to bring the Green Bank model to developing countries in support of scaling up climate finance to achieve national climate

goals. CGC's International work includes continued development of the Green Bank Network, building a framework for Green Bank creation in emerging markets to help meet climate goals, supporting a collaborative partnership with non-profits and inter-governmental agencies advancing the Green Bank model, and creating new partnerships for Green Bank creation and funding around the world. As a CGC Board member, Andrea has been a part of the CGC family for many years and has led strategic campaigns focused on energy, large-scale land conservation, and sustainable forestry over a 30-year career. Prior to joining CGC, Andrea led efforts to create a Green Bank in Vermont (ongoing), served as founding Executive Director of the Energy Action Network, a systems-based effort to transform Vermont's energy economy to one based on efficiency and renewables, Co-authored Vermont's first Comprehensive Energy Plan under Governor Madeleine Kunin. And served as policy director for the Biomass Energy Resource Center with a focus on community-scaled biomass and atmospheric carbon research.

Rob Youngs, CGC - Rob Youngs is the International Program Director at CGC. Rob oversees Green Bank development opportunities internationally, including emerging opportunities in Morocco, Colombia and Malaysia. Rob is team lead on the global Green Bank Network, an international platform launched at COP21 for sharing Green Bank best practices, with members including UK Green Investment Bank (UKGIB), Australia's Clean Energy Finance Corporation, and Japan's Green Finance Organization (GFO). Rob represents CGC at international conferences and events including the annual UNFCCC COP, the OECD's annual Green Investment Financing Forum (GIFF) and others. Prior to joining the international team at CGC, Rob worked on key recommendations for the Nevada Green Bank. Before joining CGC, Rob worked on research for the Connecticut Green Bank, and developed an energy efficiency strategic plan for the city of Los Angeles. Rob also spent four years working on technical standards under the California emissions trading system and developed program rules for new energy projects to receive carbon finance.

Christopher Scott, Director of Clean Energy Finance, CGC –CGC's Director of Clean Energy Finance is responsible for designing and implementing clean energy finance products to support CGC's growing network of Green Banks in the US and internationally. Experience includes evaluating the market opportunity for various financial products—such as credit enhancements, warehousing structures, and loans—to meet the market and operational needs of developing Green Banks. This position also works to engage the capital and other channel partners necessary to bring these products to market. Specific capacities include: Developing the mechanics of clean energy finance products for Green Banks, including structures and terms; Creating quantitative assessments of risk and expected financial performance of clean energy products and institutions; Pitching potential investors—including impact investors, foundations, pension funds, endowments, and commercial capital providers—on investment opportunities with Green Banks; Conducting market assessments to identify new product opportunities and strategies; Working closely with CGC staff to roll out new financial products as part of Green Bank consulting or implementation projects.

Green Cape – Green Cape is a South African non-profit organization that drives the widespread adoption of economically viable green economy solutions from the Western Cape and elsewhere in South Africa. We work with businesses, investors, academia and government to help unlock the investment and employment potential of green technologies and services, and to support a transition to a resilient green economy. Green Cape is a not-for-profit organization established in 2010 by the Western Cape Government as a Special Purpose Vehicle to support the development of the green economy in the region. To support development of the CFF and initial capitalization, Green Cape will provide a market analysis on target sectors including analysis of project pipeline and demand for CFF services. Green Cape will also support development of impact indicators for the CFF.

C.5. Market Overview (if applicable)

Describe the market for the product(s) or services including the historical data and forecasts.

Overview of capital landscape in the green economy

Although commercial banks in South Africa are well capitalized, they are typically restrained to tenors of 7 – 8 years, with cleantech infrastructure projects (such as solar PV) typically require a 15 – 20 year payback. These shorter tenors can be attributed to Basel III constraints, as well as the misconceptions around technological risk, where solar PV assets can operate reliably for approximately 20 years.

Historically, there was a shortage of demand for “green” finance, however, after the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) began in 2011, this created demand for loans large enough to justify the transaction costs. This provided a packaged, low-risk opportunity for banks and DFIs to invest significant

quantities of money into large-scale utility projects. In addition, recent Green Bond issuances (detailed under 1.2 Debt Capital Market) added to capital provisions for large tickets sizes.

As noted in an investigation into green finance for the Western Cape Government³⁷, this means that the burden of smaller, early-stage investments falls on angel investors, venture capitalists and private equity players, which contains a notably smaller capital pool that must be shared across all sectors. Consequently, cleantech entrepreneurs must compete against entities at an equivalent stage of risk, while requiring a significant quantum of capital, resulting in funders diversifying their resources in a number of other sectors. South Africa does not, in the conventional sense that is illustrated in the figure below, have a significant private equity, venture capital or angel investor community.

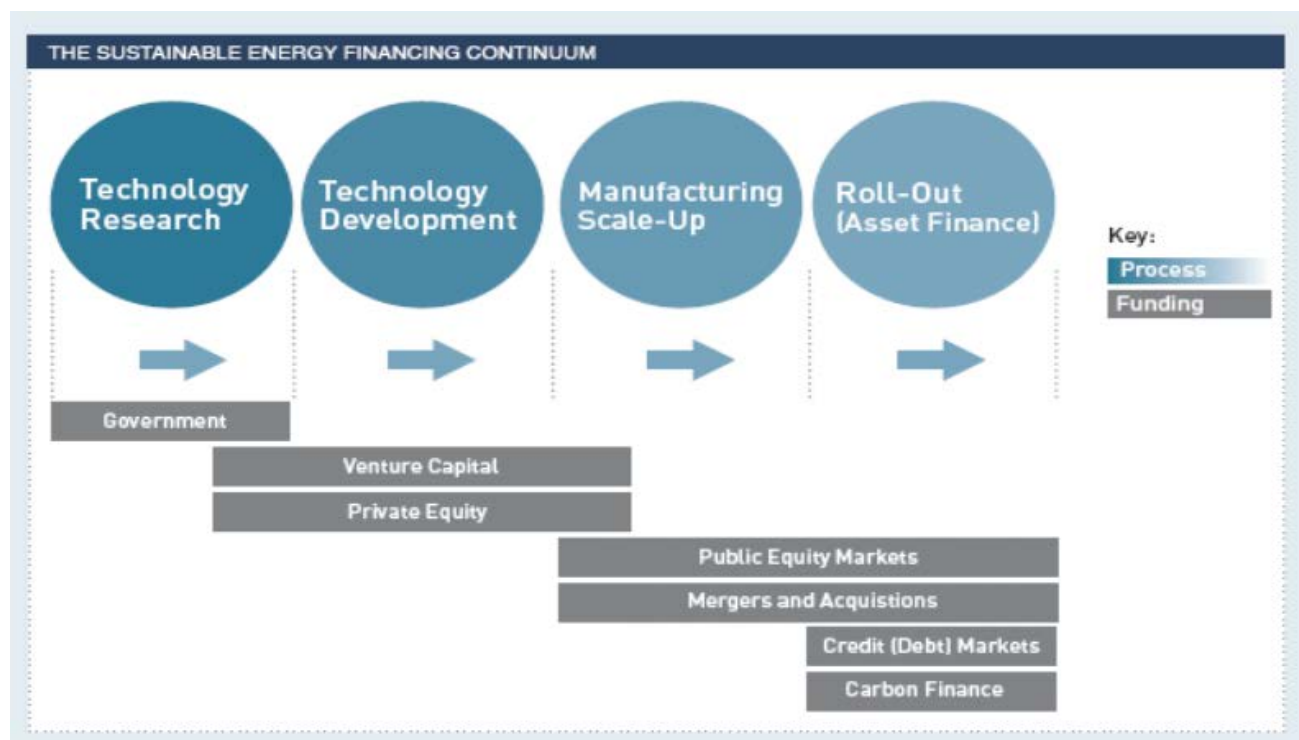


Figure 18:

Neither does it have a large number of companies touting “ethical investment” - whereas the United Kingdom, for example, has over 700 such companies. There is no equivalent of Europe’s Triodos Bank³⁸ in South Africa and ethical banks have a negligible investment footprint in the country. Development Finance Institutions (DFIs) in South Africa operate in almost exactly the same market space as commercial banks. These features, in conjunction with the limited flows of finance to green economy projects, suggest that there are clear gaps in the financial landscape.

Describe the competitive environment including the list of competitors with market shares and customer base and key differentiating factors (if applicable).

The CFF will complement existing climate-friendly finance capacity by providing an institution specifically focused on scaling up private investment in small and medium-sized independent energy production through off-grid, micro-grid and IPP via emerging independent businesses. This will be additive to existing energy policy in South Africa (affecting the entire SADC region), which is focused on large scale power contracts via the IRPPP. It will respond to a stated need of the region’s commercial banks to reduce risk and make climate-friendly projects more bankable.

³⁷ *Green Finance Investment Case, Department of Economic Development, Western Cape Government, 2013*

³⁸ Triodos finances organic food and farming businesses, pioneering renewable energy enterprises, recycling companies and nature conservation projects.

Existing energy policy and investment in South Africa has focused on implementing the Integrated Resource Plan, South Africa's blueprint for the energy mix up to 2030, through large scale power contracts under the terms of the REIPPPP and contracts with Eskom for large scale renewable energy generation. Beyond the recent challenges to the REIPPPP due to institutional capacity constraints, corruption and the inability to sign new contracts, there is a significant gap in market capacity around scaling up investment in smaller projects and independent power production for distributed generation. These issues affect the broader Southern African region too, as Eskom is the largest member of the SAPP, and the majority of the region is reliant on South Africa for power supply through long-term supply contracts and day ahead market trading.

Many of the climate mitigation and adaption companies emerging in Southern Africa are startups promoting new technology that banks regard as venture and development capital in nature. There is limited equity available from project sponsors and developers, and given the perception of these businesses as risky venture start-ups, commercial banks are reluctant to gear beyond a debt:equity ratio of 50%. In addition, banks generally limit the tenor of these loans to 5-7 years. The CFF will address these market barriers by playing a catalytic role using a blended finance approach with a focus on projects that are commercially viable but not yet bankable in the private sector without credit enhancement and related financial or development support. This focus will include resilience and adaptation projects such as local waste water treatment plants; water reticulation equipment the harvesting of rain water etc. Key added-value interventions to complement the role that commercial banks can already play will include: longer loan tenors; first loss and sub-ordinated tranches; project aggregation; guarantees; and providing a dedicated, focused and efficient public financing entity.

Provide pricing structures, price controls, subsidies available and government involvement (if any).

N/A

C.6. Regulation, Taxation and Insurance (if applicable)

Provide details of government licenses or permits required for implementing and operating the project/programme, the issuing authority, and the date of issue or expected date of issue.

Describe applicable taxes and foreign exchange regulations.

Foreign exchange regulations are applicable whereby equipment will be imported during implementation in foreign currency. The administration of exchange control is performed by the South African Reserve Bank. The Reserve Bank has delegated some of its powers to deal with exchange control related matters to commercial banks.

Any income or withholding tax applicable to: (i) the repayments to DBSA, and/or (ii) the Reflowed Funds paid by DBSA to the GCF will be borne by projects. Therefore, the GCF's reflows will not be subject to withholding tax.

Provide details on insurance policies related to project/programme.

In terms of insurance, the DBSA requirements to projects for insurance will be consistent with international best practice standards for project financing transactions. Specific policies will be required and reviewed for each project during the due diligence stage. Such policies will be part of the security package for the project lenders, including GCF.

C.7. Institutional / Implementation Arrangements

Please describe in detail the governance structure of the project/programme, including but not limited to the organization structure, roles and responsibilities of the project/programme management unit, steering committee, executing entities and so on, as well as the flow of funds structure. Also describe which of these structures are already in place and which are still pending. For the pending ones, please specify the requirements to establish them.

Describe construction and supervision methodology with key contractual agreements.

Describe operational arrangements with key contractual agreements following the completion of construction. If applicable, provide the credit analysis of key counterparties of key contractual agreements and/or structural mitigants to cover the counterparty risks.

CFF Mandate

The Climate Finance Facility (CFF) Programme implemented by DBSA will serve South Africa and the broader Southern African region and is tasked with financing low-carbon and climate resilient infrastructure projects in partnership with private lenders to catalyze greater overall climate-related investment. The CFF will focus on projects that are commercially viable but not yet bankable in the private sector without credit enhancement and related financial or development support. The CFF will finance clean energy projects, water projects that have climate benefits, and other revenue-generating climate-related projects. The CFF will draw on multiple dedicated funds, develop innovative structures and products, and support projects across Southern Africa. This effort will also support the DBSA's broader goal of "greening" the entire organization.

By design, the CFF is intended to incentivize the private sector to scale up private investment in low carbon energy and clean water infrastructure. The CFF will operate based on Green Bank principles, as a public-purpose finance institution dedicated to addressing critical market gaps and driving public and private climate investment. This increased investment will help fill the current investment shortfall to achieve Southern Africa's NDCs and provide adequate supplies of sustainable clean drinking water. The CFF will be designed to maximize total investment, using limited public funds to leverage far greater private investment and private sector leverage will occur at the project-level. The target of the CFF is to reach an overall portfolio leverage ratio of 1:5 (project leverage ratios will vary within this range). That is, for every Rand that the CFF puts into an individual transaction, it will be looking for approximately 5 Rand of private investment. The DBSA will create the CFF using best practices at existing Green Banks that have successfully catalyzed markets using blended finance to achieve similar leverage ratios without crowding out the private sector.³⁹⁴⁰ This leverage ratio will be supported by several key elements designed to ensure that private capital is crowded into desired markets (vs. crowded out) including:

- Investment Criteria – CFF investment criteria will focus on selecting and creating transaction structures that maximize leverage ratios across project types and asset classes. Investment criteria also require deals that would be unable to access financing absent CFF's involvement in the transaction.
- Metrics – The CFF will track total project costs enabled by CFF (Cumulative ZAR) and total committed funds from all investment partners (cumulative ZAR)
- Reporting – The CFF will provide quarterly reports on the private mobilizing ratio of total project costs to CFF Funds

CFF Investment Criteria

The Investment Criteria would include the eligibility criteria to be developed under the Operations Manual for the programme which will be aligned to the mandate of the GCF. In this regard, all CFF projects must demonstrate that they contribute to low-carbon infrastructure, climate-related goals and/or expansion of clean drinking water supplies. In

³⁹ NY Green Bank 2016 Business Plan. <https://greenbank.ny.gov/-/media/greenbanknew/files/2016-NYGB-Business-Plan.pdf>

⁴⁰ Coalition for Green Capital (2017), "National Green Banks in Developing Countries: Scaling Up Private Finance to Achieve Paris Climate Goals. Coalition for Green Capital." <http://coalitionforgreencapital.com/2017/07/18/paper-green-banks-in-developing-countries/>

addition, projects will be required to demonstrate that they are focused on technically and economically feasible transactions where there is market interest but limited capital availability due to specific financing gaps and barriers. Developers will also be required to show that project cannot be fully financed by the private sector. Projects will be required to show that they involve financial participation by one or more private sector financial parties and that it will have some transformative effect on markets in terms of scale, improved private sector participation, confidence in clean energy investments, or other aspects. Investment Criteria include the following:

- Transactions must contribute to low-carbon infrastructure, climate-related goals and/or expansion of clean drinking water supplies and priorities. Projects must demonstrate consistency with UN Sustainable Development Goals and meet climate objectives as determined by designated specialists within the DBSA.
- Projects must contribute to market transformation and demonstrate that they can materially and sustainably expand markets in terms of scale, improved private sector participation, confidence in clean energy investments, or other aspects.
- Projects will be required to demonstrate technically and economically feasible transactions where there is market interest but the project has not been able to secure financing from the commercial market due to specific financing gaps and barriers.
- Transactions must demonstrate leverage and the ability to crowd-in commercial investment. Each Rand invested by the CFF must be matched by approximately 3-5 Rand from the private sector.

The CFF will be a sustainable programme designed to provide an ongoing capacity to address market barriers and deliver financing techniques that combine public and private capital at scale. In this way, as proven by the Green Bank model, the CFF will use concessional and market rate capital to maximize public benefit over time.

CFF Eligibility and Evaluation Criteria

As part of its Climate Strategy, the European Investment Bank (EIB) developed a List of eligible sectors and eligibility criteria for climate action. The CFF will utilize this list as a starting point for its launch and early operation. These eligibility criteria will be refined as needed through development of the CFF Operational Manual and as informed by the early projects of the CFF. In addition, these criteria will be regularly reviewed to maximize progress towards the CFF mandate to finance low-carbon and climate resilient infrastructure projects in partnership with private lenders to catalyze greater overall climate-related investment.

The EIB climate eligibility criteria have been brought into line with the Joint Methodologies For Tracking Climate Finance developed by the Multilateral Development Banks (MDBs), and the Common Principles for Climate Finance Tracking developed by the joint climate finance group of MDBs and the International Development Finance Club (IDFC). The list of eligibility criteria noted here has been adapted to suit the priority sectors of the CFF, and, as noted above, will be further refined as needed

Table 10: Climate Change Mitigation Activities

Climate Mitigation Category	Sub-Categories and Eligible Activities 2 1F1F
Renewable Energy	Electricity generation – Wind, geothermal(*), solar, biomass(*), biogas(*), ocean, hydropower(*), renewable energy power plant retrofits. (* only with net GHG emissions savings demonstrated.
	Heat production or other renewable energy applications – solar, wind, geothermal(*) and bioenergy(*). (* only with net GHG emissions savings demonstrated.
	Measures to facilitate integration of renewable energy into grids – storage systems, information and communication technology (ICT), smart-grid and mini-grid for renewable energy.
	ICT - Use of solar panels or other renewables instead of diesel generators or other carbon-intensive energy sources to power telecommunication masts
Energy Efficiency	EE in Industry in existing facilities – industrial EE or resource efficiency improvements, installation of co-generation plants(*) and more efficient facility replacement of a same capacity, older facility (old facility retired).

	(*) In all cogeneration projects, it is required that EE is substantially higher than separate production of electricity and heat.
	EE improvements in existing commercial, public and residential buildings – EE improvement in lighting, appliances and equipment, cogeneration (*), EE retrofit of existing buildings - architectural or building changes targeted at reduced energy consumption.
	(*) In all cogeneration projects, it is required that EE is substantially higher than separate production of electricity and heat. And, the CFF will not count as climate mitigation any cogeneration plant with coal in fuel mix
	Vehicle fleet EE – existing vehicles, rail or boat fleet retrofit or replacement i.e. EE achieved through reduced fuel consumption or use of lower carbon fuels, electric or hydrogen technologies. For vehicle fleet EE, excludes fleets dedicated to transport of fossil fuels.
	EE in new commercial, public and residential buildings – use of highly efficient architectural designs, EE appliances and equipment, and building techniques that reduce building energy consumption exceeding available standards and complying with high EE certification or rating. For new buildings only - an envelope approach is taken rather than a component based approach.
	Energy audits.
Waste & Waste Water	Portion of treatment of wastewater that reduces methane emissions (only if net GHG emission reductions can be demonstrated and if not a compliance requirement to meet, for example, a performance standard or safeguard requirement).
	Portion of waste management projects which capture or combust methane emissions.
	Waste to Energy projects.
	Waste collection, recycling and management projects that recover and re-use materials and waste as inputs into new products or as a resource. Only if net GHG emissions are demonstrated.
	Low-carbon clean drinking water production projects
Transport	Urban mass transit, bicycle and pedestrian transport.
	Transport oriented development including integration of transport and development planning leading to a reduction in use of passenger cars.
	Transport demand management measures dedicated to reduce GHG and other emissions, excluding general traffic management.
	Inter-urban transport – rail or waterways transport ensuring a modal shift of freight and/or passengers from road or air, to rail or waterways. Waterways taken to include inland waterways and short sea shipping, excluding international shipping. Excluding dedicated transport of fossil fuels.

Climate change adaptation eligibility criteria - general provisions:

- Adaptation finance eligibility criteria follows a location-specific, case specific and process-based
- Approach. Granularity: the specific costs of relevant components, sub-components, elements that contribute to increasing climate resilience are counted, rather than the cost of the entire project of which they are part.
- When a project has another main objective, adaptation finance is estimated based on the incremental costs related to climate change adaptation.
- Project activities which fulfil the following three design process criteria:
 - Set out a context of climate vulnerability (climate data, exposure and sensitivity), considering both the impacts from climate change as well as climate variability-related risks;
 - Include a statement of purpose or intent to address or improve climate resilience to differentiate between adaptation to current and future climate change and normal good practice; and,

- Must be linked to the context of climate vulnerability (e.g. socio-economic conditions and location) and contribute directly to climate resilience.

CFF Project Approval Process

- As shown in the illustrations above on the CFF structure, CFF projects will first be assessed and reviewed by the CFF steering committee. The Portfolio Management team and Steercom will ensure that proper due diligence and risk assessment has been performed, and that only projects that meet CFF eligibility are accepted into the fund.
- As part of the second screening, the DBSA as part of its AE role will among others carry out the following:
 - o Prepare environmental and social impact analysis report in line with the bank's environmental and social safeguards (including ensuring that the project is within allowed environmental impact classification, i.e. only up to category B)
 - o Prepare M&E plan, develop development results template (DRT) and project level log frame
 - o Assessment of the financial viability of the project
 - o The Climate Finance Unit (CFU) of DBSA ensures that a robust climate rationale is in place for all projects (in line with CFF investment criteria)

DBSA's investment committee will make a final decision to approve or reject based on the assessment above by the project steering/advisory committee. It is envisaged that the CFF's Project Steering Committee will include representatives from CFF's funders and if required and agreed to by the funders, an independent expert.

CFF Origination Channels

The CFF will address project pipeline by relying on a diversity of channels including:

1. RFP Process – The CFF will create a “market-responsive” investment window via an open-ended Request for Proposal (RFP) process through which private sector investors/developers may bring projects directly to the CFF for consideration. The CFF will issue an open-ended Request for Proposal (RFP) defining its eligibility criteria under the various funds it can access.
2. CFF Marketing and Outreach – The CFF will conduct targeted marketing and outreach to support market and project pipeline development in South Africa and other Rand-based economies. Market outreach will focus on relevant sectors including: off-grid projects, micro-grid/urban solar farm projects, industrial and commercial solar for self-generation and wheeling on the grid to MSMEs, industrial and commercial water projects, local Community water projects, and residential and commercial energy efficiency investments. Market outreach strategies will include
 - Outreach to developers and project sponsors to explain the role and capacity of the CFF and how it addresses significant financing barriers.
 - Outreach to Commercial Banks and DFI's to encourage referral of projects that meet CFF criteria and are not currently bankable at sustainable terms.
 - Outreach to Municipalities to explain the role of the CFF around micro-grid/urban solar farm projects and local Community water projects.
 - Support for the CFF RFP solicitation process as noted above
 - Creation and maintenance of CFF website to support visibility and understanding of the CFF
3. DBSA Origination Team - The DBSA's SPU and PPU will work with the Origination Unit to identify companies and projects that will be eligible for funding from the CFF from South Africa and other nations in the SADC region. *DBSA to add structural description of the DBSA Origination Unit and how it will support pipeline development and deal flow into CFF for both South Africa markets and the SADC region*
4. DBSA Project Preparation Unit - The CFF will be closely tied to the DBSA's Project Preparation Unit (PPU) whose purpose is to develop projects from pre-feasibility to bankability. The PPU can refer projects that it has brought to bankability to the CFF. These projects may currently be in the PPU pipeline, could come to PPU from the DBSA origination teams, or may have previously been directly presented to the CFU and then referred to PPU for further development. *DBSA to add structural description of the DBSA PPU and how it will support pipeline development and deal flow into CFF for both South African markets and the SADC region.*
5. Climate Lab - The CFF will launch structured finance products as part of a “programmatic” effort to fill previously-identified gaps in the private capital markets, designed in partnership with the Innovation Hub and the new CPI “Climate Lab” to be hosted at DBSA headquarters. The DBSAs Innovation hub is working with the Climate Policy

Initiative to create a Climate Lab for Africa to accelerate well-designed financial instruments that can unlock billions for energy efficiency, renewable energy, sustainable transport, climate smart agriculture, and curbing deforestation, while also reducing private investors' risks and improving their financial returns. As a public-private partnership, the Africa Lab brings together and catalyzes broader government and private sector efforts to scale up climate finance. The Africa lab will be modeled on CPIs established success through the existing India, Brazil and Global Climate Labs. The Climate Lab will be organized around an established five part process including the following elements:

- Call for Ideas – Proposals will be solicited through an open call for innovative sustainable investment solutions
- Selection - Lab Members will select the most promising ideas based on criteria (summarized below)
- Development - Selected ideas for Climate Labs benefit from analysis, stress-testing, and guidance from experts and investors. CPI "Fire Awards" finalists are invited to prepare business pitches.
- Endorsement & Launch - Lab Members vote to launch the ideas for piloting, based on their innovation, actionability, financial sustainability, and catalytic potential. CPI "Fire Awards" winners are announced.
- Implementation – The selected ideas move into action and be deployed through partnerships by the CFF.

6. DFIs – DFIs may review and refer deals to the CFF based on their position in the South African market and SADC region. Based on initial outreach to several, it is clear that DFIs want to increase the scale and pace of their green investments (per their commitments to Paris goals), but have trouble investing in small to medium projects. The CFF could help source and perform due diligence on small or medium projects for either direct investment or co financing with the CFF.

In addition, Commercial Banks receiving inquiries to finance projects suitable for CFF financing can directly reach out to the CFF upon reviewing potential projects. Commercial Banks can submit projects to the CFF that could be commercially viable but not yet bankable in the private sector, without credit enhancement and related financial or development support. Based on initial outreach to RMB, Standard Bank, Investec, ABSA and Nedbank, it is anticipated that there is a significant pipeline of projects that could be made bankable via two main instruments: subordinated debt / first-loss and credit enhancements such as tenor extension. Examples of projects across key sectors that have surfaced include off-grid, urban solar farms, waste to energy, and local water treatment plants. More detail on deal flow and project pipeline from commercial banks will come early in 2018, based on the next stage of planned outreach.

C.8. Timetable of Project/Programme Implementation

The CFF project implementation timetable is provided below to provide an overview of project components, timeline and schedule of project deliverables. DBSA and CGC have made significant progress developing the CFF concept to this point. DBSA's Board has approved the CFF and indicated intention to provide R650 million towards initial capitalization and R1 million toward technical start up support; the South African DEA has indicated support and intent to provide a "letter of no objection" to the GCF; several major DFIs have indicated interest in capitalization; South Africa's five major private commercial Banks have expressed support for the initiative; and initial outreach to private project developers has also indicated need and receptivity. A market overview is in process via project consultants and the project team is in the process of determining target markets, staffing structures, governance, credit procedures, affiliates, stakeholders and operating model. Work has begun on development of a business plan, organizational structure and other elements and project development funding has been secured through Convergence Finance and the Climate Works Foundation.

Table 11: CFF Implementation Timetable - See Annex for CFF Implementation Timetable

Climate Finance Facility Implementation Timeline																								
Operating Stage	2018				2019				2020				2021				2022				2023			
Year																								
Quarter	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Component 1: Operationalization and project management of CFF																								
1.1 Unit establishment within DBSA																								
Lead successful operationalization and launch of the CFF unit within the DBSA; • Design and put in place CFF portfolio management unit team; • Develop strategic plan, • Form CFF Steering Committee; • Oversee legal and accounting arrangements; • Create budget template, and bank accounts; and other tasks as required																								
1.2 Development of CFF Operational Manual																								
• Manual will include detailed investment criteria, eligibility criteria for each sector targeted by CFF, monitoring and impact measurement guidelines; • The operational manual will stipulate how CFF investment and eligibility criteria will be fully integrated into the investment appraisal process of CFF; • Eligibility criteria will include a focus on the water sector.																								
1.3 Annual Work Plan																								
Develop budget plan that considers the financial sustainability of the facility and a balanced portfolio of programs;																								
1.4 Project Pipeline Development & Financing Programs																								
Portfolio Management team to develop first round of CFF financial products; • Design and launch an open-ended Request for Proposal (RFP) process to support development of a robust project pipeline for the CFF; • Interface with DBSA origination channels and the new CPI "Climate Lab" to be hosted at DBSA; • Oversee ongoing market research to continually develop new opportunities, align programs to emerging markets, and maintain competitiveness with other financing options.																								
1.5 Communications & Marketing																								
• Develop, manage, and maintain organizational brand and external presence of the CFF and execute a communications and outreach effort that promotes the objectives of the CFF working with the DBSA marketing team; • Lead direct outreach to project developers, commercial banks, DFI's and other relevant stakeholders; • Build the reputation and brand of the CFF with donors, foundations, corporations, the media and public officials and agencies; • Establish strategic partnerships with lenders, contractors, developers, government agencies, utilities, business and industry associations, and community groups.																								
Component 2: Project financing of climate change mitigation and adaptation projects																								
2.1 Mitigation Financing																								
• DBSA will enter into loan agreements for mitigation projects (Renewable Energy, Waste to Energy, Energy Efficiency, Low Emission Transport)																								
2.2 Adaptation Financing																								
• DBSA will enter into loan agreements for adaptation projects (Water Efficiency)																								
Annual Performance Report			Inception Report		APR				APR				APR					APR			APR			
NB: GCF funding focussed on implementation and operating phase of CFF.																								

D.1. Value Added for GCF Involvement

Please specify why the GCF involvement is critical for the project/programme, in consideration of other alternatives.

As highlighted earlier, South Africa has made substantial commitments to responsibly address climate change. Achieving these will require collaboration across public and private sector, and innovative funding channels to scale high-impact projects. The proposed financial structure will allow DBSA to use GCF support to “crowd in” private sector funders and support climate mitigation/adaptation interventions across the target sectors on a significant scale.

The GCF funding will enable CFF to overcome the following key barriers:

- Lack of appropriate financing mechanisms
- High capital cost
- Better rate: After access to finance, the cost of finance is the biggest obstacles to MSMEs in capital formation. We want to use the GCF funding to help reduce cost of borrowing for MSMEs. Lower financing costs (and longer tenors) help facilitate lower end-user prices (e.g. PPAs) that are more attractive to customers, spurring increased uptake.
- Private funding: By blending the GCF funding with private investor funding, we will enable private funder to lower their risk exposure per transaction. Increased participation of private sector would over time lead to sustainable private sector support without GCF participation.

The CFF will complement existing climate-friendly finance capacity by providing an institution specifically focused on scaling up private investment in small and medium-sized independent energy production through off-grid, micro-grid and IPP via emerging independent businesses. This will be additive to existing energy policy in South Africa (affecting the entire SADC region), which is focused on large scale power contracts via the IRPPP. It will respond to a stated need of the region’s commercial banks to reduce risk and make climate-friendly projects more bankable.

Existing energy policy and investment in South Africa has focused on implementing the Integrated Resource Plan, South Africa’s blueprint for the energy mix up to 2030, through large scale power contracts under the terms of the REIPPPP and contracts with Eskom for large scale renewable energy generation. Beyond the recent challenges to the IRPPP due to institutional capacity constraints, corruption and the inability to sign new contracts, there is a significant gap in market capacity around scaling up investment in smaller projects and independent power production for distributed generation. These issues affect the broader Southern African region too, as Eskom is the largest member of the SAPP, and the majority of the region is reliant on South Africa for power supply through long-term supply contracts and day ahead market trading.

Many of the climate mitigation and adaption companies emerging in Southern Africa are startups promoting new technology that banks regard as venture and development capital in nature. There is limited equity available from project sponsors and developers, and given the perception of these businesses as risky venture start-ups, commercial banks are reluctant to gear beyond a debt:equity ratio of 50%. In addition, banks generally limit the tenor of these loans to 5-7 years. The CFF will address these market barriers by playing a catalytic role using a blended finance approach with a focus on projects that are commercially viable but not yet bankable in the private sector without credit enhancement and related financial or development support. This focus will include resilience and adaptation projects such as local waste water treatment plants; water reticulation equipment the harvesting of rain water etc.

Key added-value interventions to complement the role that commercial banks can already play will include: longer loan tenors; first loss and sub-ordinated tranches; project aggregation for smaller projects so they can better access commercial financing, securitization and project aggregation; guarantees; and providing a dedicated, focused and efficient public financing entity.

GCF support for this initiative will therefore be quite transformative in releasing significant amount of funds from commercial banks to support mitigation and adaptation projects mostly spearheaded by emerging companies.

D.2. Exit Strategy

Please explain how the project/programme sustainability will be ensured in the long run, after the project/programme is implemented with support from the GCF and other sources, taking into consideration the long-term financial viability demonstrated in [E.6.3](#). This should include a description of strategies for longer term maintenance of physical assets (if applicable).

In this section, the accredited entity is expected to provide a brief description of the expected performance of the proposed project/programme against each of the Fund's six investment criteria. Activity-specific sub-criteria and indicative assessment factors, which can be found in the Fund's [Investment Framework](#), should be addressed where relevant and applicable. This section should tie into any request for concessionality made in [section B.2](#).

E.1. Impact Potential

Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas

E.1.1. Mitigation / adaptation impact potential

Specify the mitigation and/or adaptation impact, taking into account the relevant and applicable sub-criteria and assessment factors in the Fund's [investment framework](#).

Climate impact potential

- This project will demonstrate that businesses and public sector can implement various climate mitigation/adaptation initiatives in many aspects of their everyday activities including Energy Efficiency & Demand Side Management, Renewable energy and Sustainable Water Management.
- Reduction of CO₂ emissions through displacement of fossil-fueled electricity use at the user level by implementing energy efficiencies technology, installation of clean and renewable energy sources such as solar PV as shown in the tables below:

Impact Potential Methodology:

The key focus of the calculations in this model is to ensure that 5 specific criteria are resolved in each sector (listed below). Data was collated through stakeholder engagements, desktop research (local and international), expert input and years of local market experience. Values have been summed and averaged based on a specific target markets (e.g. commercial, industrial) to ensure minimal outliers have been included, whilst ensuring the investment inputs are flexible to ensure the best investment:impact ratio is achievable.

1. Market sizing and growth from a 2015 baseline to 2030 per sector (capacity, ZAR value, energy output)
2. Jobs created and beneficiaries (direct and indirect)
3. Male: Female split
4. tCO₂ emissions
5. Business impact

Through current installed capacity, price points and energy output per sector one is able to determine a baseline (2015) of a sector. This figure is extrapolated through existing market trends and research projections to determine a market potential (2030).

Further analysis is completed on [existing and past projects](#) to determine the number of jobs created, size of projects and male and female split to determine an average across the sector. From these outputs it is possible to make calculated assumptions on market sizing in the years to come.

One of the key outputs (tCO₂ emissions) is determined by the only SA utility, Eskom, which provides a carbon emission factor of 1.03kg CO₂ emissions/kWh. This value has been multiplied by the kWh generated by each sector per annum to provide a total CO₂ saving for the various sectors. The figure indicates an avoided tCO₂ emissions given the current SA energy mix, which is in excess of 85% coal.

The beneficiary multiplier has been sourced through research reports on Africa, whilst the water multiplier has been calculated based on the projected shortfall of water in South Africa (WWF 2016 report) relative to an investment of 100ML/Day.

Table 12: – Climate Mitigation Opportunities in Energy and Transport applications in South Africa

	Baseline 2015 (MW Installed or Used)	CFF Target % Portfolio ⁴¹	Potential CFF MW Installation Capacity	Potential MWh generated or saved per annum through CFF	Potential Number and type of Jobs created	Potential Number and type of Female Jobs created	CO ₂ eq reduction per annum
Rooftop Solar PV projects							
Commercial	4 MW installed	6% ⁴²	44 MW	81 192	540 installation jobs	243	83,627
Industrial	17 MW installed	25%	191 MW	324,766	2160 installation jobs	972	334,509
Energy Efficiency projects							
Commercial	15,139,180 MW used	2%	N/A	172,342	104 (Insulation, lighting and windows)	47	177,513
Industrial (including mining and transport)	138,415,360 MW used	20%	N/A	1,551,085	930 ⁴³ (EE motors and mechanical insulation)	418	1,597,617
Waste to Energy projects							
South Africa	12MW	10%	22	175,431,690	457	206	180,695
Clean Transport							
Clean transport ⁴⁴ (EV's and fuel Vehicle fuel economy and energy source as a result of CFF support)		7%	N/A	N/A	N/A	N/A	N/A

⁴¹ Estimated CFF targets subject to review on a continuous basis upon implementation. Commercial and Industrial sectors are split only for the purposes of illustrating relative impact of the 2 subsectors based on current trends.

⁴² Baseline = 2015

Commercial (13%) and industrial (57%) of total rooftop PV according to DoE (2016) therefore actual split within the C&I sector is 19:81

⁴³ Because of increased efficiencies of motors and technology, it's expected that significant jobs could be lost here in the future

⁴⁴ Due to the lack of sufficient data on clean transport in South Africa, it is not yet clear what the scale of this impact would be. The tonnes CO₂ eq reduction estimate here is based on potential impact of energy efficiency initiatives in the transport sector. The clean transport sector is an emerging sector in the CFF target countries, hence the potential impact cannot be readily estimated at this stage. EV market is in very early stage. Opportunities exist in local manufacturing of EV components, charging infrastructure roll out (public and private) and development of secondary markets.

Table 13 – Climate Adaptation Opportunities in the water sector in South

	Baseline 2015 Water Use SA per sector (KL)	CFF Target % Portfolio ⁴⁵	Potential CFF Installation Capacity (KL/Day)	Potential Number and type of Jobs created	Potential Number of Female Jobs created ⁴⁶	Resilience impact
Water treatment ⁴⁷						
Commercial	445,500,00 0	20	90,000	119	53	Increased resilience to water shortages and security of water supply
Industrial	162,000,00 0	10	10,000	13	6	

In terms of potential private sector solar and energy efficiency projects which could be supported by the CFF, both the commercial and industrial sectors will be targeted in the identified countries. The commercial sector comprises a vast range of business customers - harbours, airports, business parks, commercial buildings, shopping centres, entertainment complexes, distribution centres, cold storage facilities, hotels and lodges, hospitals, educational institutions and many more. According to ESKOM, the commercial sector consumes approximately 14% of South Africa's electricity – the four biggest end uses are:

- Heating, ventilation and air-conditioning (HVAC) systems - 26%.
- Lighting (18%)
- Motors (14%)
- Water heating (8%)

South Africa's industrial sector is a huge consumer of electricity - around the clock, every day of the week, all year long. Running a wide range of manufacturing, refining, resource extraction and mining operations and processes, the industrial sector consumes 60% of South Africa's electricity. The electricity consumption patterns in the other CFF targeted countries are similar in terms of the split between commercial and industrial use although the percentage contribution per country varies relative to domestic use. As such, the potential impact in the commercial and industrial sectors reflected in Tables 12 and 13 serves as a guide as to which of these sub-sectors presents the most significant opportunities for CFF. CFF would not necessarily be prescriptive in allocating capital based on the % split illustrated in tables 12 and 13 bearing in mind that CFF is earmarked for private sector projects irrespective of industrial or commercial applications. In the same regard, the potential impact of CFF in the water sector focuses on private sector (commercial and industrial), hence municipal use is not reflected. Studies done by the Department of Water and Sanitation (DWA), South Africa show that the vast majority of water in South Africa for example is used in agriculture, with over 60% of all available water going into the sector for irrigation. As much as 30% of water in SA is for urban and rural use (including domestic use), while the rest is split among industrial, power generation and afforestation uses. It is therefore envisaged that the industrial sector poses the greatest opportunity for CFF to create significant impact in the water sector in respect of water treatment and production which is independent of municipal supply.

The various assumptions informing the impact data shown in Tables 12 and 13 are outlined in the attached impact model created for the CFF.

⁴⁶ Based on percentage of female employees in water treatment companies

⁴⁷ Data is not readily available on water efficiency statistics in the commercial and industrial sectors.

With regards to energy efficiency projects, it should also be noted that due to the variance of technology, energy efficiency implementations yields varying results. Below we illustrate the variance in outcomes from different projects implemented in recent years through Industrial Energy Efficiency initiatives by various companies in South Africa.

Toyota South Africa was one of the Industrial Energy Efficiency (IEE) Project's first demonstrated successes. The Durban plant began energy efficiency improvements in 2009 and joined the IEE Project in 2010. To date, 103 projects have been implemented at the Toyota Durban plant, saving energy valued in excess of R 24.86 million. (26,668 tonnes of CO₂ saved)

ArcelorMittal Saldanha steel works in the Western Cape is the largest single energy saver in the IEE Project's four-year history. The introduction and implementation of interventions to manage and optimise energy usage at this plant required an investment of a mere R 500 000, but led to gross financial savings of more than R 89.6 million within the first year (2011). By 2013, savings in excess of 176 GWh, with a financial saving of R 171 million, have been realised.

Johnson Matthey South Africa, manufacturers of auto catalysts, joined the IEE Project as a candidate plant for the implementation of an ISO 50001-based energy management system. Several energy efficiency improvement projects were implemented. Following an investment of only R 666 000, energy savings worth R 7.7 million were realised – 9,425,084 kWh. (9,020 tonnes of CO₂ saved)

Sappi Cape Kraft is a paper mill based in Cape Town, and is the only plant in South Africa which uses only 100% recycled paper as its input raw material. In 2012/13 five projects were undertaken, resulting in a total saving of 944 445 kWh of electricity and 540 553 kWh of steam energy – equivalent to a saving of R 894,000 for an investment of R 70,000. (CO₂ emissions savings of 1 416 tonnes)

King Shaka International Airport in Durban, uses vast amounts of electricity. Over the period 2010 – 2013, three projects were undertaken, resulting in a total energy saving of 1 932 576 kWh, valued at R 2.76 million. With an investment of R 400 641, the payback period was 1.8 months. (CO₂ reduction of 1 850 tonnes)

The table below presents more information on the climate impact potential of potential projects which could be supported by the CFF in the organic waste sector.

Table 14: A sample of organic waste solutions in Western Cape and potential 5 year projections (2016)

Waste hierarchy	Family	Current tonnes/year	Tonnes/year projected	Input	Output
Reuse	Waste to food	720	7 200	Pre-Consumer Obsolete Stock	Recyclables
Recycle	Waste to feed	36 000	126 000	Pre-Consumer Post-Consumer	Protein Feed Oil Compost
	Waste to soil	145 000	145 000	Green Waste	Compost
		14 400	36 000	Pre-Consumer Post-Consumer	Compost
		720	720	Animals Pre-Consumer Post-Consumer	Non Sellable Compost
		3 600	12 600	Pre-Consumer Post-Consumer	Compost Vermicast
	Waste to energy	108 000	108 000	Pre-Consumer Post-Consumer Abattoir	CO2 Methane Sludge Waste Water
		0	108 000	Manure Pre-Consumer Post-Consumer	Biogas Sludge Waste Water
		36 000	41 400	Pre-Consumer Post-Consumer Abattoir	Biogas Sludge Waste Water
	Total	344 440	584 920		

Table 14 aims to illustrate the climate impact potential of potential projects which could be supported by the CFF in the organic waste sector including waste to energy. Waste to Energy is earmarked by CFF is an important target sector for financial support. At least 10% of the CFF portfolio is envisaged to come from the waste to energy sector. Although the table includes other waste beneficiation outputs which may not be directly relevant to CFF, it was included as a basis for comparing different uses of the waste in order to outline the value proposition from a climate mitigation perspective of using organic waste to produce energy compared to other uses of waste. For organic waste, conversion to energy has the greatest potential for climate mitigation impact based on the number tonnes/year of waste projected (257 500 tonnes /year) for this application as reflected in Table 14. This translates to ~226 000 tonnes of CO2 saved per year (assuming 0.88 Tons CO2 saved per Ton of organic material Recycled/Composted). It is envisaged that CFF could fund a significant component of this impact which is reflected in the log frame.

Table 15: Job creation potential in CFF targeted sectors

Market	Potential Job Creation			Number of Businesses Impacted
	Total	Female	Male	
Solar PV	28525	10269	18256	15740
C&I EE	8969	2691	6278	137942
Waste to Energy	1065	403	662	86
Water Efficiency and Treatment	503	101	402	1398

When applicable, specify the degree to which the project/programme avoids lock-in of long-lived, high emission or climate-vulnerable infrastructure.

One of the main energy challenges in South Africa is due to historical environmentally unfriendly structural design. Utility scale electricity infrastructure require large cost on distribution lines. Compounding the problem is the fact that the infrastructure is ageing. Embedded generation and micro-grids can solve this problem whilst also bringing the following additional benefits:

- Due to the small size of these individual projects, they are perfect for MSMEs in particular to undertake. Thus whilst providing electricity, there will be added benefits of MSME development, job creation and skills building
- There are no distribution costs for the energy or water produced (or saved) on site

- They are quick to install, thereby bringing delivery of basic services to underdeveloped markets, and the poor

As noted earlier in the FP, the strategic context above has a particular focus on South Africa since the pipeline is expected to be dominated by projects in South Africa during the initial phase of the implementation of CFF. It therefore follows that the impact created through the implementation of the CFF would be most significant in South Africa due to the significant size of the CFF targeted sectors relative to the other ZAR zone countries. As such, the input parameters assessed are only for South Africa where accurate data could be collated through stakeholder engagements, desktop research (local and international), expert input and years of local market experience. Although these parameters are based on the South African context, it should be noted that the parameters may not vary significantly amongst the ZAR zone countries due to the similarities in service providers used during the implementation of infrastructure projects. In fact, a large number of South African based contractors have been involved in the construction of Renewable Energy infrastructure in the other ZAR zone countries.

It should be further noted that during the reporting phase for specific projects supported by CFF, project sponsors will be expected to provide country specific parameters depending on the nature of the projects and the specific impacts being assessed at the time.

As the CFF develops and becomes active in the other ZAR economies, we anticipate refining the impact metrics to reflect the evolving scope of the CFF and it is envisaged that CFF can build a robust baseline for the other ZAR countries.

E.1.2. Key impact potential indicator (Please refer to annex for CFF Impact model)

Provide specific numerical values for the indicators below.

See attached impact model for impact potential calculations and assumptions.

GCF core indicators	Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (Mitigation only)	Annual	2 373 963
		Lifetime	29 727 942
	- Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience); - Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)	Total	Direct: 22 732 Female direct (%): 30% Indirect: 443 652 Female indirect (%): 51.7%
		Percentage (%)	0.8% of total population impacted
Other relevant indicators	Refer log frame in section H		
	Notes on Adaptation core indicators:		
	Context: Total average direct job per drop converted to (jobs/ML/Day) – Western Cape (largest province in South figures used as an average for the country).		
	- Non Agriculture – 1500 Jobs/ML/Day - Agriculture – 85 Jobs/ML/Day		
Other relevant indicators	Most of the water projects supported by CFF will go into non-agricultural sectors, but 1500 Jobs/ML/Day is deemed to be high, as many of these companies do not really have water related business. A conservative number of 226 jobs /ML / Day is suggested based on companies with highest reliance on water (Companies that would have a high probability of shutting down without access to water)		
	Other relevant indicators		
Other relevant indicators	Expected increase in the number of small, medium and large low-emission power suppliers, and installed effective capacity		

	- 260 - Expected strengthening of adaptive capacity and reduced exposure to climate risks 10ML/day saved through efficiency projects 40 ML/day treated for reuse 50ML/day new sourced
<i>Describe the detailed methodology used for calculating the indicators above.</i> <i>Describe how the project/programme's indicator values compare to the appropriate benchmarks (i.e. the indicator values for a similar project/programme in a comparable context).</i> See note above on impact methodology and see attached impact model in Annex.	
E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)	
<i>Describe how the proposed project/programme's expected contributions to global low-carbon and/or climate-resilient development pathways could be scaled-up and replicated including a description of the steps necessary to accomplish it.</i> The Climate Finance Facility and Green Bank Model as a Replicable Model for Emerging Economies National Green Banks—structured as either new purpose-built institutions or adaptations of existing institutions – can help redefine the global climate finance architecture to address critical market gaps and drive public and private climate investment.⁴⁸ Green Banks are typically not banks, but rather public-purpose finance institutions or vehicles dedicated to green investment, embodying the pure focus and local market-oriented approach needed to help local private investors bridge the investment shortfall. Green Banks are designed to maximize total investment, using limited public funds to leverage far greater private investment. By developing innovative finance and market development solutions, Green Banks address barriers that currently restrict clean energy market growth.	

⁴⁸ Coalition for Green Capital (2017), “National Green Banks in Developing Countries: Scaling Up Private Finance to Achieve Paris Climate Goals.”

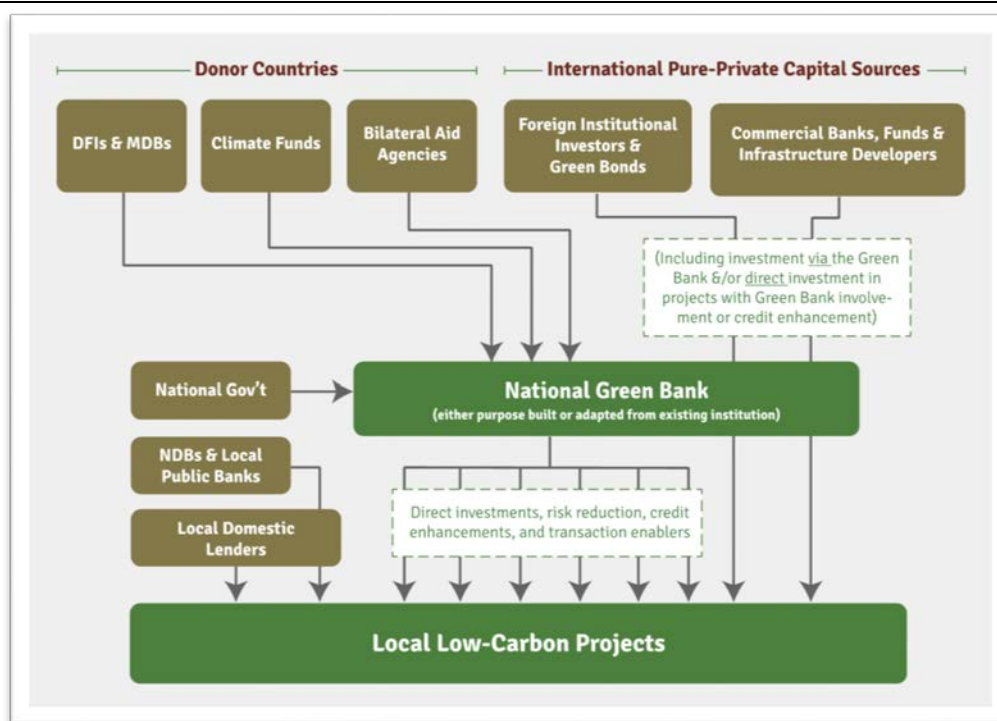


Figure 21

Green Banks provide financing in various forms, targeting commercially viable clean energy technologies, that for a host of reasons cannot attract debt at a cost of capital or tenor that allows customers to move forward with projects. Green Banks fill market gaps, and pair their capital with private investors to “crowd-in” capital and make clean energy markets more efficient. Green Banks are self-sustaining facilities, meaning they provide their capital at a cost commensurate with risk and sufficient to generate revenue to operate the facility on a break even basis. Key elements of the Green Bank model include:

- Focused institutions, created to mitigate and adapt to climate change
- Use of blended finance to de-risk & catalyze private investment with innovative funding structures
- Use of debt, warehousing, credit-enhancing and related instruments to enable cost-effective long-term sustainable financing
- Designed to support commercially viable and sustainable projects
- Complement existing funders and programs
- Market-oriented and flexible adjusting offerings, products and partnership structures to suit the project needs.

Green Banks have proven successful at driving clean energy investment and there are an increasing number of Green Banks and similar entities in development around the world. Collectively these institutions have financed billions of dollars of clean energy projects with innovative financing structures, leveraging multiple private dollars per public dollar of financing. Founded at COP21 in Paris, the Green Bank Network (GBN) is a membership organization formed to foster collaboration and knowledge exchange among existing Green Banks, enabling them to share best practices and lessons learned. Members include the Australian Clean Energy Finance Corporation, the Japan Green Fund, Malaysia Green Technology Corporation, the Connecticut Green Bank, NY Green Bank, and Green Investment Group (formerly the UK Green Investment Bank) and is supported by two non-profit organizations, the Coalition for Green Capital (CGC) and the Natural Resources Defense Council (NRDC). To date, the founding members of the Green Bank Network⁴⁹ have financed more than \$29 billion in clean energy projects in 6 years average operation. This investment has flowed into a range of clean energy technologies including solar, offshore wind, and building efficiency and resulting in 12 million tonnes of

⁴⁹ Green Bank Members: UK Green Investment Bank, Clean Energy Finance Corporation of Australia, Green Technology Financing Scheme of Malaysia, Green Finance Organization of Japan, Connecticut Green Bank, New York Green Bank

avoided CO₂ emissions annually. GBN member use a range of financing techniques including debt, subordinated debt, equity, and credit enhancements, delivered through a number of innovative structures, such as property tax-based financing, equipment leases, aggregation, fund investment and others. Together, they have demonstrated a wide range of products and market development solutions that can be deployed in partnership with the private sector. Operating Green Banks include the Australian Clean Energy Financing Corporation, Connecticut Green Bank, New York Green Bank, Japanese Green Finance Organization, Malaysia Green Technology Corporation, Rhode Island Infrastructure Bank, California CLEEN Center, Hawaii Green Infrastructure Authority, and the Montgomery County Green Bank.

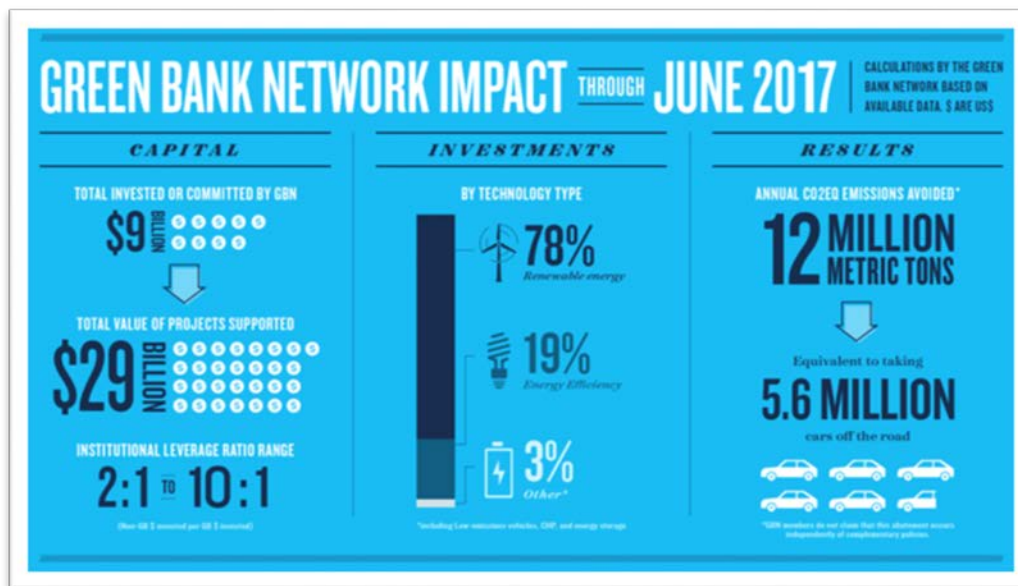


Figure 22

Knowledge by Example: DBSA's Climate Finance Facility as Proof-of-Concept Model for Emerging Economies

The CFF will provide a globally significant proof-of-concept for developing a Green Bank Programme through an existing institution in the developing country context – a more effective path for a wide range of middle and lower income nations vs. creating new stand-alone institutions. Many entities are already taking note of the DBSA's initiative as a replicable model for their regions.⁵⁰ For example, Findeter, a major National Development Bank in Colombia has indicated strong interest in the Green Bank model and specifically learning from and potentially replicating the DBSA's approach. Other emerging market countries have expressed similar interest. The OECD has featured the DBSA's approach in several high profile conventions such as the 2017 Global Investment Finance Forum and a Green Bank side-event at COP23 where the DBSA's CFF was featured prominently. The UNFCCC has indicated strong interest in the Green Bank model and the DBSA initiative, as it relates to expansion of green finance through the East African Development Bank, West African Development Bank, Infrastructure Development Bank of Zimbabwe, and others. GIZ has expressed interest in the Green Bank model and the CFF as a significant model for developing economies. The Turkish Industrial Development Bank has indicated interest in the CFF and Green Bank model. ADB has also expressed interest in the CFF model, and using it as an example when piloting new country-level "Green Finance Catalyzing Facilities" in the SE Asia region in 2018.⁵¹ In terms of replicability and chances of a successful proof-of-concept initiative, it is also important to note that the DBSA has a reputation as one of the strongest institutions in South Africa with established financial expertise, sound organizational best practice, credibility with the DFI and donor community, trust and credibility with major banks and project investors. DBSA is well positioned to deliver on a transformative agenda as envisaged by the creation of the CFF.

The CFF and the Green Bank model are strongly aligned with the mission of the Green Climate Fund to strengthen the capacity of developing nations to scale up climate finance and support their efforts to respond to the challenge of climate

⁵⁰ Responsible Investor (2017), "South African DFI Becomes Dedicated Green Investment Bank as Countries Seek to lever Private Finance". https://www.responsible-investor.com/home/article/oe_db/

⁵¹ Personal interview w/ Mehta, author of "Catalyzing Green Finance," Asian Development Bank, Aug 2017 <https://www.adb.org/publications/green-finance-catalyzing-facility>

change. Specifically, Green Banks add a critical element to the climate finance architecture to better position national and sub-national entities to directly catalyze the flow of climate finance into low-emission and climate-resilient development, and to drive a paradigm shift in the global response to climate change. To drive this shift, the CFF and Green Banks are designed to crowd-in private investment, leverage public capital and address market barriers to open new investment opportunities.

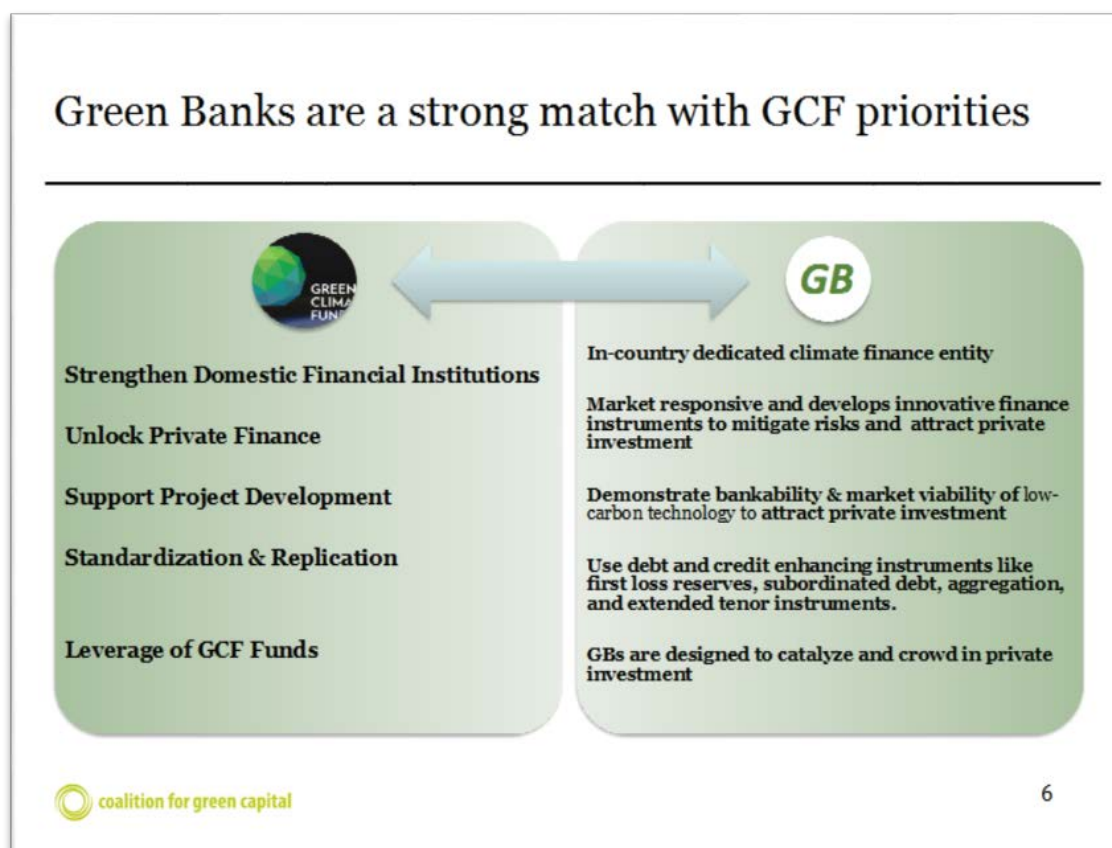


Figure 23

E.2.2. Potential for knowledge and learning

The DBSA's Climate Finance Facility will contribute to knowledge and learning regarding application of the Green Bank model in developing countries as a means to leverage and scale-up climate investment in the following ways:

Engaging DFIs

Through efforts to capitalize the CFF, the DBSA is in the forefront of engaging Development Finance Institutions in engaging with, evaluating and testing the Green bank model in the developing country context. Through the CFF initiative, several leading DFIs will test the Green Bank model as a way to complement their existing programs and increase opportunities for DFI investments, by increasing specialized intermediation and risk mitigation, and offering tailored solutions to fill existing gaps. At this point, several DFIs are actively considering capitalizing the CFF and, based on this experience will be well positioned to extend this investment model to other developing nations in support of increased climate investment. DFIs are expected to play a significant role driving climate investment and in addressing the global climate investment gap. To achieve the goals laid out in Paris, annual investment must be \$1,276 billion. To keep global warming below 2 degrees, annual investment must be \$2,019 billion. The gap between the current pathway of investment is \$9.1 trillion and the gap between the current pathway and the investment level needed to prevent 2 degrees of warming

is \$27.6 trillion.⁵² In 2014, DFIs financed \$131 billion of climate projects, including both mitigation and adaptation⁵³ while the gap between the real annual investment level in 2015 and the level needed to achieve the Paris commitments is \$759 billion.⁵⁴ Based on OECD projections of increases in developed countries public climate finance commitments as matched one-to-one with private capital (at the high end of OECD projections), only 5.8% of the annual climate investment gap that exists between current investment levels and the “Meet NDC” scenario, and only 2.9% of the annual gap between current and two-degree target investment will be met.⁵⁵

The existing DFI system will not be able to fill this investment shortfall alone.⁵⁶ Not only does more public capital have to flow into climate investment, but that public capital has to achieve far greater leverage from the private sector. Further, the development needs of nations around the world are immense and diverse. While it is wise for DFIs to increase the number of investments in low-carbon infrastructure and make all other investments through a “climate lens” to ensure that agriculture, healthcare and civil society development efforts are carried out in a climate-friendly way, these activities can be quite different from achieving the levels of direct investment needed in decarbonization, or even adaptation, projects. In addition, given that the basic institutional and financial designs of most DFIs are appropriately designed to minimize risk-taking, maintain high credit rating and diversified portfolios and introduce change slowly, changes of this magnitude and speed may be hard to achieve. By design, international DFIs work through partnerships with governments, National Development Banks (NDBs) and other local actors all of which must address the significant in-country market gaps described above. Local programs and institutions dedicated to increasing private and DFI investment in green will be critical to scale up investment to the necessary levels and act as effective local intermediaries.

Demonstrating new approaches to reducing risk

The Climate Finance Facility is designed to address market gaps and demonstrate new approaches to driving public and private climate investment. By developing innovative finance and market development solutions, the CFF will support the capacity of commercial banks to address barriers that currently restrict clean energy market growth. In this way, the CFF will complement existing market actors and prove new markets and financial strategies by absorbing risk through credit enhancement such as subordinated debt, first loss and tenor extensions. Complementarity and coordination with the region’s commercial banks will support development of new approaches to reducing risk and building demand.

Evaluating how the Green Bank Model Meets Green Climate Fund Goals

The CFF will provide a tangible opportunity for the Green Climate Fund to test the premise that the Green Bank model can drive a paradigm shift in the global response to climate change by re-aligning the global climate finance architecture around improved nation-specific climate investment capacity. Specifically, the CFF will present an opportunity to evaluate how the Green Bank model addresses the following GCF goals:

- Strengthen Domestic Financial Institutions – The CFF will work within and help evolve an existing national finance institution (the DBSA) and support the overall “greening” of this major national organization.
- Unlock Private Finance – The CFF is designed to create and test a new platform for in-country dedicated climate finance that works in partnership with private lenders to catalyze greater overall climate-related investment. The CFF will focus on projects that are commercially viable but not yet bankable in the private sector without credit enhancement and related financial or development support, and transactions will be required to demonstrate the

⁵² IEA (2016), “World Energy Outlook”, <http://www.worldenergyoutlook.org/publications.weo-2016/>

⁵³ CPI (2015), “Global Landscape of Climate Finance 2015,” see Figure 3. <http://climatepolicyinitiative.org/wp-content/uploads/2015/11/Global-Landscape-of-Climate-Finance-2015.pdf>

⁵⁴ IEA (2016), “World Energy Outlook”, <http://www.worldenergyoutlook.org/publications.weo-2016/>

⁵⁵ OECD (2017), “2020 Projections of 17 Climate Finance Towards the USD 100 Billion Goal”: Technical Note, Éditions OCDE, Paris. DOI: <http://dx.doi.org/10.1787/9789264274204-en>

⁵⁶ According to the IEA methodology, the “Current Policy” scenario already accounts for these commitments. Under this assumption, the annual gap between the Current Pathway and the Meet NDC scenario is \$364 billion. So even in the most favorable case, this collective pledge only fills 12.6% of the gap between Current and NDC scenarios or only 4.2% of the gap between the Current and Two Degree scenarios.

ability to crowd-in private sector investment for projects that are technically and economically feasible but lack access to capital due to specific financing gaps and barriers.

- Support Project Development – The CFF will provide an opportunity to develop effective approaches to supporting projects that contribute to market transformation and demonstrate that they can materially and sustainably expand markets in terms of scale, improved private sector participation, confidence in clean energy investments, or other aspects. The CFF will focus on developing and sharing strategies that support technically and economically feasible transactions where there is market interest but a lack of financing from the commercial market due to specific financing gaps and barriers.
- Standardization & Replication – The CFF will be specifically designed as a platform for continued climate investment through replicable approaches and development of new financial instruments. It will launch structured finance products as part of a “programmatic” effort to fill previously-identified gaps in the private capital markets, designed in partnership with the Innovation Hub and the new CPI “Climate Lab” to be hosted at DBSA headquarters. The DBSAs Innovation hub is working with the Climate Policy Initiative to create a Climate Lab for Africa to accelerate well-designed financial instruments that can unlock billions for energy efficiency, renewable energy, sustainable transport, climate smart agriculture, and curbing deforestation, while also reducing private investors’ risks and improving their financial returns. As a public-private partnership, the Africa Lab will catalyze broader government and private sector efforts to scale up climate finance. The Africa lab will be modeled on CPIs established success through the existing India, Brazil and Global Climate Labs.
- Leverage of GCF Funds – The CFF will provide an opportunity to test and share best practice around how to best support transactions that demonstrate leverage and the ability to crowd-in commercial investment. Each Rand invested by the CFF must be matched by approximately 3-5 Rand from the private sector.

E.2.3. Contribution to the creation of an enabling environment

Describe how proposed measures will create conditions that are conducive to effective and sustained participation of private and public sector actors in low-carbon and/or resilient development that go beyond the program.

The CFF will complement existing climate-friendly finance capacity by providing an institutional platform specifically focused on scaling up private investment in small and medium-sized independent energy production through off-grid, micro-grid and IPP via emerging independent businesses. This will be additive to existing energy policy in South Africa (affecting the entire SADC region), which is focused on large scale power contracts via the REIPPPP. It will respond to a stated need of the region’s commercial banks to reduce risk and make climate-friendly projects more bankable. Existing energy policy and investment in South Africa has focused on implementing the Integrated Resource Plan, South Africa’s blueprint for the energy mix up to 2030, through large scale power contracts under the terms of the REIPPPP and contracts with Eskom for large sale renewable energy generation. Beyond the recent challenges to the REIPPPP due to institutional capacity constraints, corruption and the inability to sign new contracts, there is a significant gap in market capacity around scaling up investment in smaller projects and independent power production for distributed generation. These issues affect the broader Southern African region too, as Eskom is the largest member of the SAPP, and the majority of the region is reliant on South Africa for power supply through long-term supply contracts and day ahead market trading.

Many of the climate mitigation and adaption companies emerging in Southern Africa are startups promoting new technology that banks regard as venture and development capital in nature. There is limited equity available from project sponsors and developers, and given the perception of these businesses as risky venture start-ups, commercial banks are reluctant to gear beyond a debt:equity ratio of 50%. In addition, banks generally limit the tenor of these loans to 5-7 years. The CFF will address these market barriers by playing a catalytic role using a blended finance approach with a focus on projects that are commercially viable but not yet bankable in the private sector without credit enhancement and related financial or development support. This focus will include resilience and adaptation projects such as local waste water treatment plants; water reticulation equipment the harvesting of rain water etc. Key added-value interventions to complement the role that commercial banks can already play will include: longer loan tenors; first loss and sub-ordinated tranches; project aggregation; guarantees; and providing a dedicated, focused and efficient public financing entity. It is also worth noting that the CFF will complement the credit enhancements provided by Africa Green Co. which address the current weak financial position of utilities and limited choice of an alternate buyer in case of utility default by mitigating credit risks associated with the current lack of creditworthy offtakers, through guarantees and grant funding. By contrast,

the CFF focuses on scaling up private sector investment in projects and businesses that mitigate or adapt to climate change and clean water infrastructure projects from commercial banks and private investors through credit enhancement and risk reduction by providing two main instruments: subordinated debt / first-loss, and credit enhancements primarily in the form of tenor extension and potentially other credit support to small and medium sized projects.

Describe how the proposal contributes to innovation, market development and transformation. Examples include:

- *Introducing and demonstrating a new market or a new technology in a country or a region*
- *Using innovative funding scheme such as initial public offerings and/or bond markets for projects/programme*

Market Transformation

The CFF will support market transformation by supporting a shift from dependence on centralized utility-based generation to more diversified power generation via embedded and privately owned renewable sources of energy and clean water supplies. Countries in the Southern Africa region face large climate mitigation and adaptation challenges that are closely linked to socio-economic and environmental challenges in the energy, transport and water sectors. In the geographic area the DBSA covers, including 14 southern African countries, over 60% of people are without access to clean, regular and sustainable water. Electricity generation is still 90% fossil fuel based and over 50% of people in the SADC geographic area have no access to electricity. Those segments of the population that do have access to electricity suffer from a recent history of power shortages. And grid connected users are facing steep tariff hikes, which could be commercially devastating⁵⁷.⁵⁸ As noted earlier, power production in South Africa affects the entire SADC region as Eskom is the largest member of the Southern Africa Power Pool ("SAPP"), and the majority of the SADC region is reliant on South Africa for power supply.

To address these problems, private actors are looking to reduce reliance on Eskom and develop their own sources of electricity. This also applies to the water sector where climate change and public sector institutional challenges result in water disruption to the private sector. While there are a number of companies emerging in Southern Africa looking to develop businesses around climate mitigation and adaption, many are startup in nature or have limited track records. The technologies being promoted by these private sector companies (off-grid, mini-grid solar, urban distributed solar farms, electric and water efficiency) are well tested in other markets but are unfamiliar to Southern Africa and commercial banks, or are novel applications of known technologies. In addition, commercial banks look for equity investment of as high as 50% to support these ventures, which is not feasible for private sector developers and investors. This lack of available and affordable financing creates a barrier to rapid deployment of these types of privately owned and embedded climate-friendly projects. The proposed CFF will address these market barriers by providing two main instruments: subordinated debt / first-loss, and credit enhancements primarily in the form of tenor extension and potentially other credit support (e.g. loan loss reserves products) to "crowd in" the necessary private capital, focused on infrastructure projects that mitigate or adapt to climate change.

The CFF will support market transformation by opening the market and stimulating demand incentivizing the private sector to develop distributed privately owned renewably-sourced energy and clear water to meet increasing demand for clean water and affordable power. It will focus particularly on the commercial and industrial sectors but also on making the agricultural and residential sectors more self-sufficient and less dependent on the public sector for these utilities. This also extends to poorer areas where local water and off/micro grid power systems can be constructed and are affordable. Energy is more straightforward to fund in the near term because it has the most readily identifiable source of repayment through various off-take structures. These include leasebacks, PPAs; concessions etc. Therefore, energy projects will likely be a larger share of CFF investments (60-70%) in the early stage with the remaining 30% of investments expected to go into water projects. In the water sector, PPPs are starting to be more prevalent in Southern Africa as a way to involve the private sector and make projects bankable, so this sector will be an increasing target as the CFF reaches steady state. Example CFF investments are listed below. Also as noted earlier, the CFF will be designed to maximize total investment, using limited public funds to leverage far greater private investment and private sector leverage will occur at the project-level. The target of the CFF is to reach an overall portfolio leverage ratio of 1:5 (project leverage

⁵⁷ <https://www.businesslive.co.za/bd/companies/energy/2017-09-18-eskom-tariff-hike-business-chambers-prepare-to-fight/>

⁵⁸ <https://citizen.co.za/business/1732425/eskom-tariff-increase-could-cost-us-r1bn-sibanye/>

ratios will vary within this range). That is, for every Rand that the CFF puts into an individual transaction, it will be looking for approximately 5 Rand of private investment. The DBSA will create the CFF using best practices at existing Green Banks that have successfully catalyzed markets using blended finance to achieve similar leverage ratios without crowding out the private sector.⁵⁹⁶⁰

The CFF will support market transformation through stimulating private investment in a range of products as illustrated below:

Table 16: Example CFF investments

Investment/Project	Description	Level of progress
Off-grid projects	Off-grid solar PV (household solar and micro-grids) for remote areas to displace existing carbon-based sources of power including wood; paraffin and diesel.	DBSA has been approached by developers and sponsors of off-grid projects to serve local communities and using “kits” to provide basic power resources in South Africa/SADC countries.
Micro-grid/urban solar farm projects	Micro-grid projects of approx. 5 MW located in rural, urban or semi-urban areas powered by solar and/or wind are increasingly viable due to falling costs for renewables.	Working with a specific sponsor developer who has completed feasibility and secured the necessary land in the Johannesburg area. USTDA has just funded a feasibility study on this, based on the Johannesburg area.
Industrial and commercial solar for self-generation up to 1 MW and for wheeling on the grid to MSMEs	Facilitate the ability of private companies to produce renewable energy as independent power producers (IPPs) and to sell that power directly to small and medium-sized businesses to reduce dependence on Eskom. Examples include rooftop on warehouses, distribution centers, shopping centers etc.	There are now multiple companies in South Africa that are building self-generation RE where roll-out is restricted by lack of finance due to off-take risk and lack of track record.
Water – Industrial & Commercial	Industrial and commercial self-supply via boreholes, water treatment plants, rain harvesting and efficient reticulation equipment.	Multiple companies in Southern African region that are building water resilience projects where roll-out is restricted by lack of finance due to off-take risk.
Water – Local Communities	Increase availability of clean water through facilitating local waste treatment plants, wells, boreholes and rain harvesting. This will extend to rural farming communities.	Multiple companies in Southern African region that are building water resilience projects where roll-out is restricted by lack of finance due to off-take risk.
Distributed RE generation and residential and MSME energy efficiency investments	PACE, PAYS or GOF finance products to address issues of scale, transaction costs and repayment security	Instruments developed and funding secured from commercial funders, however some form of credit enhancement needed to achieve market penetration at scale.
Transportation – EV’s	Publically owned EV fleets have initial potential to show significant growth in South Africa.	EV market is in very early stage. Opportunities in local manufacturing of EV components, charging infrastructure

⁵⁹ NY Green Bank 2016 Business Plan. <https://greenbank.ny.gov/-/media/greenbanknew/files/2016-NYGB-Business-Plan.pdf>

⁶⁰ Coalition for Green Capital (2017), “National Green Banks in Developing Countries: Scaling Up Private Finance to Achieve Paris Climate Goals. Coalition for Green Capital.” <http://coalitionforgreencapital.com/2017/07/18/paper-green-banks-in-developing-countries/>

		roll out (public and private), skills support and development of secondary markets.
E.2.4. Contribution to regulatory framework and policies		
<i>Describe how the project/programme strengthens the national / local regulatory or legal frameworks to systematically drive investment in low-emission technologies or activities, promote development of additional low-emission policies, and/or improve climate-responsive planning and development.</i> The development and increased participation of MSMEs in the radical transformation to a low carbon and climate resilient developmental trajectory for the South African economy is a national strategic priority. As such, several regulatory and policy instruments have been put in place and institutional arrangements reconfigured to ensure optimal participation of small scale players in the economy. National strategies such as the National Development Plan, Industrial Policy Action Plan, and Green Economy Accord, the New Growth Path as well as defined guides including the Shared Economic Infrastructure Facility Guidelines and Guidelines for Reducing Municipal Red Tape, all have the common goal of ensuring that MSMEs play a part in growing the economy of South Africa. While the CFF is not a regulatory initiative, it will directly support the South African mandate that 30% of private and public sector procurement come from MSMEs to support local job creation. At the moment public and private sector entities are not able to achieve this target to due largely to lack of financially viable MSMEs and the CFF will enable them to achieve these goals. Lessons learned from implementing the CFF will also form part of the baseline data and knowledge bank that will assist South Africa towards making further headway towards scaling-up low-carbon and climate resilient initiatives in which MSMEs are particularly involved. The CFF will also serve as a flagship mechanism for DFIs on the African continent regarding scaling up private investment in low carbon infrastructure. In terms of the Green Bank model, this project has globally significant proof-of-concept value to a range of middle and lower income nations seeking to address market barriers and quickly scale up blended finance approaches to support the high levels of private investment required by Paris commitments. As such, it will inform the extent to which the current regulatory environment promotes the Green Bank model as a means to strengthen local finance institutions and build local capacity to unlock private finance. As noted, several nations are already expressing interest in replicating the proposed DBSA model (e.g. Rwanda, Colombia, Mexico, Zimbabwe).		

E.3. Sustainable Development Potential		
Wider benefits and priorities		
E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact		
The commercial- and industrial-focused energy, waste to energy and water projects funded through the facility, will contribute to South Africa's Sustainable Development Goals in the following ways: Environmental benefits The projects will help strengthen South Africans' resilience and adaptive capacity to climate-related hazards and natural disasters (SDG 13) by building on a distributed model of waste management, renewable energy generation and water treatment. They will also help reduce GHG emission of commercial and industrial sectors (SDG 9). Projects will help ensure greater water use efficiency and use of alternative water in the economy (SDG 12), which lower the consumption of potable municipal water; this, in turn, makes more water available to the broader economy, including the vulnerable. The solar PV and energy efficiency projects funded will help reduce GHG emissions by reducing reliance of the commercial and industrial sectors on the centralised coal-based electricity grid (SDG 7). In addition, the GHG (in particular methane) reductions from the biogas/waste-to-energy projects will divert waste away from landfill or uncontrolled decomposition (SDG 12), while contributing further to help reach South Africa's GHG-reduction goals. By		

2016, less than 1% of South Africans used solar energy directly for cooking, lighting and heating. The energy projects in particular will help contribute to South Africa's goals around the provision of sustainable energy for all (SDG 7), by helping to: i) increase the share of renewable energy in the global energy mix by 2030; ii) reduce energy intensity measured in terms of primary energy and GDP; iii) increase the proportion of electricity produced from renewable sources (6089,8 GWh in 2016); and iv) increase the percentage of the population that uses solar energy as their main source of energy.

Economic benefits

South Africa's SDG Indicator Baseline Report (2017), sets the target to achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors (SDG 8.2). It also promotes inclusive and sustainable industrialization (SDG 9) and, by 2030, aim to significantly raise industry's share of employment and gross domestic product, in line with national circumstances, and double its share in least developed countries. The projects that will be funded will contribute to these targets and to SDG 8, by creating jobs in manufacturing, construction and the service industries, and in project-related value chains. The projects will also contribute to business and GDP growth through efficiencies, business growth, green sector growth and FDI. Another goal is to, by 2030, upgrade infrastructure and retrofit industries to make them sustainable, with increased resource-use efficiency and greater adoption of clean and environmentally sound technologies and industrial processes (SDG 9). The projects will contribute significantly to this goal, by targeting funding at commercial and industrial sectors with a focus on more sustainable and resource efficient energy, waste to energy and water technologies.

Social benefits

Through the creation of jobs, by reducing unemployment and through economic growth, the projects will contribute to reducing poverty (SDG 1) and hunger (SDG 2) in South Africa. The energy projects targeted at the commercial and industrial sectors, will contribute to greater food security (SDG 2) by improving and lowering the cost of cold chain management, which in reduce food waste and lowers food prices. The water efficiency and treatment projects will help build resilience to the current and future droughts in a number of ways. By providing commercial farmers with technologies to help reduce their demand on municipal water, the projects will help build resilience to drought (SDG 9). The projects will contribute to adaptation and drought-resilience by providing technologies and efficiencies that reduce municipal water demand from industry and business, which in turn make more water available for the sustainable provision of water and sanitation to vulnerable urban and rural human settlements (SDG 6). This will also help contribute to SA's SDG target of universal provision of safely managed drinking water by 2030 (only 75.4% of rural people had safely managed drinking water in 2016).

Benefits for women (gender-sensitive development impact) – Refer to Gender Action Plan

As highlighted in E.4.1., South African women are disproportionately vulnerable to climate change impacts. The projects will have a series of benefits for women and will help i) contribute to greater gender equality (SDG 5), ii) build resilience and iii) reduce vulnerabilities in the following ways:

- The projects will create jobs in South Africa's manufacturing, construction and service industries, and in project-related value chains. This will reduce poverty (SDG 1) and hunger (SDG 2), and enable women, and in particular single mothers, to care for their children. As highlighted in E.4.1., between July to September 2017 women held 597k jobs in the manufacturing sector in SA in September 2017, vs 1,749k jobs held by men that same period. Even the total increase in jobs will benefit women, but the the project will set goals to proportionally increase the number of women in the relevant sectors and value chains.
- The manufacturing opportunity in local lamination of solar PV panels provides a significant opportunity for gender-sensitive development. There are examples of current facilities with 80% of employees in PV lamination facilities being women.
- A critical SDG goal is to, by 2030, achieve access to adequate and equitable sanitation and hygiene for all and to end open defecation, paying special attention to the needs of women and girls and those in vulnerable situations. By 2016, 92.3% of urban and 62.1% of rural people had access to safely managed sanitation services. CFF financed water supply and water efficiency projects will support greater sustainability of water sanitation services for women and others, especially in times of drought. A sustainable supply of water, and access to safely managed drinking water and hand washing facilities (SDG 6), even in droughts, will also improve the health of women and children, especially in rural or informal areas.

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

Describe the scale and intensity of vulnerability of the country and beneficiary groups, and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc).

Southern Africa's is vulnerability to the impacts of climate change

Climate models predict a slow drying trend across southern Africa, with a greater likelihood of quick impact events like heat waves and floods, and adverse weather events like droughts. The current version of South Africa's National Climate Change Adaptation Strategy (Government of SA, 2017) highlights the following:

- **Rising temperatures:** According to the strategy, "South Africa is plausibly committed to relatively large increases in near-surface temperatures, even under high-mitigation futures" (p.71). In fact, the warming trend across southern Africa has been double the global average over the past fifty years. This is also the case in SA, with a few local exceptions in the interior which have shown lower rates of warming, and even cooling in some cases.
- **More extreme weather events:** Droughts, floods, heatwaves and fires are expected to increase compared to the period up until 2000. Droughts or dry spells are expected to increase across SA, with a few regional exceptions, no matter which mitigation scenario is reached. Heat waves are more likely, and there is already evidence that dry spells last longer and that rainfall intensity is increasing. Until 2013, South Africa managed to avoid the drying effect of El Nino conditions since 1991/92 and recorded above-average rainfall which resulted in more floods and storms. However, since then, large parts of the country have experienced droughts and other extreme events (see next section).
- **Higher rainfall variability and/or less rainfall:** While future climate models predict a drying trend with fairly high certainty, studies based on historical data fail to agree on whether a drying trend is already visible. What is evident from the national strategy is that rainfall will be heavily influenced by which mitigation measures are taken - under low mitigation scenarios, most of the country is set to experience a drying trend, but rainfall predictions improve under higher mitigation scenarios.

The poor, women, young people, children, businesses, rural and urban human settlements and the electricity sector are most vulnerable to the impacts of climate change.

South Africa's National Climate Adaptation Strategy provides an overview of key vulnerabilities and their biophysical, economic and social impacts (see table below). Of particular importance to this proposal, are vulnerabilities related to access to water, agricultural productivity (affecting economic growth and food security), vulnerable energy systems, and market volatility.

Table 17: Key vulnerabilities and social impacts

KEY VULNERABILITY	BIOPHYSICAL	ECONOMIC	SOCIAL
Unreliable and uncertain access to water	x	x	x
Risk to agricultural productivity and livestock		x	x
Human safety from climate related extreme events			x
Poor service delivery in human settlements	x	x	x
Vulnerable energy system and infrastructure		x	x
Diminished labour force productivity through exposure and health		x	x
Supply and demand volatility of the market		x	
Carbon intensity and dependence of the economy		x	x
Ecosystem and conservation estate	x	x	x

According to the National Adaptation Strategy, the following are key vulnerable groupings:

- **The rural and urban poor** are the most vulnerable to climate change effects, the effects on their livelihoods and their environment, which in turn exacerbate social inequalities.
- **Women, young people and children** are highlighted in the strategy as groups on which climate change will have disproportionate impacts.
- **Farmers**, especially those who rely on rain-fed agriculture, and those without access to capital and resources to adapt to, say, droughts.
- **Businesses** and their supply chains may be affected on multiple fronts given that climate change can affect multiple sectors in a value chain. For instance, businesses who receive supplies from sectors which rely on raw materials or inputs (agricultural products, water, energy) may be particularly affected by climate change. Business operations may also be affected by droughts (which reduce water availability for operations and sanitation), floods and heatwaves. Consumer spending patterns may also change which may benefit one sector at the detriment of others.
- **Human urban settlements**, especially informal ones are vulnerable to droughts and floods (which affect the availability of running water and sanitation), and fires.
- **Rural settlements**, are especially vulnerable to fires and floods, and water shortages, of which one effect is reduced quantity and quality of subsistence farmers' crops.
- **Electricity supply sector** may be affected by droughts, given that 90% of SA's electricity is generated by coal-fired plants, which are extremely water intensive. Energy and other public infrastructure may also be affected by extreme weather

A number of structural factors underpin these vulnerabilities, including poverty and unemployment and gender inequality.

The poor are most vulnerable to the impacts of climate change, and in 2015, over 25% of South Africans fell under the food poverty line. South Africa uses three poverty measures, the food poverty line (FPL; R531 in 2017), the lower-bound poverty line (LBPL; R758 in 2017), and the upper-bound poverty line (UBPL; R1138 in 2017). The table below highlights that by 2015, 21.9 million people (40% of the population) were 'LBPL poor'. This means that they did not have command over enough resources to purchase or consume both adequate food and nonfood items and were therefore forced to sacrifice food to obtain essential non-food items. 13.8 million people (25.2% of the population) fell under the food poverty line (FBL), which means that those individuals were unable to purchase or consume enough food to supply them with the minimum per-capita-per-day energy requirement for adequate health.

Table 18: Poverty headcounts and the number of poor persons (2006 – 2015)

Poverty headcounts	2006	2009	2011	2015
Percentage of the population that is UBPL poor	66,6%	62,1%	53,2%	55,5%
Number of UBPL poor persons (in millions)	31,6	30,9	27,3	30,4
Percentage of the population that is LBPL poor	51,0%	47,6%	36,4%	40,0%
Number of LBPL poor persons (in millions)	24,2	23,7	18,7	21,9
Percentage of the population living in extreme poverty (below FPL)	28,4%	33,5%	21,4%	25,2%
Number of extremely poor persons (in millions)	13,4	16,7	11,0	13,8

The same publication reports that the number of subsistence farmers 1 723 000, a group vulnerable to the vagaries of the weather, given that most subsistence farmers live in rural areas and are reliant on rain to farm successfully.

Unemployment contributes significantly to poverty (and vice versa), and according to Stats SA, between July & September 2017 unemployment had increased by 2% from the previous year. In addition:

- 36.8% of adults in South Africa were unemployed, using the expanded unemployment rate.
- 14 885 000 people were unemployed.
- More women than men were unemployed at 8 579 000.
- Most people unemployed were between the ages 25-34, a total of 4 581 000.

Gender inequality is another factor that contributes to vulnerability among women.

Women are disproportionately vulnerable to climate change impacts because of 1) their social roles as (often, single,) mothers and 2) because they are poorer and have fewer resources and skills available to adapt. Women are often the ones responsible for cooking, and in rural areas for collecting biofuels such as wood, waste and charcoal for cooking. They also suffer the health effects of cooking indoors, when using harmful fuels.

In addition, South African women also lag behind men in educational attainment, which in turn affects their ability to find employment. "Women [also] bear a disproportionate burden of unemployment, constitute the majority of casual or contract workers, generally occupy low-wage job positions, and are poorly represented in senior and top management positions".

- *Unemployment:* In South Africa, as already noted, more women are unemployed than men.
- *Status as single parents:* Stats SA reports that in 2017 most live births in SA were by women aged 20–29 years old, with over 60% of those births registered by a single mother, without a father's details. This highlights the fact that the majority of children in South Africa are raised by single mothers, who earn a lot less than single fathers.
- *Income inequality:* According to the 2011 gender statistics report from Stats SA, 72.9% of women receive a monthly income of less than R1,250 compared to 52.1% for men. Only 11.8% of women receive a monthly income of R3,206 or more compared to 25.1% of men.
- *Educational attainment:* According to the 2016 national community survey, females constitute the largest proportion among those who had no schooling (55,3%) compared to males (44,7%). That said, more females (95%) aged 15-34 are literate compared to males (92.8%) and more females are currently in post secondary education. It should be kept in mind that educational attainment in South Africa is low, with only 64.8% of people under 25 having completed high school (grade 12).
- *Women's employment in the clean energy, water and waste sectors:* One global survey of 90 renewable energy companies by the International Renewable Energy Agency (IRENA), found that the average share of women employed by those companies was 35%. An informal study of 11 solar PV companies who are GreenCape members, show a 34% rate of employment for women in those companies. The US, German and Spanish solar markets employ roughly 25% in their workforce, more than the conventional energy industry.

The table below illustrates that men dominate agriculture, mining, manufacturing, utilities, and construction industries in South Africa. With a greater focus on creating jobs for women in these industries, the impact on GDP can be significant - one McKinsey report found that by growing the proportion of women in formal employment, economic output can increase by as much as 45%.

Table 19: Employment statistics by industry and sex – South Africa

	Jul-Sep 2016	Oct-Dec 2016	Jan-Mar 2017	Apr-Jun 2017	Jul-Sep 2017
	Thousand	Thousand	Thousand	Thousand	Thousand
Both sexes	15 833	16 069	16 212	16 100	16 192
Agriculture	881	919	875	835	810
Mining	438	421	447	434	446
Manufacturing	1 683	1 727	1 790	1 799	1 749
Utilities	118	131	145	148	153
Construction	1 491	1 483	1 505	1 395	1 365
Trade	3 198	3 222	3 207	3 265	3 286
Transport	915	961	965	954	988
Finance	2 323	2 329	2 378	2 395	2 463
Community and social services	3 499	3 571	3 569	3 560	3 616
Private households	1 281	1 299	1 319	1 311	1 313
Other	5	5	11	3	3
Women	6 873	7 031	7 182	7 078	7 125
Agriculture	255	305	278	264	252
Mining	54	62	60	61	53
Manufacturing	562	554	627	603	597
Utilities	35	37	31	31	43
Construction	164	174	199	174	158
Trade	1 533	1 555	1 573	1 578	1 563
Transport	177	189	199	171	190
Finance	967	970	1 009	1 003	1 008
Community and social services	2 126	2 186	2 199	2 172	2 248
Private households	996	995	1 001	1 018	1 010
Other	4	5	7	3	3
Men	8 960	9 037	9 030	9 022	9 067
Agriculture	627	614	597	571	558
Mining	384	359	387	373	393
Manufacturing	1 121	1 173	1 163	1 196	1 152
Utilities	83	95	114	116	110
Construction	1 327	1 309	1 306	1 221	1 206
Trade	1 664	1 668	1 634	1 686	1 722
Transport	738	772	766	783	798
Finance	1 356	1 359	1 369	1 392	1 455
Community and social services	1 373	1 385	1 370	1 388	1 369
Private households	285	304	319	294	303

The impacts of anthropogenic climate change may already be evident in South Africa, in the form of the droughts which have plagued the country since 2013.

In the past five years alone, large parts of South Africa have been experiencing **droughts**, with various provinces being declared disaster areas:

- North West province (2013); declared disaster in July 2013, and July 2015
- KwaZulu-Natal province, declared disaster in December 2014
- Free State declared disaster September 2015

- Limpopo and Mpulanga provinces declared disaster zones in November 2015
- Mpumalanga, Limpopo, KwaZulu-Natal, North West and the Free State provinces all declared disaster zones by February 2016
- Western Cape declared a disaster zone in May 2017 and is experiencing its worst drought in 80 [check] years.

By September 2017, 8 287 out of 22 502 (37%) of settlements in South Africa were experiencing moderate, severe or extreme droughts, with the Eastern and Western Cape provinces most affected, as shown in the table below.

Table 20: Drought affected settlements in South Africa

	Moderate	Severe	Extreme	Total	Settlements
EC	1593	2460	1023	5076	9076
FS	56	0	0	56	321
GT	0	0	0	0	2579
KZN	1226	205	24	1455	3620
LP	178	26	0	204	2692
MP	46	29	1	76	773
NW	0	0	0	0	1299
NC	6	7	40	53	545
WC	60	86	1221	1367	1597
TOTAL	3165	2813	2309	8287	22502

The commercial- and industrial-focused energy, waste and water projects funded through the CFF, will help to a) address the above vulnerabilities and b) will create resilience to future impacts in the ways outlined in the table below.

Table 21: CFF contribution to Sustainable Development Goals (see more in section E.3.1.).

CFF Finances Projects	Impact, i.e. how vulnerabilities will be addressed & resilience will be built	Link to SDGs (for more, see E.3.1.)
<i>Provision of clean energy to the commercial and industrial sectors</i>	• <i>Create jobs</i> in solar PV manufacturing and the renewable energy value chain, in particular jobs for women. • <i>Grow businesses and the economy</i> through development of the renewable energy value chain, and through savings in commercial and industrial sector electricity bills. • <i>Reduce GHG emissions</i> by reducing reliance on the coal-based electricity grid. • Contribute to <i>more resilient and sustainable cities</i> and businesses. • <i>Contribute to greater food security</i>, by improving and lowering the cost of cold chain management, which reduces food waste, which in turn reduces food prices.	SDG 7, affordable and clean energy; SDG 13, climate action; SDG 8, decent work and economic growth; SDG 1, no poverty; SDG 5, gender equality; SDG 2, zero hunger.
<i>Provision of energy efficiency technology and solutions to the commercial and industrial sectors</i>	• <i>Create jobs</i> in energy efficiency value chains, in particular jobs for women. • <i>Grow businesses and the economy</i> through development of the energy efficiency value chains, and through savings in commercial and industrial sector electricity bills.	SDG 12, responsible consumption and production; SDG 8, decent work and economic growth; SDG 1, no poverty; SDG 5, gender equality.

	• <i>Reduce GHG emissions</i> by reducing usage of coal-based electricity. • Contribute to <i>more resilient and sustainable cities</i> and businesses.		
<i>Biogas and waste-to-energy</i>	• <i>Create jobs</i> in waste-to-energy value chains, in particular jobs for women. • <i>Grow businesses and the economy</i> through development of bio-based energy value chains, and through savings in electricity bills. • <i>Ensure greater resource efficiency</i> in the economy. • <i>Reduce GHG, in particular methane emissions</i>, by providing solutions for biowastes. • <i>Reduce waste to landfill</i>, resulting in a cleaner and safer environment for all. • <i>Reuse/recycling of waste</i> • <i>Chemical recovery</i> from wastes, which improve the environment for all. • <i>Contribute to greater food security</i>, by improving and lowering the cost of cold chain management, which reduces food waste, which in turn reduces food prices.	SDG 12, responsible consumption and production; SDG 8, decent work and economic growth; SDG 1, no poverty; SDG 5, gender equality.	
<i>Water efficiency and treatment</i>	• <i>Create jobs</i> in water efficiency and treatment value chains, in particular jobs for women. • <i>Grow businesses and the economy</i> through development of water value chains, and through savings in water bills. In turn business and economic growth addresses poverty. • <i>Build resilience to drought</i> of all vulnerable groups, including commercial and subsistence farmers, and cities' sanitation infrastructure, through the provision of technologies that reduce demand for potable municipal water and which can help these groups cope with droughts and/or dry spells. In turn, these help to safeguard local economies, jobs and especially the most vulnerable. • <i>Ensure greater water use efficiency and use of alternative water</i> in the economy, which lower the consumption of potable municipal water. This ensures more water is available to vulnerable urban and rural human settlements, to the electricity sector and to the business sector. • <i>Build business resilience and job resilience</i> for those in water-intensive sectors like agriculture and construction, through a more distributed and resilient model of water provision. • Contribute to <i>more resilient and sustainable cities</i> and businesses.	SDG 12, responsible consumption and production; SDG 6, clean water and sanitation; SDG 8, decent work and economic growth; SDG 1, no poverty; SDG 5, gender equality.	

E.4.2. Financial, economic, social and institutional needs

Describe how the project/programme addresses the following needs:

- *Economic and social development level of the country and the affected population*
- *Absence of alternative sources of financing (e.g. fiscal or balance of payment gap that prevents from addressing the needs of the country; and lack of depth and history in the local capital market)*
- *Need for strengthening institutions and implementation capacity.*

Economic and social development

As noted in earlier sections of this application, the CFF will support the economic and social development of South Africa and the southern African region with a focus on vulnerable populations through energy, waste and water projects. As funded through the CFF, these projects will help address vulnerabilities and create resilience to future impacts. CFF financed lean energy projects will create jobs in the solar PV sector, grow businesses through lower and more stable energy costs, reduce GHG emissions through reduced reliance on the coal-based electricity grid, increase climate resilience of cities and businesses, and contribute to food security through improvements to cold-chain food management. CFF financed energy efficiency projects will support job creation in efficiency based value chains, grow businesses through energy savings and price stability, reduce GHG emissions by reducing use of coal-based electricity, and contribute to more resilient and sustainable cities and businesses. CFF financed water efficiency and supply projects will create jobs in water efficiency and treatment value chains, grow businesses and the economy through savings in water bills, *build resilience to drought* for all vulnerable groups, increase water availability to vulnerable urban and rural settlements, and increase the resilience and sustainability of businesses and cities.

Financing and strengthening institutional capacity

Also as noted in earlier sections of this application, the CFF will complement existing climate-friendly finance capacity by providing an institution specifically focused on scaling up private investment in small and medium-sized independent energy production through off-grid, micro-grid and IPP via emerging independent businesses. This will be additive to existing energy policy in South Africa (affecting the entire SADC region), which is focused on large scale power contracts via the IRPPP. It will respond to a stated need of the region's commercial banks to reduce risk and make climate-friendly projects more bankable. Existing energy policy and investment in South Africa has focused on implementing the Integrated Resource Plan, South Africa's blueprint for the energy mix up to 2030, through large scale power contracts under the terms of the REIPPPP and contracts with Eskom for large sale renewable energy generation. Beyond the recent challenges to the REIPPPP due to institutional capacity constraints, corruption and the inability to sign new contracts, there is a significant gap in market capacity around scaling up investment in smaller projects and independent power production for distributed generation. These issues affect the broader Southern African region too, as Eskom is the largest member of the SAPP, and the majority of the region is reliant on South Africa for power supply through long-term supply contracts and day ahead market trading.

Many of the climate mitigation and adaption companies emerging in Southern Africa are startups promoting new technology that banks regard as venture and development capital in nature. There is limited equity available from project sponsors and developers, and given the perception of these businesses as risky venture start-ups, commercial banks are reluctant to gear beyond a debt:equity ratio of 50%. In addition, banks generally limit the tenor of these loans to 5-7 years. The CFF will address these market barriers by playing a catalytic role using a blended finance approach with a focus on projects that are commercially viable but not yet bankable in the private sector without credit enhancement and related financial or development support. This focus will include resilience and adaptation projects such as local waste water treatment plants; water reticulation equipment the harvesting of rain water etc. Key added-value interventions to complement the role that commercial banks can already play will include: longer loan tenors; first loss and sub-ordinated tranches; project aggregation; guarantees; and providing a dedicated, focused and efficient public financing entity. It is also worth noting that the CFF will complement the credit enhancements provided by Africa Green Co. which address the current weak financial position of utilities and limited choice of an alternate buyer in case of utility default by mitigating credit risks associated with the current lack of creditworthy offtakers, through guarantees and grant funding. By contrast, the CFF focuses on scaling up private sector investment in projects and businesses that mitigate or adapt to climate change and clean water infrastructure projects from commercial banks and private investors through credit enhancement and risk reduction by providing two main instruments: subordinated debt / first-loss, and credit enhancements primarily in the form of tenor extension and potentially other credit support to small and medium sized projects.

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

Please describe how the project/programme contributes to country's identified priorities for low-emission and climate-resilient development, and the degree to which the activity is supported by a country's enabling policy and institutional framework, or includes policy or institutional changes.

Since 1994, South Africa has achieved far-reaching political, economic and social changes, and has shown an increasing commitment to sustainable development. Along with its involvement in international negotiations, it has developed its own national framework for a shift to a green economy. South Africa recognises sustainable development as a human right in the Bill of Rights of its 1996 Constitution (Republic of South Africa, 1996) and also committed to achieving the Millennium Development Goals, which include environmental sustainability as a target (United Nations, 2000). The country is a Party to both the Kyoto Protocol and the United Nations Framework Convention on Climate Change (UNFCCC) and has made commitments under the Cancun Agreement for its greenhouse gas emissions to “peak, plateau and decline”, with reductions in emissions compared to a “business as usual” scenario of 34% in 2020 and 42% in 2025.

The South African framework which guides the development of a low carbon and climate resilient economy consist of several policies. The Table below provides a chronological overview of these policies and measures, summarising their relevant goals, current (mid-2012) progress and the nature and level of civil society involvement in their establishment.

Table 22: Overview of policies and measures

Policies and measures	Main goals	Progress (mid-2012)	Civil society involvement
Framework for Environmental Fiscal Reform (NT, 2006)	Provides principles and guidelines for fair and effective environmental taxes	Taxes and levies have been implemented on plastic bags, incandescent light bulbs, ecosystem restoration costs related to water use, liquid fuel, non-renewable electricity and new vehicle carbon dioxide emissions performance	A paper on carbon tax was published in 2010 for public consultation
Innovation Plan (DST, 2008)	Includes “safe, clean, affordable and reliable energy supply” and climate change as priorities	Support for innovation in electric vehicles, fuel cells and carbon capture and storage, but cancellation of the country's largest clean energy R&D programme (the Pebble Bed Modular Reactor) and delay in the implementation of renewable energy demonstration projects (e.g. solar tower)	Limited
Medium-Term Strategic Framework 2009-2014 (NPC, 2009)	Notes the need for sustainable livelihoods and sustainable resource management and relates these to various other policy areas including energy, water, housing, technology and competitiveness	Numerous policy responses implemented in line with the Medium-Term Strategic Framework, particularly the NSSD, the creation of an enabling environment for renewable energy, several water management projects and the National Climate Change Response.	Limited
Industrial Policy Action Plan (the dti, 2010, 2011 and 2012)	Specifically targets growth in green industries, focusing on solar water heaters (SWHs), but also includes other solar and wind energy, biofuels, electric vehicles and organic farming	Around 200 000 SWHs installed by mid-2012 and a procurement process started for around ZAR 120 billion worth of large-scale renewable electricity generation	Some consultation via NEDLAC
New Growth Path (EDD, 2010)	Targets the growth of a green economy, resulting in 400 000 new and additional jobs	Enabling regulation passed in other departments	Government, business and civil society signed the Green Economy Accord in 2011, which details (mostly already-existing) support measures
Integrated Resource Plan 2010-2030 (DoE, 2011)	Limits emissions from electricity generation to 275 Mt per year. Expects renewable energy to make up 42% of all new electricity generation over the next 20 years	Procurement of renewable energy under the plan is on track, but procurement of nuclear energy has been delayed by at least a year	Large public participation, including inputs on modelling parameters and a first draft of the plan

National Climate Change Response (Republic of South Africa, 2011)	Endorses and quantifies South Africa's greenhouse gas emissions limits/commitments. Aims to grow green jobs while limiting job loss in unsustainable industries	South Africa had already made a (voluntary) emissions commitment and approved an emissions-limited energy plan (Integrated Resource Plan 2010-2030) prior to the publication of the policy	Significant public consultation, starting with the multi-stakeholder Long-Term Mitigation Scenarios process in 2007
National Strategy for Sustainable Development (DEA, 2011)	A large variety of indicators and goals spanning social, economic and environmental issues, but no budgets, timelines or responsibilities	Strategy published and annual publication of sustainability indicators (different from the ones in the strategy)	Public consultation on a draft NSSD since 2009.
National Development Plan (NPC, 2011b)	The NDP is very specific about goals and focuses on energy and carbon: □ greenhouse emissions to peak in 2025 and introduce carbon budgeting □ an economy-wide price for carbon and incentives for energy efficiency and managing waste better. □ 5 million SWHs by 2030 □ vehicle emission standards, zero-emission building by 2030 □ simplify the regulatory regime for contracting about 20 000 MW of renewable energy by 2030	Greenhouse gas emissions may already be higher than levels committed to for stabilization in 2025. A carbon tax with exemptions is expected in 2013. SWH installations stand at just over 200 000 in 2012 (compared to the targeted 1 million by 2014/2015) Tax on carbon dioxide emissions of motor vehicles and new building energy efficiency regulations implemented Procurement has started on the first 3.7 GW of electricity supply from renewable energy out of a plan for 17.8 GW by 2030.	The NPC is guided by nominated Commissioners from outside of government (and business for the most part) and consulted publicly on the Development Plan based on an initial publication of a Diagnostic document.

The CFF selected initiatives in the water, waste and renewable energy align closely with the above national policies and plans. South Africa's national climate strategy, to transition to a lower carbon and climate resilient society, is defined in the **National Development Plan (NDP)** (NPC, 2012) and the **National Climate Change Response White Paper (NCCRP)** (DEA, 2011). These documents outline the long-term goals and strategy for climate change adaptation and mitigation.

The **NDP** advocates a transition to an environmentally sustainable, climate change resilient, low-carbon economy and just society, which will be enabled by:

- Coordinated planning and investment in infrastructure and services that take account of climate change and other environmental pressures;
- Implemented adaptation and national development strategies;
- **Focus on becoming a zero-waste society;**
- **Growth in the renewable energy sector;**
- **Domestic manufacturing of renewable energy technologies coupled with job creation;**
- **Reducing the country's carbon emissions;**
- Conservation and restoration of protected areas through policy and regulatory frameworks for land use; and
- **Public investment in new agricultural technologies and the development of resilient and environmentally sustainable strategies.**

The National Climate Change Response White Paper (NCCRP) (DEA 2011) has two main objectives:

- Manage expected climate change impacts
- Contribution to the global effort to stabilize greenhouse gas concentrations

The National Climate Change Response Strategy for the Water Sector (DWS, 2014) emphasizes good water management which is a critical foundation for adaptation to water- related climate change impacts.

South Africa has incrementally increased the its commitments to climate change response since 2011 as seen the figure below.

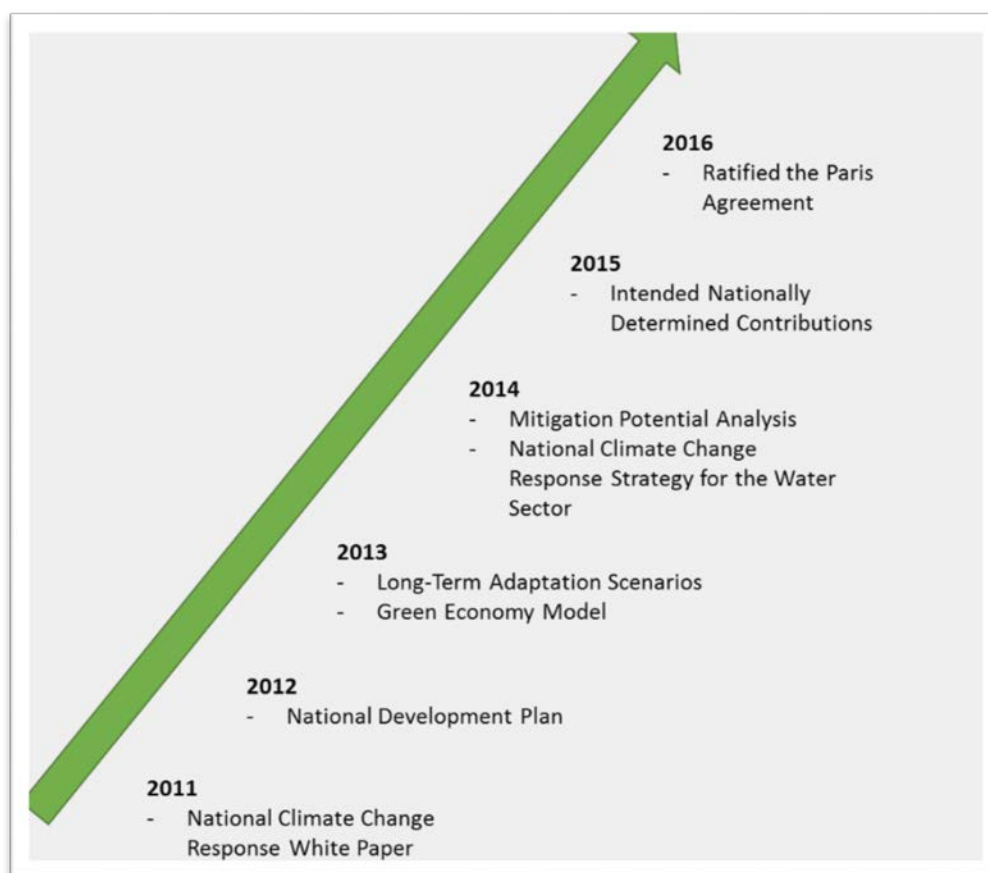


Figure 24: South Africa's climate change response 2011-2015 (DEA, 2017)

The **National Climate Change Adaptation Strategy (NAS)**, which begun in 2016, will be used as the country's **National Adaptation Plan (NAP)**. This strategy includes plans to improve:

- Improve water and energy efficiency
- Improve waste management
- Increase agricultural resilience
- Increase uptake of renewable distributed generation

Related Nationally Appropriate Mitigation Actions (NAMAs)

Diversion of Solid Waste from Landfills in Selected Municipalities

The main objective of this NAMA is the promotion of diversion of waste (especially organic waste) from landfills in local municipalities to mitigate environmental impacts such as the greenhouse gas effect.

- Diversion of organic waste away from landfills: this is the long-term approach to mitigating climate change in the waste sector;

South African Renewables Initiative

The energy sector remains the single largest contributor to the South Africa's total GHG emissions (81.7% in 2012) and any large mitigation contributions will have to come through reduced emissions from energy generation and use. The main opportunities for mitigation consist of energy efficiency, demand- side management and moving to a less emissions-intensive energy mix. Through the NDC, renewable energy sector was identified as the largest contributor to climate change mitigation. South Africa has a high level of renewable energy potential and has revised its targets to about 17,800

MW, adopted by the Government in 2013 as part of the policy adjusted IRP, a blueprint for the energy mix in the period up to 2030.

As a responsible global citizen and as a global citizen with moral as well as legal obligations under the UNFCCC and its Kyoto Protocol, South Africa is committed to contributing its fair share to global GHG mitigation efforts in order to keep the temperature increase well below 2°C. In this regard, on 6 December 2009, the President announced that South Africa will implement mitigation actions that will collectively result in 34% and 42% deviation below its “Business As Usual” emissions growth trajectory by 2020 and 2025, respectively. In accordance with Article 4.7 of the UNFCCC, the extent to which this outcome can be achieved depends on the extent to which developed countries meet their commitment to provide financial, capacity-building, technology development and technology transfer support to developing countries. With financial, technology and capacity-building support, this level of effort will enable South Africa’s GHG emissions to peak between 2020 and 2025, plateau for approximately a decade and decline in absolute terms thereafter.

The CFF programme has the potential of ensuring that the objectives set by the policies and measures noted above could be met timeously.

E.5.2. Capacity of accredited entities and executing entities to deliver

Please describe experience and track record of the accredited entity and executing entities with respect to the activities that they are expected to undertake in the proposed project/programme.

The DBSA is a leading development finance institution established in 1983, with a clear mandate to promote economic development and sustainable growth in South Africa and the African continent, with a key sectoral focus on energy, transport, water and ICT. Furthermore, the DBSA’s role is to advance the development impact in the African region by expanding access to development finance and effectively integrating and implementing sustainable development solutions. The DBSA has been accredited as a Regional Agency of the Green Climate Fund (“GCF”) to programme and disburse funds provided on approval of projects by the GCF. As an accredited entity, the DBSA may carry out a range of activities, including:

- a) developing and submitting funding proposals for projects and programmes;
- b) overseeing management and implementation of projects and programmes;
- c) deploying a range of financial instruments within their respective capacities (grants, concessional loans, equity & guarantees); and
- d) mobilizing private sector capital.

DBSA experience in implementing the National Green Fund

The Bank offers Fund Management and Institutional support to the Green Fund Programme through dedicated senior management oversight. The Green Fund also makes use of the Bank’s fiduciary, governance and environmental safeguards policies and processes. The Green Fund Programme is a South African National Government catalytic financing mechanism that aims to address market weaknesses by supporting green initiatives that will contribute to South Africa’s transition to a low carbon and resource efficient economy. The key objectives of the Fund are to support innovative and high impact green programmes and projects that will:

- Reinforce South Africa’s climate and sustainable development policy objectives through green interventions
- Build an evidence base for the expansion and financing of its green economy transition, and
- Attract additional resources to support South Africa’s green economy development.

DBSA management of projects in the water sector

To date, the DBSA has supported various water management programmes such as the Water Demand Management programme which aims to entrench a water demand culture in the SADC region, thereby contributing to the goals of regional integration and poverty alleviation through the principles of pro-poor, efficient and sustainable water utilisation. The programme is in line with the SADC Regional Strategic Action Plan on Integrated Water Resources Development and Management. Various water institutions within the SADC region implemented several projects on water demand

management. The Agency Management Services Unit established a Programme Implementation Unit, which took responsibility for in-country launch events for the confirmation of the various management structures of the programme (for publicity and political buy-in). This resulted in the establishment of the country nodes and the confirmation of the international reference group members. Water conservation initiatives supported by the DBSA included the Greater Hermanus Water Conservation Project and OR Tambo District Municipality.

DBSA experience in supporting the renewable energy sector

The DBSA has extensive experience in renewable energy in South Africa through its pioneering role of assisting the government in setting up the IPPPP office that subsequently implemented the REIPPPP. As a financier, DBSA participated in the financing of projects from bidding window 1 through to the current bidding window 4. The participation in REIPPPP is summarised below:

- The DBSA has to date funded 19 projects (1 507 MW) under South Africa's Renewable Energy Independent Power Producers Procurement Programme ("REIPPPP")
- Renewable Energy technology portfolio mix: 5 CSP projects, 9 PV projects & 5 Wind projects
- 15 of the projects are currently operational and delivering power onto the national grid. 4 remaining projects are currently under construction
- The DBSA has committed approximately R12,4 billion (USD0.83 bn) in Renewable Energy Projects in the form of Senior Debt, Mezzanine Debt and BEE Funding
- Participation by BBBEE players and local community trusts in renewable energy projects was one of the upfront requirements under the REIPPPP.
- Funding was provided by the DBSA so that Broad Based Black Economic Empowerment entities and local community trusts could be involved on an equal footing.
- So far the Bank has provided funding of approximately R1.66 billion (USD0.11 bn) to empowerment parties and local community trusts.
- The DBSA has currently committed to support up to 14 projects in bidding Round 4 of the REIPPPP should the projects achieve financial close.

In addition the DBSA continues to support renewable energy initiatives by participating as a funder in the Small Projects Independent Power Producer Procurement Programme ("SPIPPP"). The Programme is aimed at emerging, smaller power developers (less than 5 MW) with an emphasis on South African and MSME participation in Projects. In recognition of the challenges faced by small previously disadvantaged IPPs, the DBSA has made the following financing available for SPIPPPs:

- **The Global Environment Facility ("GEF") Equity Facility** of USD 15,000,000 – managed by the DBSA, the GEF Equity Facility will address the challenges of accessing equity funding by MSMEs for the implementation of the small renewable energy projects (1-5 MW), through providing equity funding at reasonable lending rates to ensure their sustainability as project developers and/or investors. The GEF capitalised equity facility targeting MSMEs under the SPIPPP, is meant to catalyse the small scale renewable energy markets and in line with South Africa's commitment to transition to a low carbon economy.
- **Infrastructure Investment Programme for South Africa ("IIPSA") Interest Subsidy** - The objective of the IIPSA Interest Subsidy is to address high cost of funding challenges experienced by SPIPPP Projects. A maximum grant facility of R80 million (USD5.3 million) is reserved by the IIPSA Secretariat to cover 10 renewable energy Projects under the SPIPPP Bid Window 1 and 2. This will also be managed by the DBSA.

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

Please provide a full description of the steps taken to ensure country ownership, including the engagement with NDAs on the funding proposal and the no-objection letter.

Please also specify the multi-stakeholder engagement plan and the consultations that were conducted when this proposal was developed.

The DBSA has engaged with a wide range of critical stakeholders to ensure country ownership, including multiple interactions with the South African Department of Environmental Affairs and NDA who have indicated interest and indicative support for the CFF, the five major South African Commercial Banks all of which have indicated support for formation of the CFF and strong interest in co-financing a robust pipeline of projects, several project developers who also expressed significant interest in the CFF as a promising mechanism to unlock private investment in their projects, and with several DFIs who are actively engaged in considering capitalizing the CFF alongside the GCF and DBSA. A summary chart of stakeholder engagement to ensure country ownership is provided below. Engagements with National Treasury and other relevant South African government departments are ongoing in order to secure a letter of no-objection.

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

E.6.1. Cost-effectiveness and efficiency

Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers; providing the least concessionality; and without crowding out private and other public investment.

The interest rates charged by DBSA through the CFF will be a function of the cost of funds invested in the Programme and the risks of the individual investments. The Programme is expected to be funded by debt from the DBSA and a number of DFIs (identified above). It is envisaged that the concessionality received from the GCF will enable DBSA to offer a final blended cost of capital which will be below commercial rates. This core funding construct is what will allow the Programme to provide two main instruments: subordinated debt / first-loss, and credit enhancements primarily in the form of tenor extension and credit to support risk mitigation and produce economically sound climate mitigation projects. The CFF is being designed to provide project financing at costs sufficient to produce affordable renewable electricity and related climate benefits at costs that can drive demand.

The DBSA itself will not have seniority – all funders will commit their capital on a pari passu basis. Funds will be repaid to funders, per their individual return requirements, on an equivalent basis. This also means all possible losses will be shared equally. All funding and returns will be completely transparent to all co-funders. Through a 2% CFF service fee and allocation of a 3.0% AE fee from the GCF (\$1.65 million USD at a \$55 million USD GCF loan) as allowed for administration, the CFF will provide for cost recovery of operating costs.

The CFF will focus only on projects that could be commercially viable but are not yet bankable in the private sector, without credit enhancement and related financial or development support (primarily subordinated debt / first-loss, and credit enhancements primarily in the form of tenor extension and other credit support). Based on numerous conversations with commercial banks in South Africa, there are many projects that are “near investable” from the commercial bank perspective, but that would be able to cross the line with some credit support from a dedicated facility like the CFF. To mitigate the risk of “crowding out” the CFF will maintain a close relationship with the existing banks, and will not invest in any projects that are already able to secure 100% of their investment from commercial banks. Developers will also be required to show that project cannot be fully financed by the private sector. Projects will be required to show that they involve financial participation by one or more private sector financial parties and that it will have some transformative effect on markets in terms of scale, improved private sector participation, confidence in clean energy investments, or other aspects. Investment Criteria include the following:

Project Pipeline

To mitigate the potential bottleneck of inadequate project pipeline, the CFF will address project pipeline by relying on a diversity of channels including:

Commercial Banks - Commercial Banks receiving inquiries to finance projects suitable for CFF financing can directly reach out to the CFF upon reviewing potential projects. Commercial Banks can submit projects to the CFF that could be commercially viable but not yet bankable in the private sector, without credit enhancement and related financial or development support. Based on initial outreach to RMB, Standard Bank, Investec, ABSA and Nedbank, it is anticipated

that there is a significant pipeline of projects that could be made bankable via two main instruments: subordinated debt / first-loss and credit enhancements such as tenor extension.

Examples of projects across key sectors that have surfaced include off-grid, industrial and commercial IPPs, urban solar farms, waste-to-energy projects and local water treatment plants. More detail on deal flow and project pipeline from commercial banks will come early in 2018, based on the next stage of planned outreach.

DBSA Origination Team - The DBSA's SPU and PPU will work with the Origination Unit to identify companies and projects that will be eligible for funding from the CFF from South Africa and other nations in the SADC region.

- *DBSA to add structural description of the DBSA Origination Unit and how it will support pipeline development and deal flow into CFF for both South Africa markets and the SADC region*

DBSA Project Preparation Unit - The CFF will be closely tied to the DBSA's Project Preparation Unit (PPU) whose purpose is to develop projects from pre-feasibility to bankability. The PPU can refer projects that it has brought to bankability to the CFF. These projects may currently be in the PPU pipeline, could come to PPU from the DBSA origination teams, or may have previously been directly presented to the CFU and then referred to PPU for further development.

Climate Lab - The CFF will launch structured finance products as part of a "programmatic" effort to fill previously-identified gaps in the private capital markets, designed in partnership with the Innovation Hub and the new CPI "Climate Lab" to be hosted at DBSA headquarters. The CFF will solicit private sector participation and partnership in these structured products through RFPs. These structured will be designed in partnership with the Innovation Hub and the Global Lab. The DBSAs Innovation hub is working with the Climate Policy Initiative to create a Climate Lab for Africa to accelerate well-designed financial instruments that can unlock billions for energy efficiency, renewable energy, sustainable transport, climate smart agriculture, and curbing deforestation, while also reducing private investors' risks and improving their financial returns. As a public-private partnership, the Africa Lab brings together and catalyzes broader government and private sector efforts to scale up climate finance. The Africa lab will be modeled on CPIs established success through the existing India, Brazil and Global Climate Labs.

DFIs – DFIs may similarly review and refer deals to the CFF based on their position in the South African market and SADC region. Based on initial outreach to several DFI's, it is clear that DFIs want to increase the scale and pace of their green investments (per their commitments to Paris goals), but have trouble investing in small to medium projects. The CFF could help source and perform due diligence on small or medium projects for either direct investment or co financing with the CFF.

CFF Marketing and Outreach – The CFF will conduct targeted marketing and outreach to support market and project pipeline development in South Africa and in the SADC region. Market outreach will focus on relevant sectors including: off-grid projects, micro-grid/urban solar farm projects, industrial and commercial solar for self-generation and wheeling on the grid to MSMEs, industrial and commercial water projects, local Community water projects, and residential and commercial energy efficiency investments. Market outreach strategies will include

- Outreach to developers and project sponsors to explain the role and capacity of the CFF and how it addresses significant financing barriers.
- Outreach to Commercial Banks and DFI's to encourage referral of projects that meet CFF criteria and are not currently bankable at sustainable terms.
- Outreach to Municipalities to explain the role of the CFF around micro-grid/urban solar farm projects and local Community water projects.
- Support for the CFF RFP solicitation process as noted above
- Creation and maintenance of CFF website to support visibility and understanding of the CFF

RFP Process – The CFF will create a "market-responsive" investment window via an open-ended Request for Proposal (RFP) process through which private sector investors/developers may bring projects directly to the CFF for consideration. The CFF will issue an open-ended Request for Proposal (RFP) defining its eligibility criteria under the various funds it can access.

Please describe the efficiency and effectiveness, taking into account the total project financing and the mitigation/adaptation impact that the project/programme aims to achieve, and explain how this compares to an appropriate benchmark. For mitigation, please make a reference to [E.6.5 \(core indicator for the cost per tCO2eq\)](#).

Due to the incredibly high solar resource, combined with the low installation costs of solar PV and energy efficiency, high leveraging ability and high baseline emissions, the average cost efficiency of the mitigation aspect of this project is very high when compared with global standards for funds of this nature. The expected cost of investment per ton of CO2 mitigated is US\$ 5.16. Comparable funds have an expected reduction of US\$8 – US\$11 of investment per ton of CO2 mitigated and South Africa's carbon tax is benchmarked at R120 per ton (~US\$10).

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

Please provide the co-financing ratio (total amount of co-financing divided by the Fund's investment in the project/programme) and/or the potential to catalyze indirect/long-term low emission investment.

- **Total project funding : USD 170.5M**
 - DBSA (AE) contribution : USD 55M plus USD 0.6m in kind contribution
 - Other DFIs contribution : USD 59M
 - GCF contribution: USD 55.6M
 - Other grant: USD 0.3m grant
- **Co-financing ratio = $170.5 / 55.6 = 3.06$**
 - **CFF leverage ratio = 1:5** (Average ratio for facility level. Project ratios will vary). This translates to USD 850 million mobilization from private sector banks and equity sponsors.

The CFF will be designed to maximize total investment, using limited public funds to leverage far greater private investment. By developing innovative finance and market development solutions, Green Banks address barriers that currently restrict clean energy market growth and effectively scale up private investment at ratios ranging from 2:1 to 10:1 depending on market sector and project specifics.

The important private sector leverage will occur at the project-level. The target of the CFF is to reach an overall portfolio leverage ratio of 1:5 (project leverage ratios will vary within this range). That is, for every Rand that the CFF funders put into an individual transaction, it will be looking for approximately 5 Rand of private investment. The DBSA will create the CFF unit using best practices at existing Green Banks that have successfully catalyzed markets using blended finance to achieve similar leverage ratios without crowding out the private sector.^{61,62} This leverage ratio will be supported by several key elements designed to ensure that private capital is crowded into desired markets (vs. crowded out) including:

- **Investment Criteria** – CFF investment criteria will focus on selecting and creating transaction structures that maximize leverage ratios across project types and asset classes. Investment criteria also require deals that would be unable to access financing absent CFF's involvement in the transaction.
- **Metrics** – DBSA will track total project costs enabled by CFF (Cumulative ZAR) and total committed funds from all investment partners (cumulative ZAR)
- **Reporting** – DBSA will provide quarterly reports on the private mobilizing ratio of total project costs to CFF Funds

Mobilization ratios will include the effects of capital recycling as capital is repaid in accordance with agreed terms for each product and counterparty. As each Rand from the CFF cycles through successive investments, benefits will compound. The effective rate of accumulation of these benefits is directly tied to the weighted average expected holding periods of specific projects. In addition, as commercial markets expand into clean energy finance needs supported by the CFF, the multiplier effect on activities and investments will continue.

The specific leverage ratios for a given project will vary at the project level, depending on the project specifics. For example, a hypothetical C&I solar project with 70:30 debt-to-equity ratio, with the CFF providing 30% of the debt would have a CFF investment to local commercial investment ratio of 1:3.8. A hypothetical biogas project with 60:40 debt-to-equity ratio, with the CFF providing 30% of the debt would have a CFF investment to local commercial investment ratio of 1:4.5. With the GCF providing 1/3 of the CFF capitalization, the hypothetical projects would have a GCF investment to local commercial investment ratio of 1:12.5 and 1:15 respectively.

Please make a reference to [E.6.5 \(core indicator for the expected volume of finance to be leveraged\)](#).

⁶¹ NY Green Bank 2016 Business Plan. <https://greenbank.ny.gov/-/media/greenbanknew/files/2016-NYGB-Business-Plan.pdf>

⁶² Coalition for Green Capital (2017), "National Green Banks in Developing Countries: Scaling Up Private Finance to Achieve Paris Climate Goals. Coalition for Green Capital." <http://coalitionforgreencapital.com/2017/07/18/paper-green-banks-in-developing-countries/>

E.6.3. Financial viability

Please specify the expected economic and financial rate of return with and without the Fund's support, based on the analysis conducted in [F.1](#).

Please describe financial viability in the long run beyond the Fund intervention.

1. The CFF with & without GCF support

The viability of the CFF is based on the catalytic and concessional support of the GCF. Without this support, the CFF will not be able to provide an effective cost of capital, support subordinated debt or extended tenors, or attract other co-funders of the Programme. Interest rates will be tied to the blended cost of capital and service fees to cover operating costs charged to funders based on assets under management. For more detail, see the financial modeling (Annex), as based on the GCF's support. (Note: the CFF was not modeled without GCF support as this is not a viable financial model.)

The GCF's participation in the CFF, which will provide long tenor and concessional terms, will allow the CFF to provide financing and credit enhancement to projects that meet CFF investment criteria. GCF's anchor investment in the CFF will facilitate the creation of the CFF, and the concessional rate will lower the blended rate the CFF offers to projects. As the lead investor, the GCF will play an essential role in bringing in capitalization by the DBSA and DFI co-investment at the fund level.

The credit enhancements the CFF provides (facilitated by GCF involvement) will help projects reach financial close in various ways, from lowering debt costs, extending tenors, allowing debt service terms (and associated PPA prices) that aid end-consumer adoption etc. See section F.1 for more details.

Please describe the GCF's financial exit strategy in case of private sector operations (e.g. IPOs, trade sales, etc.).

N/A to CFF

E.6.4. Application of best practices

Please explain how best available technologies and practices are considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.

The CFF's investment criteria will support application of best available technologies by requiring projects to demonstrate that they use technologies that are 1) effective in addressing climate-related goals and/or expansion of clean drinking water supplies; 2) market-tested and commercially viable; and 3) contribute to market transformation in terms of demonstrating the viability and commercial application of new technologies.

The CFF will promote best available clean energy and water technologies by addressing market barriers with a focus on commercial and industrial solar PV, commercial and industrial energy efficiency, commercial and industrial water treatment, and waste to energy technologies. The CFF's subordinated debt and patient, long tenor capital can effectively crowd in private sector investment into these important technology sectors where there is robust market opportunity: current market size for C&I solar PV in South Africa is R280 million and potential market size is up to R25 billion by 2030. For C&I energy efficiency, current market size is R750 and estimated potential market size is R41 billion by 2030. For water efficiency and treatment, the potential market size is R16 billion by 2030.

Building on this initial market focus, the CFF will expand into priority markets in the other Rand-based economies (Namibia, Lesotho, Swaziland) and address additional markets and best available technology in sectors including off-grid and micro-grid solar, urban distributed solar farms, energy efficiency, precision irrigation, water efficiency,

desalinization, etc. that are well tested in other markets but are unfamiliar to Southern Africa and commercial banks, or are novel applications of known technologies.

E.6.5. Key efficiency and effectiveness indicators

Estimated cost per t CO₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)

(a) Total project financing ~ US\$ 170.5M

(b) Requested GCF amount ~ US\$ 55.6M

(c) Expected lifetime emission reductions overtime	29 727 942 tCO ₂ eq
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(d) Estimated cost per tCO ₂ e (d = a / c)	US\$	5.78 / tCO ₂ e
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(e) Estimated GCF cost per tCO₂eq removed (e = b / c) US\$ 1.87 / tCO₂eq

Describe the detailed methodology used for calculating the indicators (d) and (e) above.

Total project financing (a) is based on co-contributions from DBSA and DFIs of R1 350M.

Total carbon emission reductions(c) is based on South Africa's average grid emissions of 1.03kg of CO₂/kWh.

Total energy generation is based on split outlined earlier and in Impact model.

Lifetime: solar PV 20 years, EE 10 years, Waste to energy 20years

PV yield is 1716 kWh / kWp/ annum

Average leveraging is set at a 1:4 ratio.

Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context.

Please refer to Annex for Impact model and Table 24 below

Input														
ZAR/USD	11.68													
Beneficiary multiplier	7.8	http://www.odccorp.com/documents/measuring%20rate%20employment%20of%20the%20												
Water Beneficiary Multiplier	4.110	People impacted/HA/Day												
Remotes in South Africa	31.70%	2011 Census												
Total Population	55,908,900	2016												

Input Model															
Item	Unit	Amount	Split	R	ZAR Value	Capacity installed	Unit value	MWh generated per annum	Jobs created	Female Jobs	t CO ₂ eq reduction/annum	Lifespan of projects	# of projects	Indirect beneficiaries	Female - Indirect
CA/ Solar	35%	R		3,100,000,000		256,641 kW		405,956,015	2,700	1,215	42,8137	20	1,480	21,061	10,888
CA/energy efficiency and transport	35%	R		2,900,000,000		1,722,428,971 kWh/Annum		1,722,428,971	1,084	463	2,775,121	10	3,715	8,096	4,170
Water to Energy	30%	R		1,000,000,000		23,623 kW		173,431,680	457	206	150,690	20	37	3,965	1,842
Water efficiency, treatment and production	30%	R		3,000,000,000		200,000 kJ/D			132	59	0	20	167	430,963	212,468
Clean transport	0%	R		-		0 ha			0	0	0	10	0	0	0
Total	100%	R		10,000,000,000				2,804,638,277	4,322	1,940	2,375,963		7,409	446,621	229,368

Outputs		
Beneficiaries		412,621
Indirect Beneficiaries		4,322
% of total population impacted		0.8%
ZAR/Job of GCF funding	R	130,377.46

t CO ₂ eq		
kg CO ₂ emissions/kWh		1.03
Lifespan Equivalent		28,727,942 t CO ₂ eq
GCF investment amount	R	650,000,000

Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only)

	Private sector leverage through the CFF will occur at the project-level. The target of the CFF is to reach an overall portfolio leverage ratio of 1:5 (project leverage ratios will vary within this range). That is, for every Rand that the CFF co-funders invest in an individual transaction, it will be looking for approximately 5 Rand of private investment. The DBSA will create the CFF using best practices at existing Green Banks that have successfully catalyzed markets using blended finance to achieve similar leverage ratios without crowding out the private sector.⁶³⁶⁴ Leverage ratios will vary at the project level, depending on project specifics. For example, a hypothetical C&I solar project with 70:30 debt-to-equity ratio, with the CFF co-funders providing 40% of the debt would have a CFF investment to local commercial investment ratio of 1:2.6. A hypothetical biogas project with 60:40 debt-to-equity ratio, with the CFF providing 40% of the debt would have a CFF investment to local commercial investment ratio of 1:3.1. With the GCF providing 1/3 of the CFF funding, the hypothetical projects would have a GCF investment to local commercial investment ratio of 1:7.7 and 1:9.5, respectively. Refer to section E.6.2 <i>Describe the detailed methodology used for calculating the indicators above.</i> <i>Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context.</i> See annex impact model
Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/ programme)	

⁶³ NY Green Bank 2016 Business Plan. <https://greenbank.ny.gov/-/media/greenbanknew/files/2016-NYGB-Business-Plan.pdf>

⁶⁴ Coalition for Green Capital (2017), "National Green Banks in Developing Countries: Scaling Up Private Finance to Achieve Paris Climate Goals. Coalition for Green Capital."
<http://coalitionforgreencapital.com/2017/07/18/paper-green-banks-in-developing-countries/>

** The information can be drawn from the project/programme appraisal document.*

F.1. Economic and Financial Analysis

Please provide the narrative and rationale for the detailed economic and financial analysis (including the financial model, taking into consideration the information provided in [section E.6.3](#)).

Based on the above analysis, please provide economic and financial justification (both qualitative and quantitative) for the concessionality that GCF provides, with a reference to the financial structure proposed in section B.2.

Supporting Documents for Funding Proposal

- Integrated Financial Model that includes a projection covering the period from financial closing through final maturity of the proposed GCF financing with detailed assumptions and rationale; and a sensitivity analysis of critical elements of the project/programme (xls format, if applicable)

The financial modeling of the CFF is intended to demonstrate how concessional GCF financing will underpin the CFF at the facility and project levels, and how it will help unlock market barriers to support low-carbon energy and water projects.

The market for sustainable infrastructure in South Africa remains mired in risks (both real and perceived) that lead to high costs for lending, and in many cases projects that simply cannot secure financing. The high risks and pricing in the market have negative impacts on project uptake, leading to issues such as high PPA prices and escalators relative to grid prices, and/or long tenors limiting commercial lender participation and customer uptake. The GCF's concessional participation in the CFF provides a level of pricing efficiency necessary to accelerate the adoption of more clean energy and water projects that currently struggle to reach financial close. Specifically, the concessional nature of the GCF capital results in a financing structure that provides more risk protection to private capital providers and preserves greater savings to end consumers.

The GCF participation will help projects in the following ways:

- High costs of commercial debt currently make projects unattractive: The CFFs low-cost, subordinated position will help bring down the costs offered by commercial lenders (due to risk mitigation), as well as lower the final blended debt enjoyed by projects.
- Lack of sufficient debt to finalize projects: The CFF, with its mandate to invest in climate projects that are near-bankable, will be a first mover/risk absorber to offer additional financing where it is needed most. Projects that need additional debt that cannot currently be sourced in the commercial market can approach CFF to, on a case by case basis, provide the remaining finance and round out the total debt needed at the project level.
- Low IRRs making projects unattractive: Risk-adjusted pricing that is more affordable will help support higher IRRs for project equity sponsors. Equity IRRs often need to be high in nascent markets in South Africa, and these improved financial outcomes will hopefully lead to more participation by equity sponsors and entrepreneurs.
- Longer tenor that are currently unfeasible: The CFF will facilitate longer tenors for projects where necessary. Longer tenors (as well as appropriate risk-adjusted pricing for debt) will help end user prices (e.g. PPAs, or debt service + grid costs) to be lower relative to total costs before an energy or water upgrade. This helps projects be "cash flow positive" from day one, which aids in customer adoption.

The financial analysis of the CFF was developed through a bottom-up approach, wherein several expected project profile types were modeled in order to understand 1) the impact of GCF capital on the overall economics at the project level and 2) the impact of GCF capital at the fund level. The GCF capital allows for the tenor extension and subordination features that market lenders have indicated will help induce their participation and lead to better project economics for end-consumers. Projects and the fund were modeled with these structural features and the results compared to commercial lending benchmarks in South Africa.

Tenor

The commercial lending market in South Africa typically limits debt tenors to ~7 years. However, for assets we expect the CFF fund to cover, the payback terms on the assets likely extend beyond 7 years. The mismatch between asset payback time and lender underwriting terms has two implications:

- 1) Projects will not receive the requisite debt funding to fill out the capital stack and / or
- 2) The debt, if provided, will carry prohibitive risk premiums that will make projects unviable due to high prices and/or high associated debt service

The CFF is structured to have a debt term that can flexibly extend beyond the window offered by commercial lenders up to 12 years. Further, GCF's concessional participation in the CFF expands the pool of assets that can be financed within the 12-year financing window.

Subordinated Position

On a project level basis, the CFF debt (of which GCF's contribution is the central anchor) is structured as a subordinated position to commercial lenders. The subordinated position is necessary in order to catalyze commercial lender's participation in assets that are market transformational (assets that use innovative technologies, new credit off-takers, use new distributed generation business models, etc.). Fundamentally, subordination provides the risk protection commercial lenders require to provide predictable market rates. Yet by its nature, subordinated debt often carries higher rates than senior debt. The GCF's concessional participation in CFF, which will take subordinated positions, is both essential for providing risk protection, while keeping the subordinated debt expense below the senior debt.

Program's Expected Performance

Through the example of a solar PV project, it can be shown how the GCF's concessional participation impacts a typical project's economic outcomes. When GCF capital is contributed, projects are more financeable, carry lower overall risks, will be more cost competitive with existing energy options, and return more savings to consumers. For instance, a typical 750 kWdc solar PV installation would save almost ZAR 2 Million in debt expense over a 10-year period. Importantly, the GCF concessional capital also reduces the payback period on the asset. This is a self-reinforcing risk mitigation feature of the structure that protects the CFF from default risks in the tail years of an asset's life span.

The methodology used was to inspect a number of sample projects across a number of asset types that the facility is expected to fund. Asset types include Commercial Solar PV, Waste to Energy, and Water Efficiency and others. While the CFF is designed to be flexible in approaching projects and is technology agnostic (as long as climate criteria are met), several indicative assets were chosen as the most likely to benefit from the inclusion of GCF concessional capital in the near term. Those sample projects are then scaled to simulate the total facility size and modeled the CFF's expected economics. In all situations the model compares project returns with and without CFF (which includes GCF's concessional capital) to ascertain the expected benefit of the GCF capital as it relates to 1) the fund's economics 2) the project's economics 3) the consumer's savings and 4) the private investor's returns.

The model exercises, combined with discussions with developers and commercial lenders and market analysis from a leading local NGO, points to the need for tenor extension and subordination as a critical tool for lowering risk and increasing adoption, and helping fill gaps that exist in the current market. In all cases the inclusion of GCF capital improves the financial viability of the projects. Overall, the blended rates of the CFF are lower, thereby expanding the pool of assets that can be financed. The lower blended rates have the effect of reducing the Weighted Average Life (WAL) of the assets modeled for the CFF. The lower WAL helps insulate the fund from tail risk associated with project financing. Finally, the lower overall blended rates that result from GCF participation are essential to preserve savings for consumers. More savings to consumers can lead to higher adoption rates and overall deeper market penetration of renewables in South Africa and the additional Rand-based economies that the CFF will serve. Additionally, more savings and longer tenors allow for end-consumer prices (e.g. PPAs) that can be structured to be below incumbent or grid prices. This allows for projects that are cash flow positive on day one, which makes customers value proposition more clear, which has been a limiting factor in several markets in South Africa, including solar and water projects.

It should be noted that in the medium term, it is expected that the CFF's participation in nascent markets, technologies and regulatory structures, will help "lead the market" toward more appropriate pricing of risk from commercial banks in the future. This first mover/demonstration effect is expected to have knock-on effects beyond the immediate participation in a single project, and is therefore not specifically included in the attached models. While this kind of "market transformation" is difficult to quantify, it is a common goal of existing Green Banks, and has been observed in specific markets, for instance in US, UK and Australia, where early Green Bank participation (e.g. in a new market or under a new regulatory regime) has led to increased subsequent participation by commercial lenders, and at financing costs that are more appropriately risk-adjusted.

F.2. Technical Evaluation

Please provide an assessment from the technical perspective. If a particular technological solution has been chosen, describe why it is the most appropriate for this project/programme.

The focus of the CFF (using GCF funding) will be to demonstrate the commercial viability of climate mitigation/adaptation technologies and solutions, resulting in full commercial viability and corresponding private sector funding. The technologies are also all selected to align with South Africa's development plans and strategy, addressing key sustainability concerns.

Solar PV and Energy Efficiency:

This is one of the most exciting focus areas of the GCF application. The technology is economically viable, as both technologies offer fairly attractive returns over their lifespans from avoided energy costs. The key limiting factor to the uptake of PV is the mismatch between finance tenors and project lifespan. Both the EE and PV market opportunities are far larger than the current uptake suggests. These technologies are likely to be the best option for the initial rounds of CFF. Rooftop Solar PV is also a particularly attractive option in terms of job creation, with a high number jobs created per MW installed. The local lamination of PV panels also represents a significant opportunity for gender-sensitive job creation, with an expectation of ~80% female employment.

Waste to Energy

The incomes streams are typically diverse, the technology operation risks are high/unknown and feed stock risk is difficult to price. In effect this leads to messy, complex project that become extremely hard to finance. Currently these projects are being held back by the difficulty of creating a business case for a technology that not only requires an income stream from its primary product but also from ancillary income streams in order to become commercially viable. Development of financing options specifically designed to match the market conditions in South Africa are vital to unlock these markets.

Water Efficiency and Treatment

This is a complex market with a range of inherent subsidies and incentives. Primary among these are the non-cost reflective nature of the current water prices. The current water prices do not adequately reflect the availability and cost of supply, and do not provide a particularly attractive business case for private investment in water infrastructure. Predictions of future water demand however; show a gap in predicted water availability and predicted demand that indicates that there is a need for private investment in water infrastructure. While water tariffs are trending upwards, it is likely that investment in private water infrastructure projects will be required before the tariffs reach a point where there is a bankable business case for commercially financed projects.

Hypothetical Examples of Projects in early CFF pipeline

While it is difficult to provide specific examples of pipeline projects at this point in the development of the CFF, we offer the following illustrative project sketches across the key sectors as noted in this proposal including off-grid, industrial and commercial IPPs, urban solar farm, and local water treatment plants. Each example represents multiple projects. More detail on deal flow and project pipeline will come early in 2018, based on input from major commercial bank partners and project developers through the next stage planned outreach. Each of the project sketches below would result in leveraging commercial funding anywhere between 4 and 10 times depending on the amount of equity invested by the sponsor/developer.

Off-Grid

A South African based company with a four year track record in the small scale energy generation sector has prepared a business plan to roll out, initially, 18,000 off-grid systems to identified rural and under-provided municipal areas in South Africa. These affordable "kits" will be installed and maintained by the company or their appointed agents. The systems will be either single or hub systems (the latter serving several homes) with an average cost per installed systems of R11,500 (USD 900). The business plan provides for a 5 year roll out to install 300,000 systems in South Africa and Southern Africa with a particular focus on Botswana, Mozambique and Namibia. The total funding required to complete the entire program is R3.4 billion. They are venture capital in nature and would in principle be ideal for CFF funding alongside a commercial bank. They would be looking for an initial R80-100 million.

Industrial and Commercial Solar

A South African based company that has been operating for about three years that is currently funded by shareholders and a highly conditional and uncommitted bank facility has been installing and maintaining roof top solar PV systems primarily on supermarkets. They have vertically integrated and now have a contracting division, but are constrained by shareholder funding and commercial banks will not provide funding at this stage that is sustainable. The CFF could provide needed financing to open up this market segment which includes opportunities in certain Southern African countries where their customers own shopping centers and supermarkets. They have orders on their books in excess of R500million and would be looking for about R100 million.

Urban Solar Farm

The South African operation of an international group that was established to participate in the South African RE-IPP Program is now embroiled in South African/Eskom politics. In order to keep the South African operation going they are shifting focus to the private sector. With the support of USDA, the company has completed a feasibility study on the roll out of two urban solar farms with a total initial capacity of 10-15MW. The total cost of the initial project is between R200 million and R300 million, with the potential to grow into hundreds of MWs and billions of Rand. Commercial banks that were part of the feasibility study indicated that they would need credit enhancement to provide funding due to the nature of the proposed off-take structure which involves a diversified group of companies with varying financial strengths. The project would be looking for an initial R75 million from the CFF in the form of subordinated debt with low-cost, risk-adjusted return requirements to help the projects cover their debt service coverage ratio without raising PPA prices beyond what potential offtakers will bear.

Local Water Treatment Plants

DBSA has been approached by a Dutch company that has designed container based water treatment plants that were developed and used for the Rio Olympics. They have secured very limited grant money from the Dutch Government to produce only one or two containers which, fully installed and operational, cost approximately USD 5 million per unit. Each container can treat about 350,000 liters per day. The company would like to develop a PPP structure to create these containers for several smaller and more rural municipalities that are struggling to properly treat grey-water before it is released back into the watershed and re-enters municipal water systems. The CFF would be ideal to support a project of this nature for example with subordinated debt to lower the risk profile for private co-investors, or other potential offerings (for example, generated out of the Climate Lab) such as a Loan Loss Reserve product to protect against risks perceived by private investors.

Energy Efficiency in Commercial Buildings

A South African based company that provides energy savings performance contracting (ESPC) through an ESCO model has been operating for two years. They have completed 7 small projects (avg 1 million Rand) and have recently met all requirements and received certification as a "Tier 2 provider" under South Africa's new ESCO registration program. Commercial banks view them as 'business as usual' finance applications, and do not apply credit assessment criteria specifically designed for the ESCO industry. Limited collateral associated with energy efficiency projects, unfamiliarity with the ESPC model and uncertain cash flows, and small project size make it hard for the company to access additional finance to do larger projects or bundle multiple projects. To date the company has relied on expensive equity, grant funding and small and uncertain loans from one bank, which cannot increase its position. The CFF could provide needed co-financing to open up this market segment, and give added certainty to additional commercial banks investing alongside the CFF in the deal. This financing would allow the company to reach larger scale, develop larger projects, helping them achieve "tier 1" status and eventually accessing more commercial finance from South African banks.

F.3. Environmental, Social Assessment, including Gender Considerations

Describe the main outcome of the environment and social impact assessment. Specify the Environmental and Social Management Plan, and how the project/programme will avoid or mitigate negative impacts at each stage (e.g. preparation, implementation and operation), in accordance with the Fund's Environmental and Social Safeguard (ESS) standard. Also describe how the gender aspect is considered in accordance with the Fund's Gender Policy and Action Plan.

See attached ESS framework

The programme will be implemented and operated in accordance with the DBSA's Environmental and Social Safeguard Standards which is attached. The ESSS also commits DBSA to ensure it has the necessary systems in place to implement the ESSS and to periodically update and revise it. It includes a commitment from DBSA to appoint appropriately skilled people to appraise projects, ensure they meet the minimum Safeguard requirements and apply these fairly.

The DBSA's Environmental and Social Safeguard Standards (ESSS) is an extension of the DBSA Environmental Appraisal Framework and the Social and Institutional Appraisal Guidelines. In creating its ESSS the DBSA has harmonized with the work of other Global and African DFI's and especially the Global Environmental Facility (GEF). The DBSA ESSS draws extensively from existing Safeguards which have recently gone through an extensive stakeholder engagement process. The minimum safeguards set out the minimum social and environmental and gender mainstreaming standards required of DBSA clients accessing Global Environment Facility (GEF) and GCF financing. The DBSA undertakes an integrated approach to project preparation, due diligence, approval and project surveillance, monitoring and evaluation. The DBSA Environmental and Social Safeguards addresses seven operating safeguards requirements that DBSA clients are expected to meet when addressing social and environmental issues.

The first Safeguard Standard, the Environmental and Social Assessment provides an umbrella framework to assist DBSA and clients to identify potential environmental and social risks and benefits associated with the DBSA investment lending operations and partnership initiatives such as those with GCF. It is an essential tool for integrating environmental and social concerns into bankable investment projects by providing the minimum requirements that all DBSA investment lending operations must meet. This standard also covers accountability and grievance systems.

The remaining Safeguard Standards are:

2. Protection of Natural Habitats
3. Involuntary resettlement
4. Community stakeholders and vulnerable groups (including Indigenous Peoples)
5. Pest Management
6. Physical and cultural resources and
7. Safety of Dams
8. Labour Guideline

In relation to item 4 above, using a **gender mainstreaming approach**, the DBSA typically carries out an assessment of gender issues for every project and uses the findings as the basis for project design and compensation plans that lead to enhanced gender balance. When a project includes environmental and social analysis, the gender assessment may be carried out as part of this analysis. In particular, the DBSA assesses the quality and relevance of gender data and performance indicators, specific pro-gender measures, and budgetary resources allocated for equality and empowerment for any project as key criteria in investment decision-making process. This approach is very much applicable to the CFF programme since women and men are involved in climate change responses but in different ways. In fact, the Cancun Agreements acknowledge that gender equality and the effective participation of women are important for all aspects of climate change.

Climate change hazards increase or heighten existing gender inequalities thereby contributing to the greater climate change vulnerability of many women. This is largely due to persistent gender norms and widespread gender discrimination that deny women income, legal rights, access to resources while assigning them the primary role in caring for their families and providing for their livelihoods, leading to women's marginalization in many ways. Nevertheless, women are powerful

stakeholders in implementing low-carbon pathways in development countries. This makes women important agents of change in the fight against global warming and their input needs to be taken cognizance of and incorporated in climate change adaptation and mitigation equation.

In the context of the socio-economic background outlined above, the action plan below considers the following aspects as important and relevant for streamlining gender in the CFF project:

As executing entity of the CFF, DBSA should incorporate and take cognizance of gender mainstreaming matters, within the organization as an entity. The institutional and governance structures within the organization should ensure that the meaningful participation, upskilling and empowerment of women occurs within their organizational structures. The appointment of women to key strategic positions within CFF is encouraged.

At a project level, the CFF will have the responsibility of ensuring that the projects and programmes to be financed by them are gender sensitive, and encourage the recognition and empowerment of women at all project stages. CFF shall apply a gender mainstreaming approach in line with the DBSA gender requirements described in the Environmental and Social Safeguard Standards and Gender Policy. In addition, they shall apply a gender mainstreaming approach in line with the GCF requirements for gender mainstreaming.

Table 25: Gender Framework

Activities	Indicators and Targets	Timelines	Responsibilities
Impact: Gender considerations become an integral part of CFF's design Outcome: The CFF Board/Advisory group and technical team will have a gender policy that is aligned to the DBSA, GCF, SADC and respective countries' gender protocols, and integrates gender considerations in its operations through ensuring that:			
1. There is equal opportunity for policies and practices in the workplace to ensure that there is gender parity between men and women; and 2. DBSA monitors compliance of its organizational priorities and objectives with SADC gender legislation			
Output 1: Resourcing evidence on equal opportunity for all genders practices			
- Review SADC and country specific gender policies - Review alignment of national gender policies of countries to be funded with GCF and DBSA - Implement gender sensitive recruitment processes for projects to be funded by CFF	Resourcing levels by gender breakdown	On going	DBSA
Undertake market research to determine and identify women led contractors/MSMEs to be included in projects	Number of MSMEs identified	Beginning of programme cycle	DBSA
Output 2: CFF projects shall have an appreciation of SADC gender policies			
- Familiarize with best practise gender policies and practices - Mainstream gender considerations during project implementation and	Number of workshops and engagements	Ongoing	DBSA

operations for various countries Seek to develop partnerships with vocational/technical training schools, to enhance women's access to developing technical skills			
Review the needs of women led cooperatives to identify areas where capacity building is required	Number of development areas identified	Ongoing	DBSA

F.4. Financial Management and Procurement

Describe the project/programme's financial management and procurement, including financial accounting, disbursement methods and auditing.

G.1. Risk Assessment Summary

Please provide a summary of main risk factors. Detailed description of risk factors and mitigation measures can be elaborated in G.2.

As noted above, at the institutional level, the DBSA will minimize risk associated with the CFF through application of best practice to support effective governance and administration. All CFF projects will first be reviewed and approved by the DBSA Investment Committee to provide oversight for DBSA funds, and to utilize the DBSA's own Investment Committee as the first screen of eligibility and bankability. The DBSA Investment Committee will ensure that proper due diligence and risk assessment has been performed, and that only projects that meet CFF eligibility are accepted into the fund.

In terms of specific risk factors, please see section G.2 for discussion of the following factors and mitigation measures:

- Adequate capitalization
- Sustainable financial operation
- Crowding in (vs. crowding out) private capital
- Oversight and governance
- Long-term capital strategy; and
- Currency risk

G.2. Risk Factors and Mitigation Measures

Other Potential Risks in the Horizon

Selected Risk Factor 1

Description	Risk category	Level of impact	Probability of risk occurring
Securing Capitalization	Financial	High (>20% of project value)	Medium

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The DBSA will seek capitalization from a diversity of sources to mitigate capitalization risk including: DBSA (R650 million – approved), R650 million from GCF (in process), R700 million from DFIs with business in Africa (in process). Seeking capitalization from a variety of sources, lowers this risk as does outreach to a wide number of DFI's.

Selected Risk Factor 2

Description	Risk category	Level of impact	Probability of risk occurring
Crowding in vs crowding out private investment	Financial	High (>20% of project value)	Low

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The CFF will focus only on projects that could be commercially viable but are not yet bankable in the private sector, without credit enhancement and related financial or development support (primarily subordinated debt / first-loss, and credit enhancements primarily in the form of tenor extension and other credit support). The CFF Investment Criteria will require projects to demonstrate that they are focused on technically and economically feasible transactions where there is market interest but limited capital availability due to specific financing gaps and barriers. Developers will be required to show that projects cannot be fully financed by the private sector. Projects will be required to show that they involve financial participation by one or more private sector financial parties and will have some transformative effect on markets in terms of scale, improved private sector participation, confidence in clean energy investments, or other aspects. Projects must contribute to low-carbon infrastructure and climate-related goals and priorities.

Selected Risk Factor 3

Description	Risk category	Level of impact	Probability of risk occurring
CFF Oversight and Governance	Technical and operational	High (>20% of project value)	Low

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

All sub-projects will first be reviewed and assessed by a DBSA's Structured Project Unit and the CFF Operations Management Team within DBSA. The Project Steering Committee "Steercom" will provide oversight on CFF operations, ensure that proper due diligence and risk assessment has been performed, and that only projects that meet the Programme eligibility in accordance with the operational manual are recommended to the DBSA Investment Committee for final approval. The DBSA Investment Committee will make the final decision on the approval of sub-projects to be financed.

Selected Risk Factor 4

Description	Risk category	Level of impact	Probability of risk occurring
Long-term Capital Strategy	Financial	High (>20% of project value)	Medium

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The CFF's long-term capital strategies are two-fold:

- Raise more capital into the fund once it demonstrates it can effectively deploy its starting capital base and cause positive market and climate impact and transformation;
- Grow and diversify its portfolio so that a portion may be sold through a securitization or other asset sale that allows for capital recycling, repayment to CFF investors, and continued operation of CFF at a significant scale.

Selected Risk Factor 5

Description	Risk category	Level of impact	Probability of risk occurring
Generating project pipeline	Technical and operational	Medium (5.1-20% of project value)	Low
Mitigation Measure(s)			
<i>Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?</i> The DBSA will mitigate the risk of lack of project pipeline by relying on a diversity of channels including: - A “market-responsive” investment window via an open-ended Request for Proposal (RFP) process - The DBSA origination team - DBSAs Project Preparation Unit - The DBSA will launch structured finance products as part of a “programmatic” effort to fill previously-identified gaps in the private capital markets, designed in partnership with the Innovation Hub and the new CPI “Climate Lab” to be hosted at DBSA headquarters. - Commercial banks can directly reach out to the CFF upon reviewing projects the banks consider suitable for CFF support. Several commercial banks have already expressed interest in doing so DFIs may review and refer deals to the CFF.			
Selected Risk Factor 6			
Description	Risk category	Level of impact	Probability of risk occurring
Currency Risk	Financial	High (>20% of project value)	Low
Mitigation Measure(s)			
Currency Risk – The CFF will address currency risk by requiring investments to be largely in South African Rand and concessional USD financing from the GCF. The borrowers will be partly exposed to currency fluctuation risk. However, the concessional financing by GCF at lower pricing will enhance the ability of the borrowers to hedge or partly absorb currency fluctuations. <i>Please describe other potential issues which will be monitored as “emerging risks” during the life of the projects (i.e., issues that have not yet raised to the level of “risk factor” but which will need monitoring). This could include issues related to external stakeholders such as project beneficiaries or the pool of potential contractors.</i>			

** Please expand this sub-section when needed to address all potential material and relevant risks.*

H.1. Logic Framework.

Please specify the logic framework in accordance with the GCF's [Performance Measurement Framework](#) under the [Results Management Framework](#).

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level ⁶⁵					
Paradigm shift objectives					
Shift to low-emission sustainable development pathways	The DBSA's new Climate Finance Facility (CFF) will address market constraints, playing a catalytic role with a blended finance approach, to increase climate related investment in the Southern African region. The CFF will be a first-of-its-kind application based on the Green Bank model, adapted for emerging market conditions. It offers globally significant proof-of-concept value to middle and lower income nations seeking to address market barriers and quickly scale up the high levels of private investment required by Paris climate commitments. The CFF will use its capital to fill market gaps and crowd-in private investment, targeting commercially viable technologies that cannot currently attract market-rate capital at scale. It will focus on infrastructure projects that mitigate or adapt to climate change and utilize two main instruments: subordinated debt / first-loss and credit enhancements such as tenor extension to projects that are commercially viable but not currently being financed by the private sector banks. The DBSA's CFF will deliver outcomes related to each of the GCFs Investment Criteria including: 1) Mitigation of climate change via expanding private investment in climate-friendly infrastructure; 2) Providing a replicable model for building nation-specific capacity to scale up climate finance in support of Paris climate goals; 3) Supporting at least eight of the UN Sustainable Development Goals supporting: reduction in poverty, access to clean water, renewable energy generation, economic growth, development of sustainable infrastructure, reduced inequality in terms of energy access and access to clean water, sustainable cities and communities, and taking climate action; 4) Providing necessary financial capacity to meet socio-economic needs related to climate friendly and clean water infrastructure; 5) Expanding national capacity to achieve established climate goals working through local stakeholders 6) Increasing market efficiency through a leverage approach to scaling up private investment.				
	Expected Result	Indicator	Baseline	Target	Assumptions

⁶⁵ Information on the Fund's expected results and indicators can be found in its Performance Measurement Frameworks available at the following link (Please note that some indicators are under refinement):
http://www.gcfund.org/fileadmin/00_customer/documents/Operations/5.3_Initial_PMF.pdf

		Means of Verification (MoV)		Mid-term (if applicable)	Final	
Fund-level impacts						
<i>M1.0 Reduced emissions through increased low-emission energy access and power generation</i>	1.1 Tonnes of carbon dioxide equivalent (t CO ₂ eq) reduced or avoided as a result of Fund-funded projects/programmes (accumulated GHG reduction over 20 years)	Project monitoring and completion reports	0		11 976 640	CFF model is a new concept, so previous baselines on a similar facility not available. As per attached impact model, Future estimates based on ~31% of annual CFF funding being invested in clean energy production
<i>M2.0 Reduced emissions through increased access to low-emission transportation</i>	3.1 Tonnes of carbon dioxide equivalent (t CO ₂ eq) reduced or avoided as a result of Fund-funded projects/programmes (Accumulated GHG reduction over 10 years)	Project monitoring and completion reports	0		124 2590 ⁶⁶	As per attached impact model, future estimates based ~7% of annual CFF funding being invested in low emission transport. (Accumulated tonnes CO ₂ eq reduction estimate based on potential impact of energy efficiency initiatives in the transport sector)

⁶⁶ Due to the lack of sufficient data on clean transport in South Africa, it is not yet clear what the scale of this impact would be. The tonnes CO₂ eq reduction estimate here is based on potential impact of energy efficiency initiatives in the transport sector. The clean transport sector is an emerging sector in the CFF target countries, hence the potential impact cannot be readily estimated at this stage. EV market is in very early stage. Opportunities exist in local manufacturing of EV components, charging infrastructure roll out (public and private) and development of secondary markets.

<i>M3.0 Reduced emissions from buildings, cities, industries and appliances</i>	3.1 Tonnes of carbon dioxide equivalent (t CO ₂ eq) reduced or avoided as a result of Fund-funded projects/programmes (Accumulated GHG reduction over 10 years)	Project monitoring and completion reports Ad-hoc site visits	0		16 500 872	As per the impact model attached, future estimates based on ~22% of the annual CFF funding being invested in energy efficiency projects in various industries excluding transport.
<i>A2.0 Increased resilience of health and well-being, and food and water security</i>	2.3 Number of males and females with year-round access to reliable and safe water supply despite climate shocks and stresses	Project monitoring and completion reports Facility reports (ID numbers and HR/training reports) Ad-hoc site visits	0	205481	410 963	As per the impact model attached, future estimates are based on 30% of the annual CFF funding being invested in adaption with a focus water treatment, efficiency and new sources

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level						
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Project/programme Outcomes	Outcomes that contribute to Fund-level impacts					
M6.0 Increased number of small, medium and large low-emission power suppliers	6.3 MWs of low-emission energy capacity installed, generated and/or rehabilitated as a result of GCF support	Project monitoring and completion reports	0	130	260	Assumptions as detailed in the attached impact model
M8.0 Increased use of low-carbon transport	8.2 Vehicle fuel economy and energy source as a result of Fund support	Project monitoring and completion reports	0	41600 litres per year	83 200 litres per year	
M7.0 Lower energy intensity of buildings, cities, industries and appliances	7.1 Energy intensity/improved efficiency of buildings, cities, industries and appliances as a result of Fund support. – GWh/annum installed)	Project monitoring and completion reports Focus group with relevant project staff to verify energy efficiency measures	0		1603	
A7.0 Strengthened adaptive capacity and reduced exposure to climate risks	7.1: Use by vulnerable households, communities, businesses and public-sector services of Fund-	Project monitoring and completion reports	0		410 963	Installed capacity 100 000kl/day Water beneficiary

	supported tools, instruments, strategies and activities to respond to climate change and variability					multiplier 4110 people impacted (ML/day)
	Water sourced, saved or treated for reused	Project monitoring and completion reports Monthly water meter readings Focus group with relevant project staff to verify water efficiency measures	0		100 ML/day	(ML/day) 10ML/day saved through efficiency projects 40 ML/day treated for reuse 50ML/day new sourced
Project/programme outputs	Outputs that contribute to outcomes					
1. Market development	Mitigation : Renewable energy projects generation-SPV	Project monitoring reports	0		4	Number of programs (could be made up of a number projects) based on pipeline projection as per impact model
	Mitigation: Waste to energy	Project monitoring reports	0		2	
	Mitigation: Energy efficiency	Project monitoring reports	0		3	
	Mitigation: Low emissions transport	Project monitoring reports	0		1	
	Adaptation: Water efficiency	Project monitoring reports	0		1	

	Adaptation: Water treatment	Project monitoring reports	0		2	
	Adaptation: New Clean Water	Project monitoring reports	0		2	
	Percentage finance leveraged	CFF monitoring reports				
	Mitigation programmes per country: • South Africa • Lesotho • Namibia • Swaziland	CFF monitoring reports	0		• 6 • 1 • 2 • 1	based on pipeline projection but subject to variation
	Adaptation programmes per country: • South Africa • Lesotho • Namibia • Swaziland	CFF monitoring reports	0		• 2 • 1 • 1 • 1	based on pipeline projection but subject to variation
2. Value of funding	Total value of CFF funding approved	Project monitoring reports	0	84.5	169	Million USD value of based on pipeline projection as per impact model USD/ZAR exchange rate 12.37 All approved amounts will be disbursed
	Total value of CFF funds disbursed	Project monitoring reports	0	84.5	169	
	Mitigation : Renewable energy projects generation-SPV	Project monitoring reports	0	26.15	52.31	
	Mitigation: Energy efficiency and transport	Project monitoring reports	0	24.50	49.01	

	Mitigation: Waste to energy	Project monitoring reports	0	33.8	16.9	and CPs are met
	Adaptation: Water efficiency, treatment and production	Project monitoring reports	0	25.33	50.76	
	Value mitigation programmes per country: • South Africa • Lesotho • Namibia • Swaziland	Project monitoring reports			• 70.98 • 11.83 • 23.66 • 11.83	Million USD value of based on pipeline projection as per impact model USD/ZAR exchange rate 12.37
	Value adaptation programmes per country: • South Africa • Lesotho • Namibia • Swaziland	Project monitoring reports			• 20.28 • 10.14 • 10.14 • 10.14	All approved amounts will be disbursed and CPs are met
3. Economic benefits	Number of jobs created: Renewable energy projects generation- SPV • Total • Female jobs	Project monitoring reports Identification Numbers	0		• 2 700 • 1 215	51.7% female population (2011 census) See impact model for assumptions
	Number of jobs created : Energy efficiency and transport • Total • Female jobs	Project monitoring reports	0		• 1 034 • 465	

	Number of jobs created: Waste to energy • Total • Female jobs	Project monitoring reports	0		• 457 • 206	
	Number of jobs created: Water efficiency, treatment and production • Total • Female jobs	Project monitoring reports	0		• 132 • 59	
	Value of funding to MSMEs • Mitigation • Adaptation	Project and CFF monitoring reports APR			• 41.35 • 17.49	(million USD) 35% of CFF funding recipients will be MSMEs
PROJECT MANAGEMENT AND PROJECT FINANCING ACTIVITIES						
Activities	Description		Inputs		Description	
Component 1: Operationalization and project management of CFF						
1.1 Unit establishment within DBSA	• Lead successful operationalization and launch of the CFF unit within the DBSA; • Design and put in place CFF portfolio management unit team; • • Develop strategic plan, • Form CFF Steering Committee; • Oversee legal and accounting arrangements; • Create budget template, and ring-fenced trust bank accounts: and other tasks as required •		• Human Resources		Facilitate the establishment of the Climate Finance Facility as a unit within DBSA	
1.2 Development of CFF Operational Manual	• Manual will include detailed investment criteria, eligibility criteria for each sector targeted by CFF, monitoring and impact measurement guidelines; • The operational manual will stipulate how CFF investment and eligibility criteria will be fully integrated into the investment appraisal process of CFF;		• Human Resources		External consultants and CFF project team members to develop operational manual in consultation with the Green Climate Fund	

	• Eligibility criteria will include a focus on the water sector.		
1.3 Annual Work Plan and Budget	• Develop budget plan that considers the financial sustainability of the facility and a balanced portfolio of programs •	• Human Resources	
1.4 Project Pipeline Development & Financial Product Development Programs	• To develop a pipeline of appropriate projects Extended tenor • subordinated debt • project aggregation securitization	• GCF Mandate • CFF operational guidelines • Lead development of first round of CFF financial products • Market Data based on engagement with commercial banks • Design and launch an open-ended Request for Proposal (RFP) process to support development of a robust project pipeline for the CFF; • Interface with DBSA origination channels and the new CPI “Climate Lab” to be hosted at DBSA;	Oversee ongoing market research to continually develop new opportunities, align programs to emerging markets, and maintain competitiveness with other financing options. GCF mandate and operational guideline will be used to design the RfP.
1.5. Communications, Marketing and Awareness Raising	• Develop, manage, and maintain organizational brand and external presence of the CFF and execute a communications and outreach effort that promotes the objectives of the CFF working with the DBSA marketing team; • Lead direct outreach to project developers, commercial banks, DFI's and other relevant stakeholders; • Build the reputation and brand of the CFF with donors, foundations, corporations, the	• Human Resources	

	media and public officials and agencies; Establish strategic partnerships with lenders, contractors, developers, government agencies, utilities, business and industry associations, and community groups.		
Component 2: Project financing of climate change mitigation and adaptation projects			
2.1 Mitigation Financing	DBSA will enter into loan agreements for mitigation projects (Renewable Energy, Waste to Energy, Energy Efficiency, Low Emission Transport) Signing of Loans and disbursements to be in line with the eligibility criteria and the operations manual of the programme.	- Record of decision from DBSA Investment committee - Project specific business plans - Project specific appraisal reports	
2.2 Adaptation Financing	DBSA will enter into loan agreements for adaptation projects (Water Efficiency, Water Treatment, New Clean Water) Signing of Loans and disbursements to be in line with the eligibility criteria and the operations manual of the programme.	- Record of decision from DBSA Investment committee - Project specific business plans - Project specific appraisal reports	

H.2. Arrangements for Monitoring, Reporting and Evaluation

Monitoring and Evaluation of the CFF will be aligned to the DBSA Development Results Reporting Framework and the requirements of GCF as detailed GCF master accreditation agreement. The broad monitoring and reporting requirements are as follows:

- Project level monitoring
 - o Undertaken as per the project specific log frame that will be compiled for each project funded by the CFF.
 - o The project team will be required to report to the CFF biannually based on the project log frame.
- Fund level monitoring
 - o Information from projects will be collated into the APR and submitted to GCF at the end of the calendar year.
 - o The CFF project team will compile the APR based on information received on individual projects.
- Project level evaluation
 - o Mid-term and terminal project evaluations will be undertaken as per the GCF requirements.
 - o The evaluations will be undertaken by an independent evaluator/ consultant at the cost of the
- Fund level evaluation
 - o A mid-term evaluation will be undertaken on the performance of the fund as per the indicators identified in the CFF log frame.
 - o At the end of the term of the CFF a terminal evaluation will be undertaken on the funds' performance as per the log frame indicators and other qualitative information.
 - o The mid term and terminal evaluations will be undertaken by an independent consultant appointed by the DBSA

In addition the DBSA independent evaluation function, the Operations Evaluation Unit will provide advisory and oversight services to the CFF. The OEU provides a framework for the monitoring and evaluation of all the DBSA's operations.

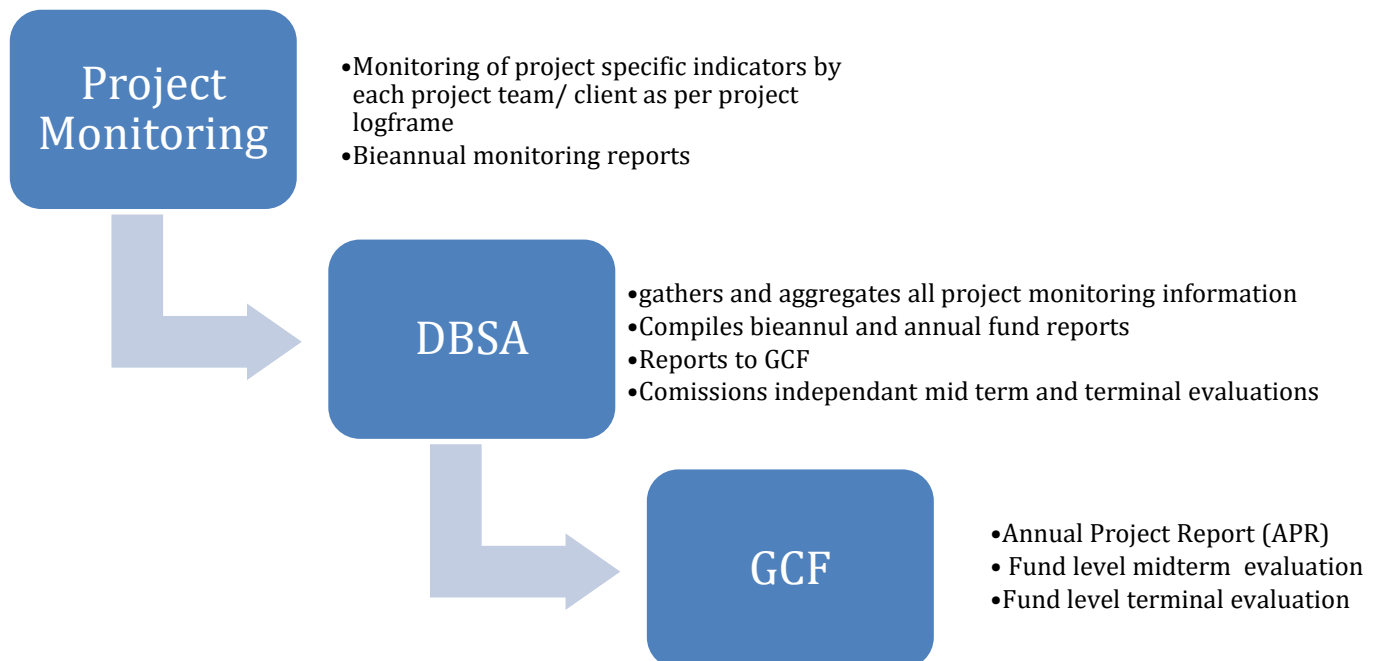


Figure 26

Monitoring Process

The monitoring plan for GCF funding takes cognisance of the timeframes and governance as well as reporting requirements. To this end, reporting at the project level (CFF funded projects) is critical for the process of Fund management reporting. An example CFF project monitoring template is attached which provides basic information on the project performance indicators to inform GCF reporting requirements as well as a means of verification for fund performance. The requirement is that every project that has been in implementation for at least 6 months or the duration of the project implementation period should ensure that monitoring is conducted diligently across the implementation of the project. In addition it is incumbent that on project completion a close out monitoring report is compiled and submitted. The figure below shows the process for M&E information flow.

Process for M&E information flow

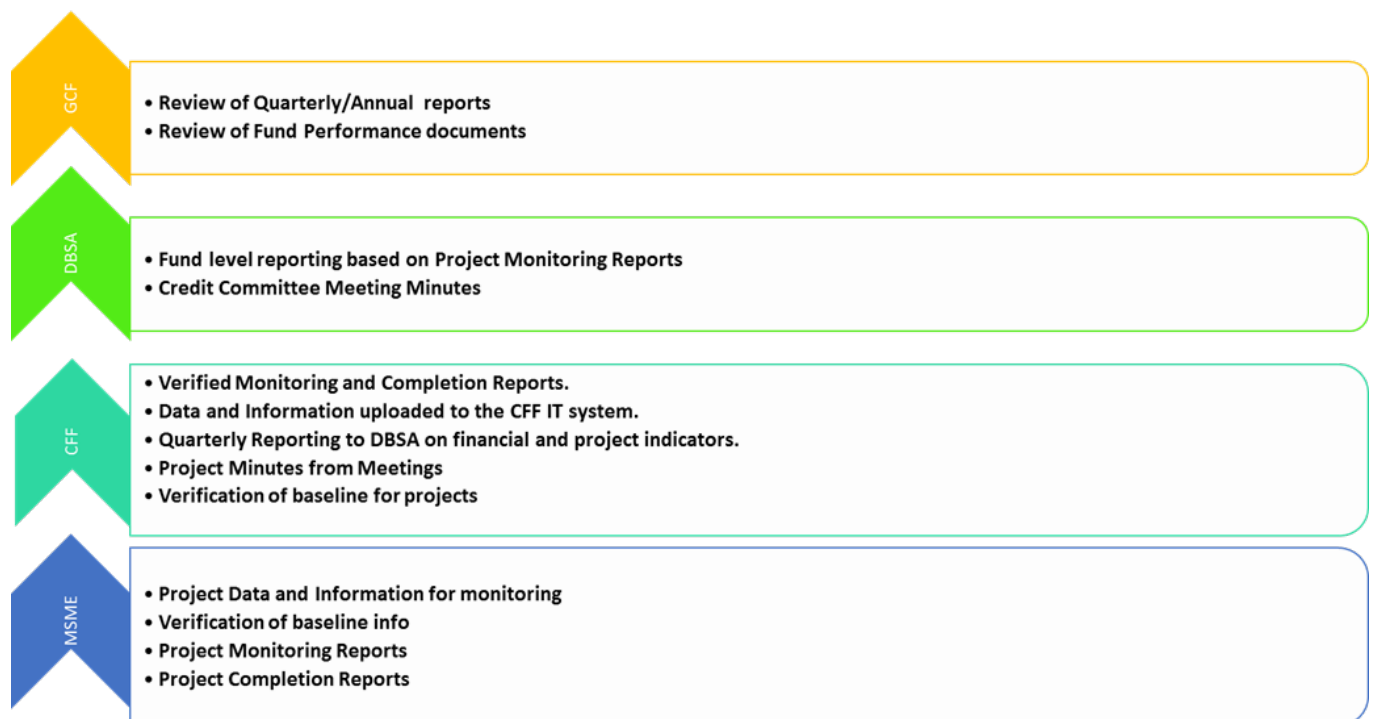


Figure 27

Project Completion Report

Project Completion Reporting (PCR) is a critical part of the project cycle and provides a wealth of information for reporting requirements. The PCR is a self-evaluation mechanism which will be utilised by the CFF, to assess the performance of projects upon completion. This mechanism is, therefore, distinct from the monitoring close out report. The monitoring close out reports provide information purely on the outputs whilst the PCR is meant to detail the outputs and immediate outcomes

of the project against its original objectives, set out in the projects planning framework as approved by the GCF and DBSA as part of the original GCF Fund agreement. A proposed PCR is attached.

Data Collection Plan

The data collection plan included herein provides an overall basis for data governance and credibility. The data collection plan summarizes information about the data sources required to populate both the DBSA and the GCF identified indicators to monitor the Fund performance. The GCF identified indicators are largely quantitative in nature. Although baseline information is vital to support the monitoring process, however in the case of CFF, the baseline is not readily available. The CFF management team in collaboration with the AE will have to create a baseline established on credible methodological process upon implementation.

It is critical to note that there are challenges with baselines for the nature of the projects being funded. The use of official statistics as generated by Statistic South Africa, Department of Energy and the Department of Trade and Industry will provide a credible source for base line data where the data exists. Country level baselines which are not readily available, may be required to be generated through audit reports and portfolio data. The table below provides a synopsis of the DBSA required Indicators as well as the data source and the credibility thereof

Table 26: Fund Level Indicators as per DBSA Requirements

Indicator	Baseline Source	Credibility
Amount of capital provided.	GCF, DBSA & CFF	MOU CFF monitoring tool ⁶⁷
Number of renewable energy projects.	Portfolio Data ⁶⁸	CFF monitoring tool
Number of energy efficiency projects	Portfolio Data	CFF monitoring tool
% reduction in Green House Gases	Department of Energy/Stats SA Audit Data for individual projects	Official Statistics
Rand value of amount disbursed.	GCF & DBSA	CFF monitoring tool
Number of MSMEs benefited.	Department of Trade and Industry	Official Statistics
Number of Female owned MSMEs.	Department of Trade and Industry	Official Statistics
Number of tons reduced Green House Gases.	Department of Energy Audit Data for individual projects	Official Statistics
Number of Rand Value savings	Audit Data for individual projects	CFF monitoring tool
Number of deals recommended to DBSA and CFF Credit Committees	Portfolio Data	CFF monitoring tool

⁶⁷ The Assumption is that the CFF monitoring tool or IT system will be managed diligently and accurately track data.

⁶⁸ Portfolio data would refer to data that is being collated by CFF for reporting. Where national baseline data is not readily available, some of the baselines will have to be generated through this process as per the nature of the projects.

Number of approved deals Rand/value and or % of facility utilised.	Portfolio Data	CFF monitoring tool
Number of projects implemented according to the implementation plan	Portfolio Data	CFF monitoring tool
Number of monitoring reports received from CFF.	Portfolio Data	DBSA Signed off Monitoring Reports
Number of projects completed within designated timeframes	Portfolio Data	CFF monitoring tool
Number of targets met by CFF as per the performance contract/partnership agreement.	Portfolio Data	DBSA

I. Supporting Documents for Funding Proposal

- ☒ NDA No-objection Letter
- ☒ Feasibility Study
- ☒ Integrated Financial Model that provides sensitivity analysis of critical elements (xls format, if applicable)
- ☒ Confirmation letter or letter of commitment for co-financing commitment (If applicable)
- ☒ Project/Programme Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.) – see *the Accreditation Master Agreement, Annex I*
- ☒ Environmental and Social Impact Assessment (ESIA) or Environmental and Social Management Plan (If applicable)
- ☐ Appraisal Report or Due Diligence Report with recommendations (If applicable)
- ☐ Evaluation Report of the baseline project (If applicable)
- ☐ Map indicating the location of the project/programme
- ☒ Timetable of project/programme implementation

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*





**LESOTHO METEOROLOGICAL SERVICES
MINISTRY OF ENERGY AND METEOROLOGY
P.O Box 14515, Maseru 100, Lesotho**

Email: mahahabisa@gmail.com

Tel: +266 22 317250

Ref: MEM/MET/7/50

05 June 2018

The Executive Director
The Green Climate Fund Secretariat
175, Art Center-daero, Yeonsu-gu
Incheon 406-840
Republic of Korea

Dear Madam / Sir,

**Re: Endorsement for DBSA funding proposal for the GCF to support the preparation of
“DBSA Climate Finance Facility”**

In my capacity as GCF Focal Point for Lesotho, I confirm that the above project proposal to be implemented in Lesotho as included in the funding proposal was submitted by DBSA to us in May 2018: (a) is in accordance with our Government's national priorities, strategies, plans and our commitments to the United Nations Framework Convention on Climate Change; (b) was discussed with relevant stakeholders; (c) is in accordance with the GCF's environmental and social safeguards and (d) is in conformity with relevant national laws and regulations.

Pursuant to GCF decision B.08/10, the content of which I acknowledge to have reviewed, I am pleased to endorse the preparation of the above project proposal with the support of DBSA and I hereby communicate our no-objection to the project as included in the funding proposal. I also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

If approved, the proposal will be prepared and implemented by the DBSA.

I acknowledge that this letter will be made publicly available on the GCF website.

Thank you.

Yours Sincerely,

A handwritten signature in black ink, appearing to read 'Mahahabisa'.

Mrs. Mabafokeng F. Mahahabisa
UNFCCC and GCF Focal Point in Lesotho



REPUBLIC OF NAMIBIA

MINISTRY OF ENVIRONMENT AND TOURISM

Tel. No. 061 – 2842568
Fax. No. 061 - 229936
E-mail: Petrus.Muteyauli@met.na

Cnr of Dr. Kenneth David Kaunda Street
& Robert Mugabe Avenue
Private Bag 13306
Windhoek
18 April 2018

The Secretariat
Green Climate Fund
G-Tower, 24-4 Songdo-dong, Yeonsu-gu Incheon
Republic of Korea

RE: funding proposal to the Green Climate Fund by the Development Bank of Southern Africa regarding “Climate Finance Facility”

Dear Sir/ Madam,

We refer to the project “Climate Finance Facility” in Namibia as included in the funding proposal submitted by the Development Bank of Southern Africa to us on April, 09, 2018.

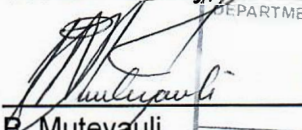
The undersigned is the duly authorized representative of the Ministry of Environment and Tourism, the National Designated Authority of Namibia. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Republic of Namibia has no objection to the project as included in the proposal;
- (b) The project as included in the funding proposal is in conformity with Namibia’s national priorities, strategies and plans;
- (c) In accordance with GCF’s environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no objection to the project as included in the funding proposal has been duly followed. We acknowledge that this letter will be made publicly available on the GCF website.

Yours Sincerely,


P. Muteyauli
GCF Focal Point for Namibia





environmental affairs

Department:
Environmental Affairs
REPUBLIC OF SOUTH AFRICA

Private Bag X 447 · PRETORIA · 0001, Environment House, 473 Steve Biko Road, Pretoria, Tel +27 12 399 9000 · Fax +27 86 625 1042

Ref: EDMS173415

Enquiries: Shahkira Parker

Tel: 012 399 9240 Fax: 012 310 3541 Email: sparker@environment.gov.za

The Green Climate Fund
Songdo Business District
175 Art center-daero
Yeonsu-gu
INCHEON 22004
REPUBLIC OF KOREA

Email: fundingproposal@gcfund.org

Dear Sir/Madam

PROPOSAL FOR THE GREEN CLIMATE FUND BY THE DEVELOPMENT BANK OF SOUTHERN AFRICA (DBSA) REGARDING THE CLIMATE FINANCE FACILITY PROGRAMME

We refer to the Climate Finance Facility Programme as included in the funding proposal submitted to us by the Development Bank of Southern Africa on 20 March 2018.

The undersigned is the duly authorised representative of the Department of Environmental Affairs, the National Designated Authority/focal point for South Africa.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the Programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- The government of South Africa has no-objection to the Programme as included in the funding proposal;
- The Programme as included in the funding proposal is in conformity with South Africa's national priorities, strategies and plans;
- In accordance with the GCF's environmental and social safeguards, the proposed project as indicated in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the Programme as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the Green Climate Fund website.

Yours sincerely

Ms Nosipho Ngcaba
DIRECTOR-GENERAL
DATE: 07/06/2018

Batho pele- putting people first



MINISTRY OF TOURISM AND ENVIRONMENTAL AFFAIRS

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Fax: +268 404-5415/404-1719/404-6438
e-mail: ps_tourism@gov.sz

P.O. Box 2652
Mbabane H100
Swaziland.

To: The Green Climate Fund (GCF)

MBABANE, 25 APRIL 2018

RE: FUNDING PROPOSAL FOR THE GCF BY THE DEVELOPMENT BANK OF SOUTHERN AFRICAN (DBSA) REGARDING “DBSA CLIMATE FINANCE FACILITY”

Dear Madam, Sir,

We refer to the programme “DBSA CLIMATE FINANCE FACILITY” in South Africa/Southern Africa Development Community as included in the funding proposal submitted by the Development Bank of Southern Africa to us on April 4th 2018.

The undersigned is the duly authorized representative of the Ministry of Tourism and Environmental Affairs, the National Designated Authority of Swaziland.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- a) The Government of Swaziland has no-objection to the programme as included in the funding proposal;
- b) The programme as included in the funding proposal is in conformity with Swaziland’s national priorities, strategies and plans;
- c) In accordance with the GCF’s environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

A handwritten signature in dark ink, appearing to be 'Emmanuel D. Dlamini', written over a light blue grid background. The signature is fluid and cursive, with a horizontal line drawn underneath it.

EMMANUEL D. DLAMINI

PRINCIPAL SECRETARY/GCF FOCAL POINT

Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	DBSA Climate Finance Facility
Accredited entity	Development Bank of Southern Africa (DBSA)
Environmental and social safeguards (ESS) category	Intermediation 2 (I2)

Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Not Applicable
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Not Applicable
Resettlement Action Plan (RAP) (if applicable)	
Date of disclosure on accredited entity's website	Not Applicable
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Environmental and Social Management Framework (ESMF)
Date of disclosure on accredited entity's website	2018-05-23
Language(s) of disclosure	English
Link to disclosure	https://www.dbsa.org/EN/About-Us/Publications/Documents/Revised%20ESS%20Framework_CFF%2021052018.pdf The ESMF mentioned in the Report is equivalent to the Environmental and Social Management System (ESMS) meeting the requirements of the IDP for Category I-2 programme.